

Space Time Algebra (STA)

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Abstract— *Space-time algebra (STA) is an associative algebra that combines scalars and vectors into a single object called a paravector. In this form, the algebra is adapted to rotations, Lorentz boosts, space-time kinematics, and relativistic field theory. This manuscript develops a common STA notation for proper velocity and momentum, Maxwell electrodynamics, and Klein–Gordon, Dirac, and Pauli-type wave equations.*

Summary of Results

Using natural units,

$$\hbar = c = 1,$$

with metric signature

$$(+, -, -, -),$$

the principal formulas developed in the manuscript are collected below.

$dX = dt + d\mathbf{r} \cdot \boldsymbol{\sigma}$	differential space-time; Eq. (174),
$ds^2 = \langle\langle dX \rangle\rangle^2 = dX dX^*$	space-time metric; Eq. (175),
$\gamma = \frac{dt}{ds} = \frac{1}{\sqrt{1 - \mathbf{v}^2}}$	Lorentz factor; Eq. (176),
$\frac{dX}{ds} = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma})$	proper velocity; Eq. (176),
$m \frac{dX}{ds} = E + \mathbf{P} \cdot \boldsymbol{\sigma}$	proper momentum; Eq. (177),
$F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$	electromagnetic field; Eq. (312),
$\partial = \partial_t + \boldsymbol{\sigma} \cdot \boldsymbol{\nabla}$	partial operator; Eq. (311),
$\partial\partial^* = \partial_t^2 - \boldsymbol{\nabla}^2$	d'Alembertian; Eq. (311),
$\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$	potential field; Eq. (312),
$\partial\Phi = S + F^*$	potential gradient; Eq. (313),
$S = \partial_t V + \boldsymbol{\nabla} \cdot \mathbf{A}$	Lorenz scalar; Eq. (313),
$\partial F = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}$	Maxwell's equations; Eq. (315),
$\partial^* \partial F = \partial^*(\rho - \mathbf{J} \cdot \boldsymbol{\sigma})$	wave equations; Eqs. (311), (318),
$m \frac{d^2 X}{ds^2} = q \operatorname{real} \left(F \frac{dX}{ds} \right)$	Lorentz force; Eqs. (178), (321),
$(\partial\partial^* + m^2)\Psi = 0$	Klein–Gordon equation; Eq. (397),
$\Omega = \langle\langle m + \boldsymbol{\sigma} \cdot \boldsymbol{\nabla} \rangle\rangle = \sqrt{m^2 - \boldsymbol{\nabla}^2}$	frequency operator; Eq. (398),
$(\partial_t \pm i\Omega)\Psi(\pm) = 0, \quad \Psi = \Psi(+) + \Psi(-)$	frequency split; Eq. (398),
$W(Z) := \{0, Z; Z^{*R}, 0\}$	2×2 matrix; Eq. (404),
$(W(i\partial) - m\mathbf{I})\Psi = 0$	Dirac equation; Eq. (405),
$(-E^2 + \mathbf{P}^2 + m^2)\Psi = 0$	Klein–Gordon with field; Eq. (400),
$E = i\partial_t - qV, \quad \mathbf{P} = -i\boldsymbol{\nabla} - q\mathbf{A}$	energy and momentum operators; Eq. (399),
$(W(i\partial - q\Phi^*) - m\mathbf{I})\Psi = 0$	Dirac with field; Eq. (406),
$2mE\Psi = (\mathbf{P}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma})\Psi$	Pauli equation form; Eq. (409),
$\Psi = \Psi(+) + \Psi(-)$	spin-up plus spin-down split; Eq. (411),
$\Pi(\pm) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \quad \Psi(\pm) = \Pi(\pm)\Psi$	spin projector; Eqs. (15), (411),
$i\partial_t \Psi(\pm) = \left(\frac{\mathbf{P}^2 \mp q\ \mathbf{B}\ }{2m} + qV \right) \Psi(\pm)$	uniform- $\mathbf{B}$ split; Eq. (412),
$X' = T^{-1}XT$	transforms; Eq. (233),
$R = \exp(\frac{1}{2}\theta \mathbf{u} \cdot i\boldsymbol{\sigma})$	rotation; Eq. (234),
$L = \exp(\frac{1}{2}\theta \mathbf{u} \cdot \boldsymbol{\sigma})$	Lorentz boost; Eq. (235).

The manuscript develops these formulas in the following order: algebraic primitives in §II, space-time kinematics in §III, transformation theory in §IV, electromagnetism in §V, and relativistic quantum equations in §VI. Each major section then closes with a detailed local development that expands the compact formulas into explicit product rules, intermediate derivations, and special cases before the manuscript moves to the next topic. Verification, historical notes, and cautions on formulas requiring qualification appear later in §VIII–§X.

Reading Guide

1. Section II fixes notation, types, involutions, products, quadratic forms, exponentials, and projectors.
2. Section III turns those algebraic objects into intervals, proper velocity, proper momentum, and free motion.
3. Section IV introduces rotors and boosts before any field theory is used.
4. Section V applies the same algebra to Maxwell theory, gauge structure, wave equations, and the Lorentz force.
5. Section VI then develops the Klein–Gordon, Dirac, and Pauli-type formulas using the earlier machinery rather than new notation.
6. Each major section ends with a detailed development subsection that integrates the broader algebraic and physical notes into the local narrative, so definitions, derivations, and special cases appear near the formulas they support.
7. Sections VIII–X separate verification, history, and cautions from the exposition, and Appendix A records selected supporting derivations.

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I Introduction

Space-time algebra extends the familiar scalar and vector calculus of three-dimensional space into a unified algebra adapted to rotations, Lorentz transformations, and relativistic field equations. A real scalar plus a real vector forms a paravector, and the same multiplication rules that generate Euclidean dot and cross products also produce the quadratic forms used for space-time intervals and field invariants.

This perspective is consistent with the modern STA and paravector literature of Hestenes, Baylis, and Doran–Lasenby [9–11], but the notation is kept deliberately lean. Lower-case Latin letters denote real scalars unless stated otherwise, bold letters denote real three-vectors, capital letters denote paravectors or operators according to context, and complexification is carried by the pseudoscalar i .

Natural units $\hbar = c = 1$ and metric signature $(+, -, -, -)$ are used unless stated otherwise. The main exposition is simple-first: the real vector product is fixed before the full complex paravector product, and each major topic is developed in two passes, first in compact form and then in a section-local detailed expansion that writes out intermediate identities and special cases. Variable classes are restated when clarity would otherwise be lost over long spans, equation references are used throughout so later formulas can point back to earlier constructions, Appendix A records supporting derivations for selected main-text equations, and bold red notes mark formulas that require qualification or correction.

The manuscript is organized to keep the logical burden low. First the algebra is developed on its own terms; next the same algebra is reinterpreted as space-time geometry; only after that are electromagnetic and quantum equations introduced. In this way the physics applications appear as consequences of a previously established language rather than as disconnected formulas gathered under one title.

The opening summary page therefore shows the destination, while the main exposition proceeds in the opposite direction: simple rules first, composite objects second, and field equations last. Whenever a formula is likely to be unfamiliar in STA form, the text around it explains both what the formula says algebraically and why it is being introduced at that point in the narrative.

II Algebraic Toolkit

The summary above is deliberately complex-first. The exposition now resets to the simplest ingredients: basis rules, involutions, vector products, paravectors, and then the elementary functional calculus built from them.

Declared types

Object	Type
$a, b \in \mathbb{R}$	real scalars
$\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$	real vectors
$c = a + ib \in \mathbb{C}$	complex scalar
$\mathbf{C} = \mathbf{A} + i\mathbf{B} \in \mathbb{C}^3$	complex vector
$X = a + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3$	real paravector
$Y = b + \mathbf{B} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3$	second real paravector
$Z = c + \mathbf{C} \cdot \boldsymbol{\sigma} = X + iY \in \mathbb{C} \oplus \mathbb{C}^3$	complex paravector

This table is archetypal rather than global. No letter is permanently tied to one type throughout the manuscript; instead, each symbol is assigned locally when it first appears in a new context. The basic ambient spaces are $\mathbb{R}$, $\mathbb{C}$, $\mathbb{R}^3$, $\mathbb{C}^3$, and the direct-sum paravector spaces $\mathbb{R} \oplus \mathbb{R}^3$ and $\mathbb{C} \oplus \mathbb{C}^3$.

The few objects that do not belong to one of those six spaces are signaled explicitly where they arise: the basis elements σ_n are algebra generators, the differential symbols ∂ , E , $\mathbf{P}$, and Ω are operators, and the Dirac map $W(Z)$ together with $\mathbf{I}$ lives in a 2×2 block-matrix representation built from paravector entries. No additional scalar fields beyond $\mathbb{R}$ and $\mathbb{C}$ are used.

Basis, pseudoscalar, and involutions

The algebra starts with three generators σ_n for $n \in \{1, 2, 3\}$. These are basis algebra elements rather than scalars, and they behave like an orthonormal spatial basis whose multiplication carries more information than an ordinary dot product. From these generators one obtains vectors, bivectors, and the pseudoscalar i , which plays the role of the imaginary unit while remaining internal to the algebra.

The basis relations are

$$\sigma_n^2 = 1, \quad \sigma_n \sigma_m = -\sigma_m \sigma_n \quad (n \neq m), \quad i := \sigma_1 \sigma_2 \sigma_3, \quad i^2 = -1. \quad (1)$$

Using δ_{nm} and ϵ_{nmk} for indices $n, m, k \in \{1, 2, 3\}$,

$$\sigma_n \sigma_m = \delta_{nm} + \epsilon_{nmk} i \sigma_k. \quad (2)$$

The three involutions used throughout are best introduced as operators before their special values are listed. Conjugation $(\cdot)^*$ flips the sign of each basis generator σ_n while leaving product order unchanged. Reversion $(\cdot)^R$ leaves each σ_n fixed while reversing the order of generator products. The combined operation $(\cdot)^{*R}$ performs both actions, and because sign-flip and order-reversal are independent, they can be applied in either order. Each is an involution, so applying it twice returns the original element. In operator form,

$$(XY)^* = X^*Y^*, \quad (XY)^R = Y^R X^R, \quad X^{*R} := (X^R)^* = (X^*)^R. \quad (3)$$

In particular, on the generators and the pseudoscalar,

$$\sigma_n^* = -\sigma_n, \quad \sigma_n^R = \sigma_n, \quad \sigma_n^{*R} = -\sigma_n, \quad (4)$$

$$i^* = -i, \quad i^R = -i, \quad i^{*R} = i. \quad (5)$$

These agree with the standard grade-involution/reversion structure in geometric algebra treatments of STA [9, 11]. These operations will be used constantly, so it is worth emphasizing their distinct roles. Conjugation flips vector sign, reversion reverses multiplication order, and the combined operation is the one that later produces the Minkowski-type quadratic form. Introducing them at the algebraic stage prevents the interval and operator formulas from seeming arbitrary when they appear later.

Vector product identity and complex paravectors

Collect the three generators into the sigma-vector

$$\boldsymbol{\sigma} := \{\sigma_1, \sigma_2, \sigma_3\}, \quad (6)$$

which is a vector of algebraic objects rather than a vector of scalar components. With that notation fixed, ordinary three-vectors act on the algebra by contraction against $\boldsymbol{\sigma}$.

For real vectors $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$,

$$(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) = \mathbf{A} \cdot \mathbf{B} + i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (7)$$

This real-vector product is the first structural identity because it makes the scalar/vector split transparent before any complexification is introduced. The fully complex product then follows as the master formula from which the simpler cases are recovered by restriction.

For real scalars $a, b \in \mathbb{R}$ and real vectors $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$, the corresponding complex scalar $S = a + ib \in \mathbb{C}$, complex vector $\mathbf{V} = \mathbf{A} + i\mathbf{B} \in \mathbb{C}^3$, real paravectors $X, Y \in \mathbb{R} \oplus \mathbb{R}^3$, and complex paravector $Z \in \mathbb{C} \oplus \mathbb{C}^3$ are related by

$$Z = (a + ib) + (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma} = S + \mathbf{V} \cdot \boldsymbol{\sigma} = X + iY, \quad (8)$$

with

$$S = a + ib \in \mathbb{C}, \quad \mathbf{V} = \mathbf{A} + i\mathbf{B} \in \mathbb{C}^3, \quad X = a + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \quad Y = b + \mathbf{B} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3. \quad (9)$$

For $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ and $Z' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma}$ with $S, S' \in \mathbb{C}$ and $\mathbf{V}, \mathbf{V}' \in \mathbb{C}^3$,

$$ZZ' = (SS' + \mathbf{V} \cdot \mathbf{V}') + (S\mathbf{V}' + S'\mathbf{V} + i(\mathbf{V} \times \mathbf{V}')) \cdot \boldsymbol{\sigma}. \quad (10)$$

Equation (10) is the master multiplication rule: pure scalar, real vector, real paravector, and mixed products all follow by setting the unused parts to zero.

Two central quadratic forms are

$$\|Z\|^2 := \text{scalar}(ZZ^R) = |S|^2 + \|\mathbf{V}\|^2, \quad (11)$$

$$\langle\langle Z \rangle\rangle^2 := ZZ^{*R} = S^2 - \mathbf{V} \cdot \mathbf{V}. \quad (12)$$

Equation (12) is the paravector form later used for space-time intervals [10, 11]. This distinction between Euclidean and Minkowski-type quadratic forms is one of the main structural themes of the manuscript. The first measures the size of the full complex paravector in a positive way, while the second has the mixed sign structure needed for relativistic geometry. Much of the later physics is the disciplined choice of which quadratic form is appropriate in each setting.

Exponentials, projectors, and functional calculus

Exponentials, roots, and functions of simple STA objects follow directly once Eqs. (2) and (7) are fixed.

If $\mathbf{A} \in \mathbb{R}^3$ is written as $\mathbf{A} = \|\mathbf{A}\|\mathbf{u}$ with $\mathbf{u} \in \mathbb{R}^3$ and $\mathbf{u}^2 = 1$, then

$$\exp(\mathbf{A} \cdot \boldsymbol{\sigma}) = \cosh \|\mathbf{A}\| + \sinh \|\mathbf{A}\| \mathbf{u} \cdot \boldsymbol{\sigma}, \quad (13)$$

$$\exp(-i\mathbf{A} \cdot \boldsymbol{\sigma}) = \cos \|\mathbf{A}\| - i \sin \|\mathbf{A}\| \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (14)$$

The idempotents

$$\Pi(\pm) := \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}) \quad (15)$$

satisfy $\Pi(\pm)^2 = \Pi(\pm)$, $\Pi(+)\Pi(-) = 0$, and $\Pi(+)+\Pi(-) = 1$. They underlie the later discussions of functions of $\mathbf{u} \cdot \boldsymbol{\sigma}$, spin decomposition, and the matrix/quaternion bridge. Geometrically, $\Pi(+)$ and $\Pi(-)$ isolate the parts aligned and anti-aligned with the direction $\mathbf{u}$. Later these same idempotents become spin projectors, but their algebraic role is already visible here: they split an object into two sectors adapted to a chosen axis.

For a scalar-valued function $f : \{\pm 1\} \rightarrow \mathbb{C}$,

$$f(\mathbf{u} \cdot \boldsymbol{\sigma}) = f(+)\Pi(+)+f(-)\Pi(-). \quad (16)$$

This gives a compact spectral calculus for logarithms, exponentials, and projector-based decompositions. The significance of Eq. (16) is broader than the specific examples used later. Once the minimal polynomial of an STA element is known, functions of that element can be defined without leaving the algebraic framework already established.

Extended Algebraic Identities

The compact formulas above give the algebraic spine of the manuscript. The next layer records the more explicit commutator rules, involution tables, inverse formulas, logarithms, complex-product decompositions, and functional identities that are used implicitly later. They are kept here, rather than postponed to a detached catalog, so that the reader can verify every algebraic step before the geometry and physics sections begin.

Commutator Algebra

Before specializing to particular STA objects, it is helpful to record the algebraic behavior of the commutator itself, since many later identities are compact ways of saying that certain products commute or fail to commute in controlled ways. See Eqs. (2) and (7). Here $X, Y \in \mathbb{C} \oplus \mathbb{C}^3$ are generic complex paravectors, and $[X, Y] := XY - YX$ is their commutator. The following numbered identities fix the notation and formulas used in this block.

$$[X, Y] := XY - YX, \quad (17)$$

$$[Y, X] = -[X, Y], \quad (18)$$

$$[X, Y] = 0 \Rightarrow XY = YX \Rightarrow \text{commutative}. \quad (19)$$

Vector Notation, Norms, and Cross Products

The elementary vector conventions are made fully explicit before introducing the sigma basis. That choice is retained here so that later products and norms can be read against familiar Euclidean notation. The following numbered identities fix the notation and formulas used in this block.

$$\mathbf{A} = \text{3-element vector}, \quad (20)$$

$$\mathbf{B} = \text{3-element vector}. \quad (21)$$

The dot product is defined as follows.

$$\mathbf{A} \cdot \mathbf{B} = A_1B_1 + A_2B_2 + A_3B_3. \quad (22)$$

We denote the squared vector magnitude by.

$$\mathbf{A}^2 = \mathbf{A} \cdot \mathbf{A}. \quad (23)$$

The scalar norm is defined by.

For $a \in \mathbb{R}$, the following formulas apply.

$$|a| = \sqrt{a^2}. \quad (24)$$

For $a \in \mathbb{R}, b \in \mathbb{R}, c = a + ib$, the following formulas apply.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
|c| &= \sqrt{cc^*}, \\
&= \sqrt{(a+ib)(a-ib)}, \\
&= \sqrt{a^2 + b^2}.
\end{aligned} \tag{25}$$

The final numbered line is the derived relation used in the subsequent discussion.

The vector norm is defined by.

For $\mathbf{A} \in \mathbb{R}^3$, the following formulas apply.

$$\|\mathbf{A}\| = \sqrt{\mathbf{A}^2}. \tag{26}$$

For $\mathbf{A} \in \mathbb{R}^3, \mathbf{B} \in \mathbb{R}^3, \mathbf{C} = \mathbf{A} + i\mathbf{B}$, the following formulas apply.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
\|\mathbf{C}\| &= \sqrt{CC^*}, \\
&= \sqrt{(\mathbf{A} + i\mathbf{B}) \cdot (\mathbf{A} - i\mathbf{B})}, \\
&= \sqrt{\mathbf{A}^2 + \mathbf{B}^2}.
\end{aligned} \tag{27}$$

The final numbered line is the derived relation used in the subsequent discussion.

Note. The quadratic product C^2 is not the same object as the norm square $\|\mathbf{C}\|^2$.

The cross product is defined by.

$$\mathbf{A} \times \mathbf{B} = \{A_2B_3 - A_3B_2, -A_1B_3 + A_3B_1, A_1B_2 - A_2B_1\}. \tag{28}$$

$$\mathbf{B} \times \mathbf{A} = -\mathbf{A} \times \mathbf{B}. \tag{29}$$

Sigma Algebra

The sigma generators and their multiplication rules are written out here in full so that every later paravector identity can be traced back to explicit algebraic relations. See Eqs. (1) and (2). For $n, m \in \{1, 2, 3\}$, the following identities hold.

Let σ_n denote a basis algebra element.

$$\sigma_n^2 = +1, \sigma_n \sigma_m = -\sigma_m \sigma_n \text{ if } m \neq n. \tag{30}$$

The basic grade decomposition of the sigma algebra is summarized in Table 1, which is the classification used throughout the later product expansions.

Table 1: Grade decomposition of the sigma algebra.

scalar:	1
vector:	$\sigma_1, \sigma_2, \sigma_3$
bivector:	$\sigma_2\sigma_3, \sigma_3\sigma_1, \sigma_1\sigma_2$
pseudo-scalar:	$\sigma_1\sigma_2\sigma_3$

The pseudoscalar connects the sigma algebra to the ordinary imaginary unit.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
i &= \sigma_1\sigma_2\sigma_3, \\
i^2 &= -1, \\
(\sigma_1\sigma_2\sigma_3)^2 &= \sigma_1\sigma_2(\sigma_3\sigma_1)\sigma_2\sigma_3, \\
&= -\sigma_1(\sigma_2\sigma_1)(\sigma_3\sigma_2)\sigma_3, \\
&= -\sigma_1^2\sigma_2^2\sigma_3^2, \\
&= -1.
\end{aligned} \tag{31}$$

The final numbered line is the derived relation used in the subsequent discussion.

The bivector sector is given by.

Let $i\sigma$ denote the vector of bivectors.

$$= \{\sigma_2\sigma_3, \sigma_3\sigma_1, \sigma_1\sigma_2\}, \quad (32)$$

$$[i, \sigma_n] = 0, \quad (33)$$

$$i\sigma_n = \sigma_n i \quad \text{commutative.} \quad (34)$$

The Kronecker delta is defined by.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \delta_{nm} &= 1 \text{ if } n = m, \\ &= 0 \text{ else.} \end{aligned} \quad (35)$$

The final numbered line is the derived relation used in the subsequent discussion.

The Levi-Civita symbol is defined by.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \epsilon_{nmk} &= +1 \text{ if } \{1, 2, 3\}, \{2, 3, 1\}, \{3, 1, 2\} && \text{even permutation,} \\ &= -1 \text{ if } \{3, 2, 1\}, \{1, 3, 2\}, \{2, 1, 3\} && \text{odd permutation,} \\ &= 0 \text{ if any two indices are equal.} \end{aligned} \quad (36)$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \sigma_n \sigma_m &= \delta_{nm} + \epsilon_{nmk} i \sigma_k, \\ &= \epsilon_{nmk} i \sigma_k \text{ if } n \neq m. \end{aligned} \quad (37)$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \sigma_1 \sigma_2 \sigma_3 &= (\delta_{12} + \epsilon_{123} i \sigma_3) \sigma_3, \\ &= (0 + i \sigma_3) \sigma_3 = i. \end{aligned} \quad (38)$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} &\text{non-commutative algebra if } n \neq m, \\ &[\sigma_n, \sigma_m] = \sigma_n \sigma_m - \sigma_m \sigma_n, \\ &= \sigma_n \sigma_m - (-\sigma_n \sigma_m), \\ &= 2\sigma_n \sigma_m, \\ &= 2\epsilon_{nmk} i \sigma_k. \end{aligned} \quad (39)$$

The final numbered line is the derived relation used in the subsequent discussion.

The following numbered identities fix the notation and formulas used in this block.

$$[\sigma_2, \sigma_3] = 2\sigma_1\sigma_2\sigma_3\sigma_1 = 2i\sigma_1, \quad (40)$$

$$[\sigma_3, \sigma_1] = 2\sigma_1\sigma_2\sigma_3\sigma_2 = 2i\sigma_2, \quad (41)$$

$$[\sigma_1, \sigma_2] = 2\sigma_1\sigma_2\sigma_3\sigma_3 = 2i\sigma_3. \quad (42)$$

Conjugation of Sigma Products

The formulas below make conjugation of sigma products explicit and collect identities used later in the manuscript. See Eqs. (4) and (5). Conjugation changes the sign of each basis vector while preserving scalar coefficients. The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
\sigma_n^* &= -\sigma_n, \\
i^* &= -i, \\
(i\sigma_n)^* &= i\sigma_n, \\
(\sigma_n\sigma_m)^* &= \sigma_n\sigma_m, \\
(\sigma_1\sigma_2\sigma_3)^* &= \sigma_1^*\sigma_2^*\sigma_3^* \quad \text{does not flip order,} \\
&= (-\sigma_1)(-\sigma_2)(-\sigma_3), \\
&= -\sigma_1\sigma_2\sigma_3.
\end{aligned} \tag{43}$$

The final numbered line is the derived relation used in the subsequent discussion. The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
(\sigma_n\sigma_m)^* &= \sigma_n^*\sigma_m^*, \\
&= (-\sigma_n)(-\sigma_m), \\
&= \sigma_n\sigma_m.
\end{aligned} \tag{44}$$

The final numbered line is the derived relation used in the subsequent discussion. The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
(i\sigma_n)^* &= i^*\sigma_n^*, \\
&= (-i)(-\sigma_n), \\
&= i\sigma_n.
\end{aligned} \tag{45}$$

The final numbered line is the derived relation used in the subsequent discussion.

Reversion of Sigma Products

The formulas below make reversion of sigma products explicit and collect identities used later in the manuscript. See Eqs. (4) and (5). Reversion reverses the order of each basis-vector product. The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
\sigma_n^R &= \sigma_n, \\
i^R &= -i \quad (\sigma_1\sigma_2\sigma_3)^R = \sigma_3\sigma_2\sigma_1 = -\sigma_1\sigma_2\sigma_3, \\
(\sigma_n\sigma_m)^R &= \sigma_m\sigma_n, \\
&= -\sigma_n\sigma_m \text{ if } n \neq m.
\end{aligned} \tag{46}$$

The final numbered line is the derived relation used in the subsequent discussion. The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
(i\sigma_n)^R &= -i\sigma_n, \\
(i\sigma_n)^R &= \sigma_n(i)^R, \\
&= \sigma_n(-i), \\
&= -i\sigma_n.
\end{aligned} \tag{47}$$

The final numbered line is the derived relation used in the subsequent discussion.

Sign Table for the Involutions

The formulas below make sign table for the involutions explicit and collect identities used later in the manuscript. See Eqs. (4) and (5). Let 0 denote scalar.

$$1 = \text{vector}, \sigma_n, \quad (48)$$

$$2 = \text{bi-vector}, \sigma_n \sigma_m = \epsilon_{nmk} i \sigma_k, \quad (49)$$

$$3 = \text{pseudo-scalar}, i = \sigma_1 \sigma_2 \sigma_3. \quad (50)$$

The conjugation signs are summarized below.

$$0 : 1^* = 1, \quad (51)$$

$$1 : \sigma_n^* = -\sigma_n, \quad (52)$$

$$2 : (i\sigma_n)^* = i\sigma_n, \quad (53)$$

$$3 : i^* = -i. \quad (54)$$

The reversion signs are summarized below.

$$0 : 1^R = 1, \quad (55)$$

$$1 : \sigma_n^R = \sigma_n, \quad (56)$$

$$2 : (i\sigma_n)^R = -i\sigma_n, \quad (57)$$

$$3 : i^R = -i. \quad (58)$$

The combined conjugation–reversion signs are summarized below.

$$0 : 1^{*R} = 1, \quad (59)$$

$$1 : \sigma_n^{*R} = -\sigma_n, \quad (60)$$

$$2 : (i\sigma_n)^{*R} = -i\sigma_n, \quad (61)$$

$$3 : i^{*R} = i. \quad (62)$$

$$\begin{array}{c|cccc} & 0 & 1 & 2 & 3 \\ \hline ()^* & + & - & + & - \\ ()^R & + & + & - & - \\ ()^{*R} & + & - & - & + \end{array}$$

Vector Form and Explicit Sigma Products

The formulas below make vector form and explicit sigma products explicit and collect identities used later in the manuscript. See Eq. (7). The following numbered relation is recorded for later use.

$$\boldsymbol{\sigma} = \{\sigma_1, \sigma_2, \sigma_3\} \quad \text{vector of algebraic objects.} \quad (63)$$

Let the following notation be fixed.

$$\mathbf{A} = \text{3-element vector}, \quad (64)$$

$$\mathbf{B} = \text{3-element vector}. \quad (65)$$

The dot product is defined as follows.

$$\mathbf{A} \cdot \boldsymbol{\sigma} = A_1 \sigma_1 + A_2 \sigma_2 + A_3 \sigma_3. \quad (66)$$

Addition is handled componentwise.

$$\mathbf{A} \cdot \boldsymbol{\sigma} + \mathbf{B} \cdot \boldsymbol{\sigma} = (\mathbf{A} + \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (67)$$

The corresponding multiplication law is.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) &= \mathbf{A} \cdot \mathbf{B} + i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\
i &= \sigma_1 \sigma_2 \sigma_3, \\
&= \sum_n \sum_m A_n B_m \sigma_n \sigma_m, \\
&= \sum_n A_n B_n \sigma_n^2 + \sum_{n < m} (A_n B_m - A_m B_n) \sigma_n \sigma_m, \\
&= \mathbf{A} \cdot \mathbf{B}, \\
&\quad + (A_1 B_2 - A_2 B_1) \sigma_1 \sigma_2, \\
&\quad + (A_1 B_3 - A_3 B_1) \sigma_1 \sigma_3, \\
&\quad + (A_2 B_3 - A_3 B_2) \sigma_2 \sigma_3, \\
&= \mathbf{A} \cdot \mathbf{B}, \\
&\quad + (A_1 B_2 - A_2 B_1) \sigma_1 \sigma_2 (\sigma_3^2), \\
&\quad + (A_1 B_3 - A_3 B_1) \sigma_1 (\sigma_2^2) \sigma_3, \\
&\quad + (A_2 B_3 - A_3 B_2) (\sigma_1^2) \sigma_2 \sigma_3, \\
&= \mathbf{A} \cdot \mathbf{B}, \\
&\quad + (A_1 B_2 - A_2 B_1) (\sigma_1 \sigma_2 \sigma_3) \sigma_3, \\
&\quad - (A_1 B_3 - A_3 B_1) (\sigma_1 \sigma_2 \sigma_3) \sigma_2, \\
&\quad + (A_2 B_3 - A_3 B_2) (\sigma_1 \sigma_2 \sigma_3) \sigma_1.
\end{aligned} \quad (68)$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
(\mathbf{A} \cdot \boldsymbol{\sigma})^2 &= \mathbf{A}^2, \\
(\mathbf{A} \cdot \boldsymbol{\sigma})^{2n} &= (\mathbf{A}^2)^n, \\
(\mathbf{A} \cdot \boldsymbol{\sigma})^{2n+1} &= (\mathbf{A}^2)^n (\mathbf{A} \cdot \boldsymbol{\sigma}), \\
((\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}))^* &= (\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}), \\
(\mathbf{A} \cdot \boldsymbol{\sigma})\boldsymbol{\sigma} &= \mathbf{A} + i\mathbf{A} \times \boldsymbol{\sigma}, \\
\sigma_n \boldsymbol{\sigma} &= (e_n \cdot \boldsymbol{\sigma})\boldsymbol{\sigma}, \\
&= e_n + ie_n \times \boldsymbol{\sigma}.
\end{aligned} \quad (69)$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) &= \mathbf{A} \cdot \mathbf{B} + i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\
(\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{A} \cdot \boldsymbol{\sigma}) &= \mathbf{A} \cdot \mathbf{B} - i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\
[\mathbf{A} \cdot \boldsymbol{\sigma}, \mathbf{B} \cdot \boldsymbol{\sigma}] &= 2i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\
(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma} &= [\mathbf{A} \cdot \boldsymbol{\sigma}, \mathbf{B} \cdot \boldsymbol{\sigma}]/(2i), \\
\mathbf{A} \cdot \mathbf{B} &= ((\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) + (\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{A} \cdot \boldsymbol{\sigma}))/2, \\
[\mathbf{A} \cdot \boldsymbol{\sigma}, \boldsymbol{\sigma}] &= -2i\mathbf{A} \times \boldsymbol{\sigma} = 2i\boldsymbol{\sigma} \times \mathbf{A}, \\
[\mathbf{A} \cdot (i\boldsymbol{\sigma}), \boldsymbol{\sigma}] &= 2\mathbf{A} \times \boldsymbol{\sigma}, \\
[\mathbf{A} \cdot \boldsymbol{\sigma}, \boldsymbol{\sigma}] \cdot \mathbf{B} &= 2i(\boldsymbol{\sigma} \times \mathbf{A}) \cdot \mathbf{B}, \\
&= 2i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma},
\end{aligned}$$

$$= [\mathbf{A} \cdot \boldsymbol{\sigma}, \mathbf{B} \cdot \boldsymbol{\sigma}]. \quad (70)$$

The final numbered line is the derived relation used in the subsequent discussion.

Products of Real Scalars, Vectors, and Paravectors

This subsection expands the basic scalar, vector, and paravector product rules used throughout the manuscript. The intermediate steps are written out explicitly as follows.

$$\begin{aligned} X &= a + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \\ Y &= b + \mathbf{B} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \\ XY &= (a + \mathbf{A} \cdot \boldsymbol{\sigma})(b + \mathbf{B} \cdot \boldsymbol{\sigma}), \\ &= ab + a(\mathbf{B} \cdot \boldsymbol{\sigma}) + b(\mathbf{A} \cdot \boldsymbol{\sigma}) + (\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}), \\ &= (ab + \mathbf{A} \cdot \mathbf{B}) + (a\mathbf{B} + b\mathbf{A} + i(\mathbf{A} \times \mathbf{B})) \cdot \boldsymbol{\sigma}, \\ &= YX \text{ if } \mathbf{A} \times \mathbf{B} = 0. \end{aligned} \quad (71)$$

The final numbered line is the derived relation used in the subsequent discussion.

The product XY generally lies in $\mathbb{R} \oplus \mathbb{C}^3$.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} XY &\in \mathbb{R} \oplus \mathbb{R}^3 \text{ if } \mathbf{A} \times \mathbf{B} = 0, \\ [X, Y] &= 2i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\ &= 0 \text{ if } \mathbf{A} \times \mathbf{B} = 0 \quad \text{conditionally commutative.} \end{aligned} \quad (72)$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} (XY)^{\mathbf{R}} &= (Y)^{\mathbf{R}} X^{\mathbf{R}}, \\ &= \dots + i^{\mathbf{R}}(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\ &= \dots - i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\ &= \dots + i(\mathbf{B} \times \mathbf{A}) \cdot \boldsymbol{\sigma}, \\ &= YX. \end{aligned} \quad (73)$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \text{scalar}(XY) &= ab + \mathbf{A} \cdot \mathbf{B}, \\ \text{vector}(XY) &= a\mathbf{B} + b\mathbf{A} + i(\mathbf{A} \times \mathbf{B}), \\ \text{scalar}(YX) &= \text{scalar}(XY), \\ \text{vector}(YX) &= a\mathbf{B} + b\mathbf{A} + i(\mathbf{B} \times \mathbf{A}), \\ &= \text{vector}(XY) \text{ if } \mathbf{A} \times \mathbf{B} = 0. \end{aligned} \quad (74)$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \text{scalar}(X^2) &= a^2 + \mathbf{A}^2, \\ \text{vector}(X^2) &= 2a\mathbf{A}, \\ X^2 &= a^2 + \mathbf{A}^2 + 2a\mathbf{A} \cdot \boldsymbol{\sigma}, \\ &= b + \mathbf{B} \cdot \boldsymbol{\sigma}, \end{aligned}$$

$$= Y. \quad (75)$$

The final numbered line is the derived relation used in the subsequent discussion.
The following numbered identities fix the notation and formulas used in this block.

$$X = \pm\sqrt{Y}, \quad (76)$$

$$a^2 = \frac{1}{2}(b \pm \sqrt{b^2 - \mathbf{B}^2}), \quad (77)$$

$$\mathbf{A}^2 = b - a^2. \quad (78)$$

The minimal polynomial is.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} X^2 - 2aX + (a^2 - \mathbf{A}^2) &= 0, \\ \langle\langle XY \rangle\rangle^2 &= \langle\langle Y \rangle\rangle^2 \langle\langle X \rangle\rangle^2 \text{ if } \mathbf{A} \times \mathbf{B} = 0, \\ X^{-1} &= X^*/(XX^*), \\ &= (a - \mathbf{A} \cdot \boldsymbol{\sigma})/(a^2 - \mathbf{A}^2) \text{ if } a^2 \neq \mathbf{A}^2. \end{aligned} \quad (79)$$

The final numbered line is the derived relation used in the subsequent discussion.
The following numbered identities fix the notation and formulas used in this block.

$$X^{-1}X = 1, \quad (80)$$

$$d(X^{-1}) = -X^{-1}(dX)X^{-1}, \quad (81)$$

$$X^{-1}Y = YX^{-1} \text{ if } \mathbf{A} \times \mathbf{B} = 0. \quad (82)$$

Exponential of a Real Vector

The formulas below make exponential of a real vector explicit and collect identities used later in the manuscript. The intermediate steps are written out explicitly as follows.

$$\begin{aligned} X &= a + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \\ \exp(X) &= \exp(a) \exp(\mathbf{A} \cdot \boldsymbol{\sigma}), \\ \exp(X^*) &= \exp(a) \exp(-\mathbf{A} \cdot \boldsymbol{\sigma}) = \exp(X)^*, \\ \exp(-X) &= \exp(-a) \exp(-\mathbf{A} \cdot \boldsymbol{\sigma}) = \exp(X)^{-1}, \\ \mathbf{u} &= \mathbf{A}/\|\mathbf{A}\| \text{ if } \mathbf{A} \neq 0, \\ \mathbf{u} &= (u_1, u_2, u_3), \mathbf{u}^2 = 1, \\ \exp(\mathbf{A} \cdot \boldsymbol{\sigma}) &= \exp(\|\mathbf{A}\|(\mathbf{u} \cdot \boldsymbol{\sigma})), \\ &= \sum_n \|\mathbf{A}\|^n (\mathbf{u} \cdot \boldsymbol{\sigma})^n / n!, \\ &= \left(\sum_n \|\mathbf{A}\|^{2n} / (2n)! \right) + \left(\sum_n \|\mathbf{A}\|^{2n+1} / (2n+1)! \right) (\mathbf{u} \cdot \boldsymbol{\sigma}), \\ &= \cosh(\|\mathbf{A}\|) + \sinh(\|\mathbf{A}\|)(\mathbf{u} \cdot \boldsymbol{\sigma}), \\ &= \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}). \end{aligned} \quad (83)$$

The final numbered line is the derived relation used in the subsequent discussion.

With these definitions fixed, one obtains.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \gamma &= \cosh(\|\mathbf{A}\|), \\ \mathbf{v} &= \tanh(\|\mathbf{A}\|)\mathbf{u}, \end{aligned}$$

$$\begin{aligned}
\exp(\mathbf{A} \cdot \boldsymbol{\sigma})^* &= \gamma(1 - \mathbf{v} \cdot \boldsymbol{\sigma}), \\
&= \exp(-\mathbf{A} \cdot \boldsymbol{\sigma}), \\
&= \exp((\mathbf{A} \cdot \boldsymbol{\sigma})^*), \\
&= \exp(\mathbf{A} \cdot \boldsymbol{\sigma})^{-1}.
\end{aligned} \tag{84}$$

The final numbered line is the derived relation used in the subsequent discussion.

The corresponding rotation statement is.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
\exp(-i\mathbf{A} \cdot \boldsymbol{\sigma}) &= \exp(\|\mathbf{A}\|(-i\mathbf{u} \cdot \boldsymbol{\sigma})), \\
&= \cos(\|\mathbf{A}\|) + \sin(\|\mathbf{A}\|)(\mathbf{u} \cdot (-i\boldsymbol{\sigma})), \\
&= \cos(\|\mathbf{A}\|) + \sin(\|\mathbf{A}\|)(\mathbf{u} \cdot \mathbf{q}) \quad \text{see quaternions.}
\end{aligned} \tag{85}$$

The final numbered line is the derived relation used in the subsequent discussion.

Logarithm of a Real Paravector

The formulas below make logarithm of a real paravector explicit and collect identities used later in the manuscript. The following numbered identities fix the notation and formulas used in this block.

$$X = a + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \tag{86}$$

$$Y = b + \mathbf{B} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3. \tag{87}$$

Find (a, A) from (b, B) s.t.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
b + \mathbf{B} \cdot \boldsymbol{\sigma} &= \exp(a + \mathbf{A} \cdot \boldsymbol{\sigma}), \\
Y &= \exp(X), \\
&= \exp(a + \mathbf{A} \cdot \boldsymbol{\sigma}), \\
&= \exp(a) \exp(\mathbf{A} \cdot \boldsymbol{\sigma}), \\
&= \exp(a)(\cosh(\|\mathbf{A}\|) + \sinh(\|\mathbf{A}\|)(\mathbf{u} \cdot \boldsymbol{\sigma})), \\
&= b + \mathbf{B} \cdot \boldsymbol{\sigma}, \\
&= b + \|\mathbf{B}\|(\mathbf{u} \cdot \boldsymbol{\sigma}).
\end{aligned} \tag{88}$$

The final numbered line is the derived relation used in the subsequent discussion.

Let the following notation be fixed.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
\|\mathbf{A}\| &= \operatorname{arctanh}(\|\mathbf{B}\|/b), \\
\exp(a) &= \sqrt{b^2 - \mathbf{B}^2}, \\
\mathbf{u} &= \mathbf{B}/\|\mathbf{B}\|, \\
\log(Y) &= \log(\exp(X)), \\
&= X = a + \mathbf{A} \cdot \boldsymbol{\sigma}.
\end{aligned} \tag{89}$$

The final numbered line is the derived relation used in the subsequent discussion.

From these identities, the resulting parameter formulas are.

$$a = \frac{1}{2} \log(b^2 - \mathbf{B}^2), \tag{90}$$

$$\mathbf{A} = \|\mathbf{A}\|\mathbf{u} = \operatorname{arctanh}(\|\mathbf{B}\|/b)(\mathbf{B}/\|\mathbf{B}\|). \tag{91}$$

Complex Paravectors

The formulas below make complex paravectors explicit and collect identities used later in the manuscript. See Eqs. (8) and (9). The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
i &= \sigma_1 \sigma_2 \sigma_3, \\
X &= a + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \\
Y &= b + \mathbf{B} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \\
Z &= c + \mathbf{C} \cdot \boldsymbol{\sigma} \in \mathbb{C} \oplus \mathbb{C}^3, \\
&= X + iY, \\
&= (a + bi) + (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma} \quad \text{paravector.}
\end{aligned} \tag{92}$$

The final numbered line is the derived relation used in the subsequent discussion. The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
a &= \text{scalar}(X) \in \mathbb{R}, \\
\mathbf{A} &= \text{vector}(X) \in \mathbb{R}^3, \\
b &= \text{scalar}(Y) \in \mathbb{R}, \\
\mathbf{B} &= \text{vector}(Y) \in \mathbb{R}^3, \\
c &= \text{scalar}(Z) \in \mathbb{C} \quad \text{complex scalar}, \\
\mathbf{C} &= \text{vector}(Z) \in \mathbb{C}^3 \quad \text{complex vector}, \\
Z^* &= (a - bi) + (-\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\
Z^R &= (a - bi) + (\mathbf{A} - i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\
Z^{R*} &= (a + bi) - (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\
X &= \text{real}(Z) = (Z + Z^R)/2, \\
Y &= \text{img}(Z) = (Z - Z^R)/(2i), \\
c &= a + bi, \\
&= (Z + Z^{R*})/2.
\end{aligned} \tag{93}$$

The final numbered line is the derived relation used in the subsequent discussion. The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
\mathbf{C} \cdot \boldsymbol{\sigma} &= (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\
&= (Z - Z^{R*})/2.
\end{aligned} \tag{94}$$

The final numbered line is the derived relation used in the subsequent discussion.

The recovery formulas for the scalar, vector, real, and imaginary parts of the complex paravector are collected in Table 2.

Table 2: Components of Z .

$$\begin{aligned}
\text{scalar: } a + bi &= (Z + Z^{R*})/2 \\
\text{vector: } (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma} &= (Z - Z^{R*})/2 \\
\text{real part: } a + \mathbf{A} \cdot \boldsymbol{\sigma} &= (Z + Z^R)/2 \\
\text{imaginary part: } b + \mathbf{B} \cdot \boldsymbol{\sigma} &= (Z - Z^R)/(2i) \\
\text{real scalar: } a &= (Z + Z^* + Z^R + Z^{R*})/4 \\
\text{imaginary scalar: } b &= (Z - Z^* - Z^R + Z^{R*})/(4i) \\
\text{real vector: } \mathbf{A} \cdot \boldsymbol{\sigma} &= (Z - Z^* + Z^R - Z^{R*})/4 \\
\text{imaginary vector: } \mathbf{B} \cdot \boldsymbol{\sigma} &= (Z + Z^* - Z^R - Z^{R*})/(4i)
\end{aligned}$$

Products of Complex Paravectors

The formulas below make products of complex paravectors explicit and collect identities used later in the manuscript. See Eq. (10). The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
 X &= a + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \\
 Y &= b + \mathbf{B} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \\
 Z &= c + \mathbf{C} \cdot \boldsymbol{\sigma} \in \mathbb{C} \oplus \mathbb{C}^3, \\
 &= X + iY.
 \end{aligned} \tag{95}$$

The final numbered line is the derived relation used in the subsequent discussion.

Table 3: Components of ZZ' .

$$\begin{aligned}
 \text{real scalar:} & \quad aa' - bb' + \mathbf{A} \cdot \mathbf{A}' - \mathbf{B} \cdot \mathbf{B}' \\
 \text{imaginary scalar:} & \quad ab' + ba' + \mathbf{A} \cdot \mathbf{B}' + \mathbf{B} \cdot \mathbf{A}' \\
 \text{real vector:} & \quad a\mathbf{A}' - b\mathbf{B}' + a'\mathbf{A} - b'\mathbf{B} - \mathbf{A} \times \mathbf{B}' - \mathbf{B} \times \mathbf{A}' \\
 \text{imaginary vector:} & \quad a\mathbf{B}' + b\mathbf{A}' + a'\mathbf{B} + b'\mathbf{A} + \mathbf{A} \times \mathbf{A}' - \mathbf{B} \times \mathbf{B}'
 \end{aligned}$$

Table 4: Components of Z^2 .

$$\begin{aligned}
 \text{real scalar:} & \quad a^2 - b^2 + \mathbf{A}^2 - \mathbf{B}^2 \\
 \text{imaginary scalar:} & \quad 2ab + 2\mathbf{A} \cdot \mathbf{B} \\
 \text{real vector:} & \quad 2(a\mathbf{A} - b\mathbf{B}) \\
 \text{imaginary vector:} & \quad 2(a\mathbf{B} + b\mathbf{A})
 \end{aligned}$$

Table 5: Components of ZZ^* .

$$\begin{aligned}
 \text{real scalar:} & \quad a^2 + b^2 - \mathbf{A}^2 - \mathbf{B}^2 \\
 \text{imaginary scalar:} & \quad 0 \\
 \text{real vector:} & \quad -2\mathbf{A} \times \mathbf{B} \quad \leftarrow \text{different from } Z^*Z \\
 \text{imaginary vector:} & \quad 2(a\mathbf{B} - b\mathbf{A})
 \end{aligned}$$

Table 6: Components of Z^*Z .

$$\begin{aligned}
 \text{real scalar:} & \quad a^2 + b^2 - \mathbf{A}^2 - \mathbf{B}^2 \\
 \text{imaginary scalar:} & \quad 0 \\
 \text{real vector:} & \quad +2(\mathbf{A} \times \mathbf{B}) \quad \leftarrow \text{different from } ZZ^* \\
 \text{imaginary vector:} & \quad 2(a\mathbf{B} - b\mathbf{A})
 \end{aligned}$$

Table 7: Components of ZZ^R .

$$\begin{aligned}
 \text{real scalar:} & \quad a^2 + b^2 + \mathbf{A}^2 + \mathbf{B}^2 \\
 \text{imaginary scalar:} & \quad 0 \\
 \text{real vector:} & \quad 2(a\mathbf{A} + b\mathbf{B} + \mathbf{A} \times \mathbf{B}) \quad \leftarrow \text{different from } Z^RZ \\
 \text{imaginary vector:} & \quad \mathbf{0}
 \end{aligned}$$

Table 8: Components of Z^RZ .

$$\begin{aligned}
 \text{real scalar:} & \quad a^2 + b^2 + \mathbf{A}^2 + \mathbf{B}^2 \\
 \text{imaginary scalar:} & \quad 0 \\
 \text{real vector:} & \quad 2(a\mathbf{A} + b\mathbf{B} - \mathbf{A} \times \mathbf{B}) \quad \leftarrow \text{different from } ZZ^R \\
 \text{imaginary vector:} & \quad \mathbf{0}
 \end{aligned}$$

Table 9: Components of ZZ^{R*} .

$$\begin{aligned}
 \text{real scalar:} & \quad a^2 - b^2 - \mathbf{A}^2 + \mathbf{B}^2 \\
 \text{imaginary scalar:} & \quad 2ab - 2\mathbf{A} \cdot \mathbf{B} \\
 \text{real vector:} & \quad \mathbf{0} \\
 \text{imaginary vector:} & \quad \mathbf{0}
 \end{aligned}$$

Table 10: Components of $Z^{\text{R}*}Z$.

real scalar:	$a^2 - b^2 - \mathbf{A}^2 + \mathbf{B}^2$
imaginary scalar:	$2ab - 2\mathbf{A} \cdot \mathbf{B}$
real vector:	$\mathbf{0}$
imaginary vector:	$\mathbf{0}$

The following numbered relation is recorded for later use.

$$\text{scalar}(ZZ') = \text{scalar}(Z'Z) \quad \text{cyclic property.} \quad (96)$$

If $Z' = MZ^{\text{R}*}$, then the following relation holds.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \text{scalar}(ZMZ^{\text{R}*}) &= \text{scalar}(MZ^{\text{R}*}Z), \\ &= \text{scalar}(M)(Z^{\text{R}*}Z) \quad \text{since } Z^{\text{R}*}Z \text{ is scalar.} \end{aligned} \quad (97)$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \text{scalar}(ZZ^*) &= cc^* - \mathbf{C}\mathbf{C}^* \quad \text{real,} \\ &= (a^2 + b^2) - (\mathbf{A}^2 + \mathbf{B}^2). \end{aligned} \quad (98)$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \|Z\|^2 &= |c|^2 + \|\mathbf{C}\|^2, \\ &= cc^* + \mathbf{C}\mathbf{C}^*, \\ &= (a^2 + b^2) + (\mathbf{A}^2 + \mathbf{B}^2) \quad \text{non-negative real,} \\ &= \text{scalar}(Z^{\text{R}}Z), \\ &= \text{scalar}(ZZ^{\text{R}}), \\ &= \frac{1}{2}(Z^{\text{R}}Z + (Z^{\text{R}}Z)^{\text{R}*}), \\ &= \frac{1}{2}(Z^{\text{R}}Z + Z^{\text{R}*}Z^*). \end{aligned} \quad (99)$$

The final numbered line is the derived relation used in the subsequent discussion.

The following numbered identities fix the notation and formulas used in this block.

$$\|\mathbf{r} \exp(i\theta)Z\|^2 = \mathbf{r}^2 \|Z\|^2, \quad (100)$$

$$ZZ^* = \text{real scalar} + \text{complex vector}, \quad (101)$$

$$ZZ^{\text{R}} = \text{real scalar} + \text{real vector}. \quad (102)$$

Let $ZZ^{\text{R}*}$ denote complex scalar.

$$Z^*Z^{\text{R}} = Z^{\text{R}}Z^* = (Z^{\text{R}*}Z)^*, \quad (103)$$

$$Z^*Z = ZZ^* \text{ if } \mathbf{A} \times \mathbf{B} = 0, \quad (104)$$

$$Z^{\text{R}}Z = ZZ^{\text{R}} \text{ if } \mathbf{A} \times \mathbf{B} = 0, \quad (105)$$

$$(ZZ')^* = Z^*(Z')^*, \quad (106)$$

$$(ZZ')^{\text{R}} = (Z')^{\text{R}}Z^{\text{R}}, \quad (107)$$

$$(ZZ')^{\text{R}*} = (Z')^{\text{R}*}Z^{\text{R}*}. \quad (108)$$

Partial operator on products.

$$\partial = \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}, \quad (109)$$

$$\partial(ZZ') = (\partial Z)Z' + Z(\partial Z') + [\boldsymbol{\sigma}, Z] \cdot \partial_{\mathbf{r}} Z'. \quad (110)$$

Paravector Metric and Interval Form

The formulas below make paravector metric and interval form explicit and collect identities used later in the manuscript. See Eqs. (11) and (12). Here $X, Y \in \mathbb{R} \oplus \mathbb{R}^3$ and $Z \in \mathbb{C} \oplus \mathbb{C}^3$, so the displayed quadratic forms can be read against the real and complex paravector decompositions already established in the main text. The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \langle\langle \text{scalar} + \mathbf{vector} \cdot \boldsymbol{\sigma} \rangle\rangle^2 &:= \text{scalar}^2 - \mathbf{vector}^2, \\ &= (\text{scalar} + \mathbf{vector} \cdot \boldsymbol{\sigma})(\text{scalar} - \mathbf{vector} \cdot \boldsymbol{\sigma}), \\ &= (\text{scalar} - \mathbf{vector} \cdot \boldsymbol{\sigma})(\text{scalar} + \mathbf{vector} \cdot \boldsymbol{\sigma}). \end{aligned} \quad (111)$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned} X &= a + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \\ Y &= b + \mathbf{B} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \\ Z &= c + \mathbf{C} \cdot \boldsymbol{\sigma} \in \mathbb{C} \oplus \mathbb{C}^3, \\ &= X + iY. \end{aligned} \quad (112)$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \langle\langle Z \rangle\rangle^2 &= c^2 - \mathbf{C}^2 \quad \text{complex scalar}, \\ &= ZZ^{\mathbf{R}*} = Z^{\mathbf{R}*}Z, \\ &= (a + ib)^2 - (\mathbf{A} + i\mathbf{B})^2, \\ &= (a^2 - b^2 - \mathbf{A}^2 + \mathbf{B}^2) + 2i(ab - \mathbf{A} \cdot \mathbf{B}). \end{aligned} \quad (113)$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \langle\langle \mathbf{r} \exp(i\theta)Z \rangle\rangle^2 &= \mathbf{r}^2 \exp(2i\theta) \langle\langle Z \rangle\rangle^2, \\ \langle\langle X \rangle\rangle^2 &= a^2 - \mathbf{A}^2 \quad \text{real scalar}, \\ &= XX^* = X^*X. \end{aligned} \quad (114)$$

The final numbered line is the derived relation used in the subsequent discussion.

Inverse Formulas

The formulas below make inverse formulas explicit and collect identities used later in the manuscript. See Eq. (12). Here $Z \in \mathbb{C} \oplus \mathbb{C}^3$.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} Z^{-1} &= Z^{\mathbf{R}*} / \langle\langle Z \rangle\rangle^2, \\ &= Z^{\mathbf{R}*} / (Z^{\mathbf{R}*}Z). \end{aligned} \quad (115)$$

The final numbered line is the derived relation used in the subsequent discussion.
Note. The following supporting identities are used in the surrounding derivation.
If $Z^{\mathbf{R}*}Z = 0$, then the inverse does not exist.

Minimal Polynomial and Spectral Decomposition

The spectral data of a complex paravector are collected here because the later functional calculus, projector formulas, and exponential identities all depend on this algebraic factorization. See Eq. (16). The following numbered identities fix the notation and formulas used in this block.

$$Z = c + \mathbf{C} \cdot \boldsymbol{\sigma} \in \mathbb{C} \oplus \mathbb{C}^3, \quad (116)$$

$$Z^2 - 2cZ + (c^2 - \mathbf{C}^2) = 0, \quad (117)$$

$$\lambda(\pm) = c \pm \sqrt{\mathbf{C}^2}, \quad (118)$$

$$(Z - \lambda(-))(Z - \lambda(+)) = 0. \quad (119)$$

Exponentials of Complex Paravectors

The formulas below make exponentials of complex paravectors explicit and collect identities used later in the manuscript. See Eqs. (13) and (14). The intermediate steps are written out explicitly as follows.

$$\begin{aligned} X &= a + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \\ Y &= b + \mathbf{B} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \\ Z &= c + \mathbf{C} \cdot \boldsymbol{\sigma} \in \mathbb{C} \oplus \mathbb{C}^3, \\ &= X + iY. \end{aligned} \quad (120)$$

The final numbered line is the derived relation used in the subsequent discussion. The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \exp(Z) &= \exp(c + \mathbf{C} \cdot \boldsymbol{\sigma}), \\ &= \exp(c) \exp(\mathbf{C} \cdot \boldsymbol{\sigma}). \end{aligned} \quad (121)$$

The final numbered line is the derived relation used in the subsequent discussion. The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \exp(Z^*) &= \exp(c^*) \exp(-\mathbf{C}^* \cdot \boldsymbol{\sigma}), \\ \exp(Z)^{-1} &= \exp(-c) \exp(-\mathbf{C} \cdot \boldsymbol{\sigma}), \\ \exp(Z^*)^{-1} &= \exp(-c^*) \exp(\mathbf{C}^* \cdot \boldsymbol{\sigma}), \\ (\mathbf{C} \cdot \boldsymbol{\sigma})^2 &= \mathbf{C} \cdot \mathbf{C} = \mathbf{C}^2, \\ &= \mathbf{A}^2 - \mathbf{B}^2 + 2i\mathbf{A} \cdot \mathbf{B}. \end{aligned} \quad (122)$$

The final numbered line is the derived relation used in the subsequent discussion. The intermediate steps are written out explicitly as follows.

$$\begin{aligned} (\mathbf{C} \cdot \boldsymbol{\sigma})^{2n} &= (\mathbf{C}^2)^n, \\ z &= \sqrt{\mathbf{C}^2} \quad \text{complex scalar,} \\ \exp(\mathbf{C} \cdot \boldsymbol{\sigma}) &= \sum_n (\mathbf{C} \cdot \boldsymbol{\sigma})^n / n!, \\ &= \sum_n (\mathbf{C}^2)^n / (2n)! + \sum_n (\mathbf{C}^2)^n / (2n+1)! (\mathbf{C} \cdot \boldsymbol{\sigma}), \\ &= \cosh(z) + \sinh(z)(\mathbf{C} \cdot \boldsymbol{\sigma})/z. \end{aligned} \quad (123)$$

The final numbered line is the derived relation used in the subsequent discussion.

Complex Field Decomposition

The formulas below make complex field decomposition explicit and collect identities used later in the manuscript. The following numbered identities fix the notation and formulas used in this block.

$$\partial = \partial_t + \partial_{\mathbf{r}} \cdot \boldsymbol{\sigma}, \quad (124)$$

$$F = (a + ib) + (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (125)$$

$$\mathbf{a} : \mathbb{R} \times \mathbb{R}^3 \Rightarrow \mathbb{R}, \quad (126)$$

$$\mathbf{A} : \mathbb{R} \times \mathbb{R}^3 \Rightarrow \mathbb{R}^3, \quad (127)$$

$$\mathbf{b} : \mathbb{R} \times \mathbb{R}^3 \Rightarrow \mathbb{R}, \quad (128)$$

$$\mathbf{B} : \mathbb{R} \times \mathbb{R}^3 \Rightarrow \mathbb{R}^3, \quad (129)$$

$$F : \mathbb{R} \times \mathbb{R}^3 \Rightarrow \mathbb{C} \oplus \mathbb{C}^3. \quad (130)$$

Table 11: Components of ∂F .

real scalar: $\partial_t \mathbf{a} + \partial_{\mathbf{r}} \cdot \mathbf{A}$

imaginary scalar: $\partial_t \mathbf{b} + \partial_{\mathbf{r}} \cdot \mathbf{B}$

real vector: $\partial_{\mathbf{r}} \mathbf{a} + \partial_t \mathbf{A} - \partial_{\mathbf{r}} \times \mathbf{B}$

imaginary vector: $\partial_{\mathbf{r}} \mathbf{b} + \partial_{\mathbf{r}} \times \mathbf{A} + \partial_t \mathbf{B}$

Table 12: Components of $i\partial F$.

real scalar: $-\partial_t \mathbf{b} - \partial_{\mathbf{r}} \cdot \mathbf{B}$

imaginary scalar: $\partial_t \mathbf{a} + \partial_{\mathbf{r}} \cdot \mathbf{A}$

real vector: $-\partial_{\mathbf{r}} \mathbf{b} - \partial_{\mathbf{r}} \times \mathbf{A} - \partial_t \mathbf{B}$

imaginary vector: $\partial_{\mathbf{r}} \mathbf{a} + \partial_t \mathbf{A} - \partial_{\mathbf{r}} \times \mathbf{B}$

Functions of a Unit-Direction Element

The formulas below make functions of a unit-direction element explicit and collect identities used later in the manuscript. See Eqs. (15) and (16). *Let the following notation be fixed.*

$$\Pi(\pm) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \quad (131)$$

$$\Pi(\pm)^2 = \Pi(\pm), \quad (132)$$

$$\Pi(\pm)\Pi(\mp) = 0, \quad (133)$$

$$\Pi(\pm) + \Pi(\mp) = 1, \quad (134)$$

$$(\mathbf{u} \cdot \boldsymbol{\sigma})\Pi(\pm) = \pm\Pi(\pm). \quad (135)$$

If $f(\cdot)$ has a power series.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} f(\mathbf{u} \cdot \boldsymbol{\sigma}), \\ &= \frac{1}{2}(f(+) + f(-)), \\ &\quad + \frac{1}{2}(f(+) - f(-))(\mathbf{u} \cdot \boldsymbol{\sigma}), \\ &= f(+)\Pi(+) + f(-)\Pi(-). \end{aligned} \quad (136)$$

The final numbered line is the derived relation used in the subsequent discussion.

The following numbered identities fix the notation and formulas used in this block.

$$f(\mathbf{u} \cdot \boldsymbol{\sigma}) = f(+)\Pi(+) + f(-)\Pi(-), \quad (137)$$

$$\sqrt{\mathbf{u} \cdot \boldsymbol{\sigma}} = \pm\Pi(+) \pm i\Pi(-), \quad (138)$$

$$\log(\mathbf{u} \cdot \boldsymbol{\sigma}) = 2i\pi(n\Pi(+) + (\frac{1}{2} + m)\Pi(-)), \quad (139)$$

$$(z + \mathbf{u} \cdot \boldsymbol{\sigma})^{-1} = (z - \mathbf{u} \cdot \boldsymbol{\sigma}) / (z^2 - 1), \quad (140)$$

$$(i - \mathbf{u} \cdot \boldsymbol{\sigma}) / (i + \mathbf{u} \cdot \boldsymbol{\sigma}) = i\mathbf{u} \cdot \boldsymbol{\sigma}, \quad (141)$$

$$\exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}) = \cosh(\theta) + \sinh(\theta)(\mathbf{u} \cdot \boldsymbol{\sigma}), \quad (142)$$

$$\exp(i\theta \mathbf{u} \cdot \boldsymbol{\sigma}) = \cos(\theta) + i \sin(\theta)(\mathbf{u} \cdot \boldsymbol{\sigma}). \quad (143)$$

Functions of a Paravector

The formulas below make functions of a paravector explicit and collect identities used later in the manuscript. The following numbered identities fix the notation and formulas used in this block.

$$X = t + \mathbf{r} \cdot \boldsymbol{\sigma}, \quad (144)$$

$$t = \text{scalar}(X) \in \mathbb{R}, \quad (145)$$

$$\mathbf{r} = \text{vector}(X) \in \mathbb{R}^3, \quad (146)$$

$$\mathbf{u} = \mathbf{r} / \|\mathbf{r}\|. \quad (147)$$

The minimal polynomial is.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} X^2 - 2tX + (t^2 - \mathbf{r}^2) &= 0, \\ \lambda(\pm) &= t \pm \|\mathbf{r}\|, \\ 0 &= (X - \lambda(-))(X - \lambda(+)), \\ &= X^2 - \lambda(-)X - X\lambda(+) + \lambda(-)\lambda(+). \end{aligned} \quad (148)$$

The final numbered line is the derived relation used in the subsequent discussion.

The following numbered identities fix the notation and formulas used in this block.

$$a = \lambda(+), \quad (149)$$

$$b = \lambda(-), \quad (150)$$

$$\Pi(\pm) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \quad (151)$$

$$X = a\Pi(+) + b\Pi(-). \quad (152)$$

If $f(\cdot)$ has a power series,.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} X' &= f(X), \\ &= f(a)\Pi(+) + f(b)\Pi(-), \\ &= t' + \mathbf{r}' \cdot \boldsymbol{\sigma}. \end{aligned} \quad (153)$$

The final numbered line is the derived relation used in the subsequent discussion.

The following numbered identities fix the notation and formulas used in this block.

$$t' = \frac{1}{2}(f(a) + f(b)), \quad (154)$$

$$\|\mathbf{r}'\| = \frac{1}{2}(f(a) - f(b)), \quad (155)$$

$$\mathbf{r}' = \|\mathbf{r}'\|\mathbf{u}. \quad (156)$$

Summary left as projectors.

The eigenvalues are.

$$\lambda(\pm) = t \pm \|\mathbf{r}\|. \quad (157)$$

The corresponding projectors are.

$$\mathbf{u} := \mathbf{r}/\|\mathbf{r}\|, \quad (158)$$

$$\Pi(\pm) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}). \quad (159)$$

The eigenvalue equation is.

$$\Pi(\pm)X = \lambda(\pm)\Pi(\pm). \quad (160)$$

The projector decomposition is.

$$X = \lambda(+)\Pi(+) + \lambda(-)\Pi(-), \quad (161)$$

$$f(X) = f(\lambda(+))\Pi(+) + f(\lambda(-))\Pi(-), \quad (162)$$

$$\exp(X) = e^t(\cosh(\|\mathbf{r}\|) + \sinh(\|\mathbf{r}\|)(\mathbf{u} \cdot \boldsymbol{\sigma})), \quad (163)$$

$$\sqrt{X} = \pm\sqrt{a}\Pi(+) \pm \sqrt{b}\Pi(-), \quad (164)$$

$$\log(X) = \log(a)\Pi(+) + \log(b)\Pi(-), \quad (165)$$

$$X^{-1} = \Pi(+)/a + \Pi(-)/b, \quad (166)$$

$$X^n = \frac{1}{2}(a^n + b^n) + \frac{1}{2}(a^n - b^n)(\mathbf{u} \cdot \boldsymbol{\sigma}). \quad (167)$$

Product of any two functions.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} f(X)g(X') &= (f(a)\Pi(+) + f(b)\Pi(-))(g(a')\Pi(+)' + g(b')\Pi(-)'), \\ &= f(a)g(a')\Pi(+)\Pi(+)' + f(a)g(b')\Pi(+)\Pi(-)'. \end{aligned} \quad (168)$$

The final numbered line is the derived relation used in the subsequent discussion.

followed by the terms $f(b)g(a')\Pi(-)\Pi(+)' + f(b)g(b')\Pi(-)\Pi(-)'$.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \Pi(+)\Pi(+)' &= \frac{1}{4}(1 + \mathbf{u} \cdot \mathbf{u}'), \\ &+ \frac{1}{4}(\mathbf{u}' + \mathbf{u} + i(\mathbf{u} \times \mathbf{u}')) \cdot \boldsymbol{\sigma}. \end{aligned} \quad (169)$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \Pi(+)\Pi(-)' &= \frac{1}{4}(1 - \mathbf{u} \cdot \mathbf{u}'), \\ &+ \frac{1}{4}(-\mathbf{u}' + \mathbf{u} - i(\mathbf{u} \times \mathbf{u}')) \cdot \boldsymbol{\sigma}. \end{aligned} \quad (170)$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \Pi(-)\Pi(+)' &= \frac{1}{4}(1 - \mathbf{u} \cdot \mathbf{u}'), \\ &+ \frac{1}{4}(\mathbf{u}' - \mathbf{u} - i(\mathbf{u} \times \mathbf{u}')) \cdot \boldsymbol{\sigma}. \end{aligned} \quad (171)$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \Pi(-)\Pi(-)' &= \frac{1}{4}(1 + \mathbf{u} \cdot \mathbf{u}'), \\ &+ \frac{1}{4}(-\mathbf{u}' - \mathbf{u} + i(\mathbf{u} \times \mathbf{u}')) \cdot \boldsymbol{\sigma}. \end{aligned} \quad (172)$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
(1 - \mathbf{r} \cdot \boldsymbol{\sigma}) / (1 + \mathbf{r} \cdot \boldsymbol{\sigma}), \\
&= \exp(-2 \operatorname{arctanh}(\mathbf{r} \cdot \boldsymbol{\sigma})), \\
&= \exp(-2 \operatorname{arctanh}(\|\mathbf{r}\|)) \Pi(+), \\
&\quad + \exp(+2 \operatorname{arctanh}(\|\mathbf{r}\|)) \Pi(-).
\end{aligned} \tag{173}$$

The final numbered line is the derived relation used in the subsequent discussion.

III Space-Time Kinematics

With the algebraic toolkit in place, the next step is to reinterpret the same paravector structure as space-time. This is the bridge from pure algebra to geometry and mechanics.

Differential space-time and proper velocity

Specializing the real paravector $X = a + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3$ from Eq. (9) to differentials gives

$$dX = dt + d\mathbf{r} \cdot \boldsymbol{\sigma} \tag{174}$$

and the interval is the differential version of Eq. (12):

$$ds^2 = \langle\langle dX \rangle\rangle^2 = dX dX^* = dt^2 - d\mathbf{r}^2. \tag{175}$$

Here $t, s, \gamma \in \mathbb{R}$, $\mathbf{r}, \mathbf{v}, \mathbf{P} \in \mathbb{R}^3$, and $X, U \in \mathbb{R} \oplus \mathbb{R}^3$. The scalar part dt is the temporal separation and the vector part $d\mathbf{r} \cdot \boldsymbol{\sigma}$ is the spatial separation. Equation (175) is therefore not an added metric axiom but the direct spacetime interpretation of the quadratic form already introduced in Eq. (12).

If $\mathbf{v} = d\mathbf{r}/dt \in \mathbb{R}^3$ and $\gamma = (1 - \mathbf{v}^2)^{-1/2} \in \mathbb{R}$, then

$$U := \frac{dX}{ds} = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), \quad \langle\langle U \rangle\rangle^2 = 1. \tag{176}$$

Multiplying Eq. (176) by the real rest mass $m \in \mathbb{R}$ gives the proper momentum formula

$$mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}, \quad E = m\gamma, \quad \mathbf{P} = m\gamma\mathbf{v}, \tag{177}$$

which immediately yields $E^2 - \mathbf{P}^2 = m^2$. The normalization $\langle\langle U \rangle\rangle^2 = 1$ is the essential point: proper velocity is the unit time-like paravector tangent to the worldline. Multiplying by m then produces the energy–momentum paravector in a way that makes the relativistic mass shell relation appear immediately.

Proper acceleration and free motion

The proper acceleration is the real paravector

$$A := \frac{dU}{ds} = \frac{d^2 X}{ds^2}, \quad \text{scalar}(U^* A) = 0. \tag{178}$$

The orthogonality condition in Eq. (178) is the paravector analogue of the standard four-acceleration orthogonality relation. It is included here because the Lorentz-force law later in the paper should be read as a specific dynamical realization of this general kinematic structure.

A free-particle Lagrangian is

$$\mathcal{L}_{\text{free}} = -\frac{m}{\gamma}, \tag{179}$$

whose Euler–Lagrange equations reproduce inertial motion and the usual nonrelativistic expansion

$$\mathcal{L}_{\text{free}} = -m + \frac{1}{2}m\mathbf{v}^2 + O(\mathbf{v}^4). \tag{180}$$

Equation (180) is worth displaying explicitly because it shows how the relativistic action reduces to the classical kinetic expression plus an additive rest-energy constant when $\|\mathbf{v}\| \ll 1$.

Supporting Kinematic Derivations

This section now expands the interval classification, component forms of proper velocity and proper momentum, and the action-based formulas that support the compact spacetime presentation above. The intent is that every later dynamical equation can be traced back to a previously defined spacetime object without forcing the reader to jump to a separate omnibus section.

Spacetime Paravectors

At this point the paravector formalism is reinterpreted as spacetime. The following identities make that reinterpretation explicit by separating temporal and spatial parts and identifying the interval carried by the same algebraic product. See Eqs. (174) and (175). Here $dt, dt', ds, s, \gamma, m \in \mathbb{R}$, $d\mathbf{r}, d\mathbf{r}', \mathbf{r}, \mathbf{v}, \mathbf{a}, \mathbf{P}, \mathbf{f} \in \mathbb{R}^3$, and $dX, U, A, mU \in \mathbb{R} \oplus \mathbb{R}^3$. The following numbered relation is recorded for later use.

$$dX = dt + dr_1\sigma_1 + dr_2\sigma_2 + dr_3\sigma_3. \quad (181)$$

Applying conjugation gives the companion relation.

$$dX^* = dt - dr_1\sigma_1 - dr_2\sigma_2 - dr_3\sigma_3. \quad (182)$$

The space-time metric is.

$$ds^2 = dt^2 - dr_1^2 - dr_2^2 - dr_3^2. \quad (183)$$

Scalar: $dt \in \mathbb{R}$, vector : $d\mathbf{r} \in \mathbb{R}^3$.

$$dX = dt + d\mathbf{r} \cdot \boldsymbol{\sigma}, \quad (184)$$

$$dX^* = dt - d\mathbf{r} \cdot \boldsymbol{\sigma}. \quad (185)$$

Decomposing the expression into scalar and vector parts gives.

$$(dX + dX^*)/2 = dt, \quad (186)$$

$$(dX - dX^*)/2 = d\mathbf{r} \cdot \boldsymbol{\sigma}, \quad (187)$$

$$\text{scalar}(dX) = dt, \quad (188)$$

$$\text{vector}(dX) = d\mathbf{r}. \quad (189)$$

The quadratic paravector form is.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \langle\langle dX \rangle\rangle^2 &= dX dX^* = dt^2 - d\mathbf{r}^2, \\ 1. \langle\langle dX \rangle\rangle^2 &> 0 \text{ for all } dX \neq 0 \text{ if } dt^2 > d\mathbf{r}^2, \\ 2. \langle\langle dX + dX' \rangle\rangle^2 &- \langle\langle dX \rangle\rangle^2 - \langle\langle dX' \rangle\rangle^2, \\ &= (dt + dt')^2 - (d\mathbf{r} + d\mathbf{r}')^2. \end{aligned} \quad (190)$$

The final numbered line is the derived relation used in the subsequent discussion.

That is, one subtracts $(dt^2 - d\mathbf{r}^2) + (dt'^2 - d\mathbf{r}'^2)$ from the combined quadratic form.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} &= 2(dt dt' - d\mathbf{r} \cdot d\mathbf{r}'), \\ 3. \langle\langle adX \rangle\rangle^2 &= a^2(dt^2 - d\mathbf{r}^2) = a^2 \langle\langle dX \rangle\rangle^2, \\ [dX, dX'] &= [d\mathbf{r} \cdot \boldsymbol{\sigma}, d\mathbf{r}' \cdot \boldsymbol{\sigma}], \\ &= 2i(d\mathbf{r} \times d\mathbf{r}') \cdot \boldsymbol{\sigma}. \end{aligned} \quad (191)$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} ds^2 &= dt^2 - d\mathbf{r}^2, \\ &= \langle\langle dX \rangle\rangle^2 = dX dX^* = dX^* dX. \end{aligned} \quad (192)$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
ds &= \sqrt{dt^2 - d\mathbf{r}^2}, \\
&= \sqrt{1 - d\mathbf{r}^2/dt^2} dt, \\
&= \sqrt{1 - \mathbf{v}^2} dt.
\end{aligned} \tag{193}$$

The final numbered line is the derived relation used in the subsequent discussion.
Define the following object.

$$\mathbf{v} = d\mathbf{r}/dt = \{v_1, v_2, v_3\} \in \mathbb{R}^3, \tag{194}$$

$$\gamma = dt/ds = 1/\sqrt{1 - \mathbf{v}^2}. \tag{195}$$

Note. The following supporting identities are used in the surrounding derivation.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
dX^{-1} &= dX^{\mathbf{R}*} / \langle\langle dX \rangle\rangle^2, \\
&= dX^* / (dX dX^*), \\
&= dX^* / ds^2.
\end{aligned} \tag{196}$$

The final numbered line is the derived relation used in the subsequent discussion.

Timelike, Null, and Spacelike Classification

Once the interval has been written as a Minkowski-type quadratic form, its sign can be used to classify displacements. The following notes record that classification in the notation of the manuscript. Here $dX, dX' \in \mathbb{R} \oplus \mathbb{R}^3$, while $dt, dt', ds, \tau \in \mathbb{R}$ and $d\mathbf{r}, d\mathbf{r}' \in \mathbb{R}^3$. The arc length s agrees with the proper time τ , so the same scalar parameter measures elapsed time along the worldline.

(measured along a path).

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
\tau = s &= \int ds = \int d\tau, \\
&= \int \sqrt{1 - \mathbf{v}^2} dt \quad \text{time dilation,} \\
&= \int 1/\gamma dt.
\end{aligned} \tag{197}$$

The final numbered line is the derived relation used in the subsequent discussion.

Note. In SI units one has $ds = c d\tau$.

If $\langle\langle dx \rangle\rangle^2 = 0$, then the interval is *light-like*.

$$dt^2 = d\mathbf{r}^2 \Rightarrow dt = \pm \|d\mathbf{r}\|. \tag{198}$$

The inverse dX^{-1} does not exist in this case.

If $\langle\langle dx \rangle\rangle^2 < 0$, then the interval is *space-like*.

Proper Velocity

The formulas below make proper velocity explicit and collect identities used later in the manuscript. See Eq. (176). Here $\mathbf{v} \in \mathbb{R}^3$, $\gamma \in \mathbb{R}$, and $U = dX/ds \in \mathbb{R} \oplus \mathbb{R}^3$ is the proper-velocity paravector. The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
dX/ds &= (dt/ds) + (d\mathbf{r}/ds) \cdot \boldsymbol{\sigma}, \\
&= (dt + \boldsymbol{\sigma} \cdot d\mathbf{r}) / \sqrt{dt^2 - d\mathbf{r}^2},
\end{aligned}$$

$$\begin{aligned}
&= (dt/dt + \boldsymbol{\sigma} \cdot d\mathbf{r}/dt) / \sqrt{(dt/dt)^2 - (d\mathbf{r}/dt)^2}, \\
&= (1 + \mathbf{v} \cdot \boldsymbol{\sigma}) / \sqrt{1 - \mathbf{v}^2}, \\
&= \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}).
\end{aligned} \tag{199}$$

The final numbered line is the derived relation used in the subsequent discussion.

Table 13: Components of dX/ds .

$$\begin{aligned}
&\text{scalar: } \gamma \\
&\text{vector: } \gamma\mathbf{v}
\end{aligned}$$

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
\langle\langle dX/ds \rangle\rangle^2 &= \gamma^2(1 + \mathbf{v} \cdot \boldsymbol{\sigma})(1 - \mathbf{v} \cdot \boldsymbol{\sigma}), \\
&= \gamma^2(1 - \mathbf{v}^2), \\
&= 1.
\end{aligned} \tag{200}$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
dX/dt &= (dX/ds)(ds/dt), \\
&= (dX/ds)(1/\gamma).
\end{aligned} \tag{201}$$

The final numbered line is the derived relation used in the subsequent discussion.

Define the following object.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
U &:= dX/ds, \\
&= \gamma + \gamma\mathbf{v} \cdot \boldsymbol{\sigma}.
\end{aligned} \tag{202}$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
U &= U_s + U_{\mathbf{v}} \cdot \boldsymbol{\sigma}, \\
U_s &= \gamma, \\
U_{\mathbf{v}} &= \gamma\mathbf{v}, \\
ds^2 &= dXdX^*, ds > 0, \\
dX^*/ds &= dX^{-1}ds, \\
&= (dX/ds)^{-1} \Rightarrow U^{-1} = U^*.
\end{aligned} \tag{203}$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
\langle\langle U \rangle\rangle^2 &= UU^* = 1, \\
d\langle\langle U \rangle\rangle^2/ds &, \\
&= dU/dsU^* + U(dU/ds)^*, \\
&= (dU/dsU^*) + (dU/dsU^*)^{*R}.
\end{aligned} \tag{204}$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
\Rightarrow \text{scalar}(U^* dU/ds) &= 0, \\
\langle\langle dX \rangle\rangle^2 &= dX dX^*, \\
&= ((dX/ds) ds)((dX/ds)^* ds), \\
&= UU^* ds^2, \\
&= \langle\langle U \rangle\rangle^2 ds^2, \\
&= ds^2.
\end{aligned} \tag{205}$$

The final numbered line is the derived relation used in the subsequent discussion.

Proper Acceleration

The formulas below make proper acceleration explicit and collect identities used later in the manuscript. See Eq. (178). Here $\mathbf{a} \in \mathbb{R}^3$, $s \in \mathbb{R}$, and $A = dU/ds \in \mathbb{R} \oplus \mathbb{R}^3$ is the proper-acceleration paravector. The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
\mathbf{a} &= d\mathbf{v}/dt = \{a_1, a_2, a_3\} \in \mathbb{R}^3, \\
d\gamma/dt &= \gamma^3(\mathbf{v} \cdot \mathbf{a}), \\
d\gamma/ds &= \gamma^4(\mathbf{v} \cdot \mathbf{a}), \\
\partial_{\mathbf{v}}(-1/\gamma) &= \gamma\mathbf{v}, \\
\partial_{\mathbf{v}}(\gamma\mathbf{v}) &= \gamma\mathbf{I} + \gamma^3\mathbf{v}\mathbf{v}^T, \\
d(\gamma\mathbf{v})/dt &= \partial_{\mathbf{v}}(\gamma\mathbf{v})d\mathbf{v}/dt, \\
&= \gamma\mathbf{a} + \gamma^3(\mathbf{v} \cdot \mathbf{a})\mathbf{v}.
\end{aligned} \tag{206}$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
d(\gamma\mathbf{v})/ds &= (d(\gamma\mathbf{v})/dt)(dt/ds), \\
&= \gamma^2\mathbf{a} + \gamma^4(\mathbf{v} \cdot \mathbf{a})\mathbf{v}.
\end{aligned} \tag{207}$$

The final numbered line is the derived relation used in the subsequent discussion.

Recall the previously established definitions.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
U &= U_s + U_{\mathbf{v}} \cdot \boldsymbol{\sigma}, \\
U_s &= \gamma, \\
U_{\mathbf{v}} &= \gamma\mathbf{v}, \\
dU/ds &= (d\gamma/ds) + (d(\gamma\mathbf{v})/ds) \cdot \boldsymbol{\sigma}, \\
&= \gamma^4(\mathbf{v} \cdot \mathbf{a}) + (\gamma^4(\mathbf{v} \cdot \mathbf{a})\mathbf{v} + \gamma^2\mathbf{a}) \cdot \boldsymbol{\sigma}.
\end{aligned} \tag{208}$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
\mathbf{u} &= \mathbf{v}/\|\mathbf{v}\|, \mathbf{u}^2 = 1, \\
\text{par}(\mathbf{a}) &= (\mathbf{a} \cdot \mathbf{u})\mathbf{u}, \\
\text{prp}(a) &= a - \text{par}(a), \\
a^2 &= \text{par}(a)^2 + \text{prp}(a)^2, \\
(\mathbf{v} \cdot \mathbf{a})^2 &= \mathbf{v}^2 \text{par}(\mathbf{a})^2,
\end{aligned}$$

$$\begin{aligned}\langle\langle dU/ds \rangle\rangle^2 &= -\gamma^4(\mathbf{a}^2 + \gamma^2(\mathbf{v} \cdot \mathbf{a})^2), \\ &= -\gamma^4(\text{prp}(a)^2 + \gamma^2\text{par}(a)^2).\end{aligned}\tag{209}$$

The final numbered line is the derived relation used in the subsequent discussion.

Let the following notation be fixed.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}\mathbf{A} &= dU/ds, \\ &= ddX/ds^2.\end{aligned}\tag{210}$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}\mathbf{A} &= A_s + A_{\mathbf{v}} \cdot \boldsymbol{\sigma}, \\ A_s &= \gamma^4(\mathbf{v} \cdot \mathbf{a}), \\ A_{\mathbf{v}} &= \gamma^4(\mathbf{v} \cdot \mathbf{a})\mathbf{v} + \gamma^2\mathbf{a}, \\ U\mathbf{A} &= (U_s A_s + U_{\mathbf{v}} \cdot A_{\mathbf{v}}), \\ &\quad + (U_s A_{\mathbf{v}} + A_s U_{\mathbf{v}} + i(U_{\mathbf{v}} \times A_{\mathbf{v}})) \cdot \boldsymbol{\sigma}.\end{aligned}\tag{211}$$

The final numbered line is the derived relation used in the subsequent discussion.

Table 14: Components of $U\mathbf{A}$.

real scalar:	$2\gamma^5(\mathbf{v} \cdot \mathbf{a})$
imaginary scalar:	0
real vector:	$\gamma^3\mathbf{a} + 2\gamma^5(\mathbf{v} \cdot \mathbf{a})\mathbf{v}$
imaginary vector:	$\gamma^3(\mathbf{v} \times \mathbf{a})$

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}U^*\mathbf{A} &= (U_s A_s - U_{\mathbf{v}} \cdot A_{\mathbf{v}}), \\ &\quad + (U_s A_{\mathbf{v}} - A_s U_{\mathbf{v}} - i(U_{\mathbf{v}} \times A_{\mathbf{v}})) \cdot \boldsymbol{\sigma}.\end{aligned}\tag{212}$$

The final numbered line is the derived relation used in the subsequent discussion.

Table 15: Components of $U^*\mathbf{A}$.

real scalar:	0
imaginary scalar:	0
real vector:	$\gamma^3\mathbf{a}$
imaginary vector:	$-\gamma^3(\mathbf{v} \times \mathbf{a})$

Note. $\text{scalar}(U^*\mathbf{A}) = 0$.

Proper Momentum

The formulas below make proper momentum explicit and collect identities used later in the manuscript. See Eq. (177). The following numbered relation is recorded for later use.

$$mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}.\tag{213}$$

The relativistic energy is.

The intermediate steps are written out explicitly as follows.

$$E := \text{scalar}(mU),$$

$$= m\gamma. \quad (214)$$

The final numbered line is the derived relation used in the subsequent discussion.

If $\mathbf{v} = 0$, then the following relation holds.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} E &= m \quad \text{rest energy,} \\ &= mc^2 \quad \text{in SI units.} \end{aligned} \quad (215)$$

The final numbered line is the derived relation used in the subsequent discussion.

Let $E - m$ denote kinetic energy.

The relativistic momentum is.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \mathbf{P} &:= \text{vector}(mU), \\ &= m\gamma\mathbf{v}. \end{aligned} \quad (216)$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \langle\langle mU \rangle\rangle^2 &= m^2, \\ &= (E + \mathbf{P} \cdot \boldsymbol{\sigma})(E - \mathbf{P} \cdot \boldsymbol{\sigma}), \\ &= E^2 - \mathbf{P}^2. \end{aligned} \quad (217)$$

The final numbered line is the derived relation used in the subsequent discussion.

The following numbered relation is recorded for later use.

$$E^2 = \mathbf{P}^2 + m^2. \quad (218)$$

Note. For $m = 0$, the mass-shell relation reduces to $E^2 = \mathbf{P}^2$, corresponding to lightlike motion.

Relativistic Lagrangian

The formulas below make relativistic lagrangian explicit and collect identities used later in the manuscript. See Eqs. (179) and (180). The following numbered relation is recorded for later use.

$$\text{action} = \int \mathcal{L}(t, \mathbf{r}, \mathbf{v}) dt. \quad (219)$$

Euler-Lagrange equation.

$$(\partial_{\mathbf{r}} - d/dt \partial_{\mathbf{v}}) \mathcal{L} = 0, \quad (220)$$

$$\Rightarrow d/dt \partial_{\mathbf{v}} \mathcal{L} = \partial_{\mathbf{r}} \mathcal{L}. \quad (221)$$

Noether's conserved quantities.

The canonical momentum is.

$$\partial_{\mathbf{v}} \mathcal{L} = \text{constant if } \partial_{\mathbf{r}} \mathcal{L} = 0. \quad (222)$$

The Hamiltonian is.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \mathcal{H} &= \mathbf{v} \cdot (\partial_{\mathbf{v}} \mathcal{L}) - \mathcal{L}, \\ &= (\mathbf{v} \cdot \partial_{\mathbf{v}} - 1) \mathcal{L}, \\ &= \text{constant if } \partial_t \mathcal{L} = 0. \end{aligned} \quad (223)$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
d/dt\mathcal{H} &= d/dt(\partial_{\mathbf{v}}\mathcal{L}) \cdot \mathbf{v} + (\partial_{\mathbf{v}}\mathcal{L}) \cdot d\mathbf{v}/dt - d\mathcal{L}/dt, \\
&= (\partial_{\mathbf{r}}\mathcal{L}) \cdot \mathbf{v} + (\partial_{\mathbf{v}}\mathcal{L}) \cdot d\mathbf{v}/dt - d\mathcal{L}/dt \quad \text{EL}, \\
&= (\partial_{\mathbf{r}}\mathcal{L}) \cdot \mathbf{v} + (\partial_{\mathbf{v}}\mathcal{L}) \cdot d\mathbf{v}/dt, \\
&\quad - (\partial_t\mathcal{L} + (\partial_{\mathbf{r}}\mathcal{L}) \cdot \mathbf{v} + \partial_{\mathbf{v}}\mathcal{L} \cdot d\mathbf{v}/dt), \\
&= -\partial_t\mathcal{L}.
\end{aligned} \tag{224}$$

The final numbered line is the derived relation used in the subsequent discussion.
The following numbered relation is recorded for later use.

$$d/dt\mathcal{H} = 0 \text{ if } \partial_t\mathcal{L} = 0. \tag{225}$$

Free-Particle Action

The formulas below make free-particle action explicit and collect identities used later in the manuscript. *Mass-weighted proper time along its world line.*

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
\int \mathcal{L}dt &= -m\tau, \\
&= -m \int \langle\langle dX \rangle\rangle = -m \int ds, \\
&= -m \int \langle\langle dX/dt \rangle\rangle dt = -m \int (ds/dt)dt, \\
&= -m \int \langle\langle U/\gamma \rangle\rangle dt = -m \int (1/\gamma)dt, \\
&= -m \int \sqrt{1 - \mathbf{v}^2}dt.
\end{aligned} \tag{226}$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
\mathcal{L} &= -m/\gamma \quad \text{Lagrangian,} \\
\partial_{\mathbf{v}}\mathcal{L} &= m\gamma\mathbf{v} = \mathbf{P} \Rightarrow (\partial_{\mathbf{v}} \cdot \boldsymbol{\sigma})\mathcal{L} = \mathbf{P} \cdot \boldsymbol{\sigma}, \\
\mathcal{H} &= m\gamma\mathbf{v} \cdot \mathbf{v} + m/\gamma, \\
&= m\gamma(\mathbf{v}^2 + 1/\gamma^2), \\
&= m\gamma = E.
\end{aligned} \tag{227}$$

The final numbered line is the derived relation used in the subsequent discussion.
The following numbered identities fix the notation and formulas used in this block.

$$\partial_t\mathcal{L} = 0 \quad m\gamma \text{ is constant}, \tag{228}$$

$$\partial_{\mathbf{r}}\mathcal{L} = 0 \quad m\gamma\mathbf{v} \text{ is constant}. \tag{229}$$

Nonrelativistic Expansion

The formulas below make nonrelativistic expansion explicit and collect identities used later in the manuscript. The intermediate steps are written out explicitly as follows.

$$\mathcal{L} = -m/\gamma,$$

$$= -m + \frac{1}{2}m\mathbf{v}^2 + O(\mathbf{v}^4). \quad (230)$$

The final numbered line is the derived relation used in the subsequent discussion.

The momentum is.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \partial_{\mathbf{v}}\mathcal{L} &= \partial_{\mathbf{v}}(\frac{1}{2}m\mathbf{v}^2 + O(\mathbf{v}^4)), \\ &= m\mathbf{v} + O(\mathbf{v}^3). \end{aligned} \quad (231)$$

The final numbered line is the derived relation used in the subsequent discussion.

Thus the leading momentum approximation is $m\mathbf{v}$.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \text{total energy} &= \text{kinetic} + \text{rest mass energy}, \\ \mathcal{H} &= (m\mathbf{v} + O(\mathbf{v}^3)) \cdot \mathbf{v} + m - \frac{1}{2}m\mathbf{v}^2 - O(\mathbf{v}^4), \\ &= \frac{1}{2}m\mathbf{v}^2 + m + O(\mathbf{v}^4). \end{aligned} \quad (232)$$

The final numbered line is the derived relation used in the subsequent discussion.

Thus the leading energy approximation is $\frac{1}{2}m\mathbf{v}^2 + m$.

IV Rotors, Boosts, and Covariant Transformations

Exponentials appear not only as algebraic functions but also as generators of transformations. After interval structure is established, rotors and boosts are the cleanest next step because they explain how the same objects move between frames before fields and wave equations are introduced.

General transform law, spatial rotors, and boosts

A guiding principle of STA is that transformations should be expressed by the same algebra that describes the objects being transformed. Instead of introducing a separate matrix formalism at this stage, one uses exponentials inside the algebra itself to generate rotations and boosts.

A general transform law is

$$X' = T^{-1}XT. \quad (233)$$

Here T denotes a generic invertible transformation element. For a real angle or rapidity $\theta \in \mathbb{R}$ and a unit real vector $\mathbf{u} \in \mathbb{R}^3$, the specific choices used throughout the remainder of the manuscript are R for spatial rotations and L for Lorentz boosts. Two important special cases are the spatial rotor

$$R(\theta, \mathbf{u}) = \exp\left(\frac{\theta}{2} \mathbf{u} \cdot i\boldsymbol{\sigma}\right), \quad X' = R^R X R, \quad (234)$$

and the Lorentz boost

$$L(\theta, \mathbf{u}) = \exp\left(\frac{\theta}{2} \mathbf{u} \cdot \boldsymbol{\sigma}\right), \quad X' = L^* X L. \quad (235)$$

The symbols R and L are invertible algebra elements rather than vectors or paravectors, while T is reserved for the generic transform law (233). Equation (234) produces Rodrigues' rotation formula, while Eq. (235) produces the usual split into components parallel and perpendicular to the boost axis [2, 3, 9]. The formal similarity between Eqs. (234) and (235) is deliberate. Both are exponential transformations, but bivector-like generators produce trigonometric rotation structure while vector-like generators produce hyperbolic boost structure. The contrast shows how Euclidean and Lorentzian motions coexist inside one algebraic language.

Field transformation formula

Once the field F is introduced in Eq. (312), the same boost element acts on it by

$$F' = L^* F L. \quad (236)$$

This is included here, before the full electromagnetic section, because it shows that fields transform with the same algebraic machinery already used for events. That preview matters conceptually. Once the reader sees that intervals, momenta, and fields all transform by related conjugation rules, the later field theory looks like a continuation of the algebraic story rather than a change of subject.

Supporting Transformation Identities

The formulas collected here expand the general transformation law into explicit basis transforms, boost parameterizations, field transforms, and composition rules. Placing these details immediately after the core rotor and boost formulas keeps the geometric mechanism visible before the manuscript turns to electromagnetism.

General Transformation Identities

The formulas below make general transformation identities explicit and collect identities used later in the manuscript. See Eq. (233). Here T denotes a generic invertible transform element. The specific cases used later are R for spatial rotations and L for Lorentz boosts. The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
 X' &= T^{-1}XT, \\
 &= T^{-1}(t + \mathbf{r} \cdot \boldsymbol{\sigma})T, \\
 &= t + \mathbf{r} \cdot (T^{-1}\boldsymbol{\sigma}T), \\
 &= t + \mathbf{r} \cdot \boldsymbol{\sigma}'.
 \end{aligned} \tag{237}$$

The final numbered line is the derived relation used in the subsequent discussion.
The following numbered relation is recorded for later use.

$$T^{-1} = T^{\mathbf{R}*} / (T^{\mathbf{R}*}T). \tag{238}$$

For boosts and rotors, the inverse simplifies to

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
 T^{\mathbf{R}*}T &= 1 \text{ and } T^{-1} = T^{\mathbf{R}*}, \\
 T^{-1}T &= 1, \\
 d(T^{-1})T &= -T^{-1}dT, \\
 dX' &= d(T^{-1})XT + T^{-1}dXT + T^{-1}XdT, \\
 &= T^{-1}dXT.
 \end{aligned} \tag{239}$$

The final numbered line is the derived relation used in the subsequent discussion.

The two intermediate terms are $(-T^{-1}dT)(TT^{-1})XT$ and $T^{-1}X(TT^{-1})dT$.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
 &= T^{-1}dXT + (-T^{-1}dT)X' + X'(T^{-1}dT), \\
 &= T^{-1}dXT + [X', T^{-1}dT].
 \end{aligned} \tag{240}$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
 d\Theta &:= T^{-1}dT, \\
 dX' &= T^{-1}dXT + [X', d\Theta], \\
 ddX' &= d(T^{-1})dXT + T^{-1}ddXT + T^{-1}dXdT, \\
 &\quad + [dX', d\Theta] + [X', dd\Theta].
 \end{aligned} \tag{241}$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
 d(T^{-1})dXT &= (-T^{-1}dT)(T^{-1}dXT), \\
 &= -d\Theta(dX' - [X', d\Theta]).
 \end{aligned} \tag{242}$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned} T^{-1}dXdT &= (T^{-1}dXT)(T^{-1}dT), \\ &= (dX' - [X', d\Theta])d\Theta. \end{aligned} \quad (243)$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned} ddX' &= T^{-1}ddXT, \\ &\quad + 2[dX', d\Theta], \\ &\quad + [X', dd\Theta], \\ &\quad + [d\Theta, [X', d\Theta]]. \end{aligned} \quad (244)$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned} d\Theta/dt &= \Omega, \\ dd\Theta/dt^2 &= d\Omega/dt, \\ dX'/dt &= T^{-1}(dX/dt)T + [X', \Omega], \\ ddX'/dt^2 &= T^{-1}(ddX/dt^2)T, \\ &\quad + 2[dX'/dt, \Omega], \\ &\quad + [X', d\Omega/dt], \\ &\quad + [\Omega, [X', \Omega]]. \end{aligned} \quad (245)$$

The final numbered line is the derived relation used in the subsequent discussion.

Transformed Basis Elements

The formulas below make transformed basis elements explicit and collect identities used later in the manuscript. The following numbered identities fix the notation and formulas used in this block.

$$T(x) = f(x) + \mathbf{g}(x) \cdot \boldsymbol{\sigma}, \quad (246)$$

$$T^{-1} = (f - \mathbf{g} \cdot \boldsymbol{\sigma})/(f^2 - \mathbf{g}^2), \quad (247)$$

$$dT = df + d\mathbf{g} \cdot \boldsymbol{\sigma}, \quad (248)$$

$$\sigma'_n := T^{-1}\sigma_n T, \quad (249)$$

$$\sigma_n = T\sigma'_n T^{-1}. \quad (250)$$

Applying conjugation and reversion gives

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} (\sigma'_n)^* &= (T^{-1})^* \sigma_n^* T^*, \\ &= -(T^*)^{-1} \sigma_n T^*. \end{aligned} \quad (251)$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned} (\sigma'_n)^R &= T^R \sigma_n^R (T^{-1})^R, \\ &= T^R \sigma_n (T^R)^{-1}. \end{aligned} \quad (252)$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
(\sigma'_n)^{*R} &= -T^{*R}\sigma_n(T^{-1})^{*R}, \\
&= -T^{-1}\sigma_n T \text{ if } T^{-1} = T^{*R}, \\
&= -\sigma'_n \text{ if } T^{-1} = T^{*R}.
\end{aligned} \tag{253}$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
\sigma' algebra &= \sigma algebra, \\
(\sigma'_n)^2 &= T^{-1}\sigma_n T T^{-1}\sigma_n T, \\
&= T^{-1}\sigma_n^2 T = 1.
\end{aligned} \tag{254}$$

The final numbered line is the derived relation used in the subsequent discussion.
For $n \neq m$, the mixed products transform as follows.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
T^{-1}\sigma_n\sigma_m T &= T^{-1}\sigma_n T T^{-1}\sigma_m T = \sigma'_n\sigma'_m, \\
&= T^{-1}\epsilon_{nmk}i\sigma_k T = \epsilon_{nmk}i\sigma'_k.
\end{aligned} \tag{255}$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
\sigma'_1\sigma'_2\sigma'_3 &= T^{-1}\sigma_1 T T^{-1}\sigma_2 T T^{-1}\sigma_3 T, \\
&= T^{-1}\sigma_1\sigma_2\sigma_3 T = iT^{-1}T = i.
\end{aligned} \tag{256}$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
d\sigma'_n &= [\sigma'_n, T^{-1}dT], \\
d\sigma' &= [\sigma', T^{-1}dT] \quad \text{vector}, \\
Z &= Z_s + Z_v \cdot \sigma' \quad \text{paravector}, \\
\langle\langle Z \rangle\rangle^2 &= ZZ^{*R}, \\
&= (Z_s + Z_v \cdot \sigma')(Z_s - Z_v \cdot \sigma') \text{ if } T^{-1} = T^{*R}, \\
&= Z_s^2 - Z_v^2 \text{ if } T^{-1} = T^{*R}.
\end{aligned} \tag{257}$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
dZ &= dZ_s + dZ_v \cdot \sigma' + Z_v \cdot d\sigma', \\
&= dZ_s + dZ_v \cdot \sigma' + Z_v \cdot [\sigma', T^{-1}dT].
\end{aligned} \tag{258}$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
T^{-1}dT &= (fdf - gdg)/(f^2 - g^2), \\
&+ (fd\mathbf{g} - df\mathbf{g} - i\mathbf{g} \times d\mathbf{g}) \cdot \sigma / (f^2 - \mathbf{g}^2).
\end{aligned} \tag{259}$$

The final numbered line is the derived relation used in the subsequent discussion.

Define the following object.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
\Omega &:= T^{-1}(dT/dt), \\
\Omega' &:= T\Omega T^{-1} = (dT/dt)T^{-1}, \\
\text{scalar}(\sigma'_n) &= \text{scalar}(T^{-1}\sigma_n T), \\
&= \text{scalar}(\sigma_n T T^{-1}) \quad \text{cyclic property}, \\
&= \text{scalar}(\sigma_n), \\
&= 0.
\end{aligned} \tag{260}$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
&\text{scalar}(\Omega\sigma'_n), \\
&= \text{scalar}(T^{-1}(dT/dt)T^{-1}\sigma_n T), \\
&= \text{scalar}((dT/dt)T^{-1}\sigma_n T T^{-1}) \quad \text{cyclic}, \\
&= \text{scalar}((dT/dt)T^{-1}\sigma_n), \\
&= \text{scalar}(\Omega'\sigma_n).
\end{aligned} \tag{261}$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
k_n + i\omega_n &:= \text{scalar}(2\Omega'\sigma_n), \\
\Omega &= \frac{1}{2}(k + i\omega) \cdot \boldsymbol{\sigma}' + \text{scalar}, \\
[\boldsymbol{\sigma}', \omega \cdot (i\boldsymbol{\sigma}')] &= -2\omega \times \boldsymbol{\sigma}', \\
[\boldsymbol{\sigma}', \mathbf{k} \cdot \boldsymbol{\sigma}'] &= 2i\mathbf{k} \times \boldsymbol{\sigma}', \\
d\boldsymbol{\sigma}'/dt &= [\boldsymbol{\sigma}', \Omega], \\
&= i(k + i\omega) \times \boldsymbol{\sigma}', \\
&= (-\omega + ik) \times \boldsymbol{\sigma}'.
\end{aligned} \tag{262}$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
Z_{\mathbf{v}} \cdot [\boldsymbol{\sigma}', \Omega] &= (-Z_{\mathbf{v}} \times \omega + iZ_{\mathbf{v}} \times k) \cdot \boldsymbol{\sigma}', \\
dZ/dt &= dZ_s/dt, \\
&+ (dZ_{\mathbf{v}}/dt - Z_{\mathbf{v}} \times \omega + iZ_{\mathbf{v}} \times k) \cdot \boldsymbol{\sigma}'.
\end{aligned} \tag{263}$$

The final numbered line is the derived relation used in the subsequent discussion.

This interpretation also applies in non-inertial settings.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
X &= t + \mathbf{r} \cdot \boldsymbol{\sigma}', \\
dX/dt &= dt/dt, \\
&+ (d\mathbf{r}/dt - \mathbf{r} \times \omega + i\mathbf{r} \times \mathbf{k}) \cdot \boldsymbol{\sigma}'.
\end{aligned} \tag{264}$$

The final numbered line is the derived relation used in the subsequent discussion.

Energy & momentum.

$$mdX/dt = E + \mathbf{P}' \cdot \boldsymbol{\sigma}', \quad (265)$$

$$E/m = dt/dt, \quad (266)$$

$$\mathbf{P}'/m = d\mathbf{r}/dt + \mathbf{r} \times (-\omega + ik). \quad (267)$$

If $T^{-1} = T^{*\mathbf{R}}$, then the following relation holds.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \langle\langle dX \rangle\rangle^2 &= dt^2 - (d\mathbf{r} - \mathbf{r} \times \omega dt + i\mathbf{r} \times \mathbf{k} dt)^2, \\ &= (1 - (d\mathbf{r}/dt - \mathbf{r} \times \omega + i\mathbf{r} \times \mathbf{k})^2) dt^2. \end{aligned} \quad (268)$$

The final numbered line is the derived relation used in the subsequent discussion.

The following numbered relation is recorded for later use.

$$ds' := \sqrt{1 - (\mathbf{v} - \mathbf{r} \times \omega + i\mathbf{r} \times \mathbf{k})^2} dt. \quad (269)$$

Lorentz Boosts

The formulas below make lorentz boosts explicit and collect identities used later in the manuscript. See Eq. (235). Here $\theta \in \mathbb{R}$, $\mathbf{u} \in \mathbb{R}^3$ with $\|\mathbf{u}\| = 1$, and L is the boost element acting by conjugation on paravectors. The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \theta &= \operatorname{arctanh}(\beta), \\ \beta &= \tanh(\theta), \\ \gamma &= \cosh(\theta), \\ \gamma\beta &= \sinh(\theta), \\ \mathbf{u} &= (u_1, u_2, u_3), \mathbf{u}^2 = 1, \\ L(\theta, \mathbf{u}) &:= \exp(\tfrac{1}{2}\theta(\mathbf{u} \cdot \boldsymbol{\sigma})), \\ &= \cosh(\tfrac{1}{2}\theta) + \sinh(\tfrac{1}{2}\theta)(\mathbf{u} \cdot \boldsymbol{\sigma}). \end{aligned} \quad (270)$$

The final numbered line is the derived relation used in the subsequent discussion.

The following numbered relation is recorded for later use.

$$L^{-1} = \exp(-\tfrac{1}{2}\theta(\mathbf{u} \cdot \boldsymbol{\sigma})). \quad (271)$$

Note. The boost inverse satisfies $L^{-1} = L^* = L^{*\mathbf{R}}$.

Commutative over the same axis.

$$[L(\theta, \mathbf{u}), L(\theta', \mathbf{u})] = 0. \quad (272)$$

Co-linear boost.

$$L(\theta', \mathbf{u}) = L(\theta_1, \mathbf{u})L(\theta_2, \mathbf{u}), \quad (273)$$

$$\beta' = (\beta_1 + \beta_2)/(1 + \beta_1\beta_2), \quad (274)$$

$$\gamma' = \gamma_1\gamma_2(1 + \beta_1\beta_2). \quad (275)$$

Re-mapping space-time numbers.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} X &= t + \mathbf{r} \cdot \boldsymbol{\sigma}, \\ X' &= L^* X L, \\ &= t' + \mathbf{r}' \cdot \boldsymbol{\sigma}. \end{aligned} \quad (276)$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned} t' &= t \cosh(\theta) - \sinh(\theta)(\mathbf{u} \cdot \mathbf{r}), \\ &= \gamma(t - \beta(\mathbf{u} \cdot \mathbf{r})). \end{aligned} \quad (277)$$

The final numbered line is the derived relation used in the subsequent discussion.
Split the vector $\mathbf{r}$ into parts parallel and perpendicular to the boost direction.
Parallel and perpendicular to $\mathbf{u}$.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \text{par}(\mathbf{r}) &:= (\mathbf{r} \cdot \mathbf{u})\mathbf{u} && \text{parallel to } \mathbf{u}, \\ \text{prp}(\mathbf{r}) &:= \mathbf{r} - \text{par}(\mathbf{r}) && \text{perpendicular to } \mathbf{u}, \\ \mathbf{r} &= \text{par}(\mathbf{r}) + \text{prp}(\mathbf{r}), \\ \text{par}(\mathbf{r}') &= \gamma(\text{par}(\mathbf{r}) - \beta t\mathbf{u}), \\ \text{prp}(\mathbf{r}') &= \text{prp}(\mathbf{r}), \\ \mathbf{r}' &= \text{par}(\mathbf{r}') + \text{prp}(\mathbf{r}'), \\ &= \text{prp}(\mathbf{r}) + \gamma(\text{par}(\mathbf{r}) - \beta t\mathbf{u}), \\ &= \mathbf{r} + (\gamma(\mathbf{r} \cdot \mathbf{u} - \beta t) - \mathbf{r} \cdot \mathbf{u})\mathbf{u}. \end{aligned} \quad (278)$$

The final numbered line is the derived relation used in the subsequent discussion.
If $\beta \Rightarrow 0$, then the following relation holds.

$$\gamma \Rightarrow 1, t' \Rightarrow t, \mathbf{r}' \Rightarrow \mathbf{r}. \quad (279)$$

Paravector metric invariance.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \langle\langle X' \rangle\rangle^2 &= X' X'^*, \\ &= (L^* X L)(L^* X L)^*, \\ &= L^* X L L^* X^* L, \\ &= L^* X X^* L, \\ &= \langle\langle X \rangle\rangle^2 L^* L, \\ &= \langle\langle X \rangle\rangle^2. \end{aligned} \quad (280)$$

The final numbered line is the derived relation used in the subsequent discussion.

Lorentz Transformation of the Field

The formulas below make lorentz transformation of the field explicit and collect identities used later in the manuscript. See Eq. (236). Here $F \in \{0\} \oplus \mathbb{C}^3 \subset \mathbb{C} \oplus \mathbb{C}^3$, while $\mathbf{E}, \mathbf{B}, \mathbf{u}, \mathbf{v} \in \mathbb{R}^3$ and $\gamma, \theta \in \mathbb{R}$. The following numbered identities fix the notation and formulas used in this block.

$$F' = L^* F L, \quad (281)$$

$$L = \exp(\tfrac{1}{2}\theta \mathbf{u} \cdot \boldsymbol{\sigma}), \mathbf{u}^2 = 1, \quad (282)$$

$$\beta = \tanh(\theta), \gamma = \cosh(\theta), \quad (283)$$

$$F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (284)$$

$$\text{par}(\mathbf{E}') = \text{par}(\mathbf{E}), \quad (285)$$

$$\text{par}(\mathbf{B}') = \text{par}(\mathbf{B}), \quad (286)$$

$$\text{prp}(\mathbf{E}') = \gamma(\text{prp}(\mathbf{E}) + \beta \mathbf{u} \times \mathbf{B}), \quad (287)$$

$$\text{prp}(\mathbf{B}') = \gamma(\text{prp}(\mathbf{B}) - \beta \mathbf{u} \times \mathbf{E}). \quad (288)$$

These identities can be compressed into the single relation.

$$\mathbf{E}' = \text{par}(\mathbf{E}) + \gamma(\text{prp}(\mathbf{E}) + \beta \mathbf{u} \times \mathbf{B}), \quad (289)$$

$$\mathbf{B}' = \text{par}(\mathbf{B}) + \gamma(\text{prp}(\mathbf{B}) - \beta \mathbf{u} \times \mathbf{E}), \quad (290)$$

$$\partial' = L^* \partial L, \quad (291)$$

$$\partial' F' = L^*(\rho - \mathbf{J} \cdot \boldsymbol{\sigma}) L. \quad (292)$$

Spatial Rotations

The formulas below make spatial rotations explicit and collect identities used later in the manuscript. See Eq. (234). Here $\theta \in \mathbb{R}$, $\mathbf{u} \in \mathbb{R}^3$ with $\|\mathbf{u}\| = 1$, and R is the rotor element acting by conjugation on paravectors. The intermediate steps are written out explicitly as follows.

$$\begin{aligned} R(\theta, \mathbf{u}) &:= \exp(\tfrac{1}{2}\theta(\mathbf{u} \cdot i\boldsymbol{\sigma})), \\ &= \cos(\tfrac{1}{2}\theta) + \sin(\tfrac{1}{2}\theta)(\mathbf{u} \cdot i\boldsymbol{\sigma}) \quad \text{rotor.} \end{aligned} \quad (293)$$

The final numbered line is the derived relation used in the subsequent discussion.

The following numbered relation is recorded for later use.

$$R^{-1} = \exp(-\tfrac{1}{2}\theta(\mathbf{u} \cdot i\boldsymbol{\sigma})). \quad (294)$$

Note. The rotor inverse satisfies $R^{-1} = R^R = R^{*R}$.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} X' &= R^R X R, \\ &= t' + \mathbf{r}' \cdot \boldsymbol{\sigma}. \end{aligned} \quad (295)$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} t' &= t, \\ \mathbf{r}' &= \mathbf{r} \cos(\theta) + (\mathbf{u} \times \mathbf{r}) \sin(\theta), \\ &\quad + (\mathbf{u} \cdot \mathbf{r})(1 - \cos(\theta))\mathbf{u}, \end{aligned}$$

Rodrigues's formula. (296)

The final numbered line is the derived relation used in the subsequent discussion.

Differentiating the transformed quantities gives.

$$d(R^R) = (dR)^R = dR^R, \quad (297)$$

$$R^R R = 1, \quad (298)$$

$$dR^R R + R^R dR = 0, \quad (299)$$

$$dR = -R dR^R R. \quad (300)$$

The reverse symmetry can be written as.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \Omega &:= -R(dR^R/ds) = (dR/ds)R^R, \\ \Omega^R &= -\Omega, \end{aligned}$$

$$\begin{aligned}
\Omega' &= R^R \Omega R = R(dR/ds), \\
\Omega'^R &= -\Omega', \\
dR/ds &= \Omega R, \\
X' &= R^R X R \Rightarrow RX' = X R, \\
dRX' + RdX' &= dXR + XdR, \\
RdX' &= dXR + XdR - dRX', \\
&= dXR + XdR - dRX'.
\end{aligned} \tag{301}$$

The final numbered line is the derived relation used in the subsequent discussion. The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
dX'/ds &= R^R(dX/ds)R + [X', \Omega'], \\
&= R^R(dX/ds + [X, \Omega])R.
\end{aligned} \tag{302}$$

The final numbered line is the derived relation used in the subsequent discussion.

Compositions of Boosts and Rotations

The formulas below make compositions of boosts and rotations explicit and collect identities used later in the manuscript. The following numbered identities fix the notation and formulas used in this block.

$$T = LR, \tag{303}$$

$$T^{-1} = T^{*R} = R^{*R} L^{*R} = R^R L^*, \tag{304}$$

$$T = RL, \tag{305}$$

$$T^{-1} = T^{*R} = L^{*R} R^{*R} = L^* R^R, \tag{306}$$

$$T = R_1 R_2, \tag{307}$$

$$T^{-1} = T^{*R} = R_2^R R_1^R, \tag{308}$$

$$T = L_1 L_2, \tag{309}$$

$$T^{-1} = T^{*R} = L_2^* L_1^*. \tag{310}$$

V Electromagnetic Sector

With space-time and boosts in place, the next step is the first major physics payoff: the electromagnetic field, gauge structure, and Lorentz force written in compact STA form.

Fields, potentials, and Maxwell in one line

The derivative operator comes first because every later field equation is built from it. Writing time and space derivatives together as a single paravector operator mirrors the earlier unification of temporal and spatial coordinates in dX , and it is this parallelism that makes the electromagnetic formulas compact rather than merely abbreviated.

With the paravector differential operator

$$\partial := \partial_t + \boldsymbol{\sigma} \cdot \boldsymbol{\nabla}, \quad \partial^* = \partial_t - \boldsymbol{\sigma} \cdot \boldsymbol{\nabla}, \quad \partial \partial^* = \partial^* \partial = \partial_t^2 - \boldsymbol{\nabla}^2 =: \square, \tag{311}$$

and with $V, \rho, \lambda, S, q \in \mathbb{R}$, $\mathbf{E}, \mathbf{B}, \mathbf{A}, \mathbf{J} \in \mathbb{R}^3$, the electromagnetic field and potential are written as

$$F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad \Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}. \tag{312}$$

Here $\Phi \in \mathbb{R} \oplus \mathbb{R}^3$, while F is a purely vector complex paravector in $\{0\} \oplus \mathbb{C}^3 \subset \mathbb{C} \oplus \mathbb{C}^3$. Equation (312) packages electric and magnetic fields into one object F , and scalar and vector potentials into one object Φ . The purpose is not only notational compression: once these quantities are assembled into STA objects, differentiation and transformation act on their full content in a single step.

The identity

$$\partial \Phi = S + F^*, \quad S := \partial_t V + \boldsymbol{\nabla} \cdot \mathbf{A} \tag{313}$$

encodes

$$\mathbf{E} = -\partial_t \mathbf{A} - \nabla V, \quad \mathbf{B} = \nabla \times \mathbf{A}. \quad (314)$$

The compact Maxwell equation is

$$\partial F = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}. \quad (315)$$

Splitting real/imaginary scalar/vector parts of (315) reproduces

$$\nabla \cdot \mathbf{E} = \rho, \quad \nabla \cdot \mathbf{B} = 0, \quad (316)$$

$$\nabla \times \mathbf{B} = \mathbf{J} + \partial_t \mathbf{E}, \quad \nabla \times \mathbf{E} = -\partial_t \mathbf{B}. \quad (317)$$

This is standard STA/GA electrodynamics up to sign conventions fixed by (312) and (311) [9, 10, 12]. The advantage of Eq. (315) is that divergence and curl do not need to be introduced as separate primitive operators at this stage. They emerge automatically when the scalar and vector parts of the single STA equation are separated.

Wave equations, gauge, and Lorentz force

Applying ∂^* to (315) gives

$$\square F = \partial^*(\rho - \mathbf{J} \cdot \boldsymbol{\sigma}), \quad (318)$$

whose scalar part yields charge conservation,

$$\partial_t \rho + \nabla \cdot \mathbf{J} = 0. \quad (319)$$

For a real scalar gauge function $\lambda \in \mathbb{R}$,

$$\Phi' = \Phi + \partial^* \lambda, \quad F' = F. \quad (320)$$

This gauge transformation shows why the potential Φ is retained even when the field F is physically primary: Φ carries the redundancy, while F carries the gauge-invariant derivative content.

For a particle of real mass $m \in \mathbb{R}$, real charge $q \in \mathbb{R}$, and proper velocity $U \in \mathbb{R} \oplus \mathbb{R}^3$, the proper-force formula is

$$m \frac{dU}{ds} = q \operatorname{real}(FU) = q\gamma(\mathbf{v} \cdot \mathbf{E} + (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \boldsymbol{\sigma}), \quad (321)$$

which packages both relativistic power and the Lorentz three-force. The real part in Eq. (321) extracts the physically relevant sector of the product FU . The scalar term gives power transfer and the vector term gives the ordinary Lorentz force, so the standard split between energy change and momentum change is preserved inside one STA formula.

Lagrangian and energy-momentum remarks

The electromagnetic particle Lagrangian is

$$\mathcal{L} = -\frac{m}{\gamma} - q(V - \mathbf{v} \cdot \mathbf{A}), \quad (322)$$

and its Euler–Lagrange equations return $d\mathbf{P}/dt = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$. This is the expected relativistic coupling prescription: the free action from Eq. (179) is supplemented by the potential term, and the field-strength form of the force law reappears after variation.

A natural paravector candidate for field energy density and flux is

$$T := \frac{1}{2} F F^R = \frac{1}{2} (\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (323)$$

Editorial note: Eq. (323) is a useful frame-dependent energy-density/Poynting paravector, but by itself it is not the full electromagnetic stress-energy tensor. In STA language the full stress-energy object is more naturally a linear map such as $T(a) = \frac{1}{2} F a F^R$ [9, 11].

Electromagnetic Component Expansions

The detailed electromagnetic development below writes out the field decomposition, potential reconstruction, source formulas, gauge transformations, force law expansions, and variational identities that underlie Eqs. (311)–(323). Organizing those details here keeps the field theory self-contained before the manuscript moves to wave equations.

Electric and Magnetic Fields

The next step is to spell out the electromagnetic field content that was previously presented in compact STA form. These identities make the relation between the vector fields and the single field paravector fully explicit. See Eqs. (311), (312), and (315). Here $V, \rho, S, q \in \mathbb{R}$, $\mathbf{A}, \mathbf{E}, \mathbf{B}, \mathbf{J} \in \mathbb{R}^3$, $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3$, and $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma} \in \{0\} \oplus \mathbb{C}^3$. *Introduce the following vector notation.*

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}\partial_{\mathbf{r}} &= \{\partial/\partial r_1, \partial/\partial r_2, \partial/\partial r_3\}, \\ \partial_i &= \partial/\partial r_i, \\ \partial_{\mathbf{r}}^2 &= \partial_1^2 + \partial_2^2 + \partial_3^2, \\ \partial &= \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}, \\ &= \partial_t + \sigma_1 \partial_1 + \sigma_2 \partial_2 + \sigma_3 \partial_3.\end{aligned}\tag{324}$$

The final numbered line is the derived relation used in the subsequent discussion.

Introduce the following vectors.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}\mathbf{E} &= \{\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3\} && \text{electric field,} \\ \mathbf{B} &= \{\mathbf{B}_1, \mathbf{B}_2, \mathbf{B}_3\} && \text{magnetic field,} \\ \mathcal{F} &= \mathbf{E} + i\mathbf{B} && \text{electromagnetic field,} \\ F &= (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, i = \sigma_1 \sigma_2 \sigma_3, \\ &= \mathcal{F} \cdot \boldsymbol{\sigma}.\end{aligned}\tag{325}$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}F^* &= (-\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ &= -\mathcal{F}^* \cdot \boldsymbol{\sigma}.\end{aligned}\tag{326}$$

The final numbered line is the derived relation used in the subsequent discussion.

Table 16: Components of $\frac{1}{2}F^2$.

real scalar:	$\frac{1}{2}(\mathbf{E}^2 - \mathbf{B}^2)$
imaginary scalar:	$i\mathbf{E} \cdot \mathbf{B}$
real vector:	$\mathbf{0}$
imaginary vector:	$\mathbf{0}$

Table 17: Components of $\frac{1}{2}FF^R$.

scalar:	$\frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2)$	$\leftarrow$ energy density
vector:	$\mathbf{E} \times \mathbf{B}$	$\leftarrow$ Poynting vector

Introduce the source-density paravector.

$$D = \rho + \mathbf{J} \cdot \boldsymbol{\sigma},\tag{327}$$

$$D^* = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}.\tag{328}$$

Let ρ denote the scalar charge density.

Let $\mathbf{J}$ denote the current-density three-vector.

The intermediate steps are written out explicitly as follows.

$$\partial F = D^* \quad \text{Maxwell's equations,}$$

$$= \rho - \mathbf{J} \cdot \boldsymbol{\sigma}. \quad (329)$$

The final numbered line is the derived relation used in the subsequent discussion.
The following numbered relation is recorded for later use.

$$\partial^* F^* = D. \quad (330)$$

Expanding the left-hand side gives.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \partial F &= (\partial_t + \boldsymbol{\sigma} \cdot \boldsymbol{\partial}_r)((\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}), \\ &= (\boldsymbol{\partial}_r \cdot \mathbf{E}), \\ &\quad + (\boldsymbol{\partial}_r \cdot \mathbf{B})i, \\ &\quad + (\partial_t \mathbf{E} - \boldsymbol{\partial}_r \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\ &\quad + (\partial_t \mathbf{B} + \boldsymbol{\partial}_r \times \mathbf{E}) \cdot (i\boldsymbol{\sigma}). \end{aligned} \quad (331)$$

The final numbered line is the derived relation used in the subsequent discussion.

Table 18: Components of $\partial F = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}$.

real scalar:	$\boldsymbol{\partial}_r \cdot \mathbf{E} = \rho$
imaginary scalar:	$\boldsymbol{\partial}_r \cdot \mathbf{B} = 0$
real vector:	$\partial_t \mathbf{E} - \boldsymbol{\partial}_r \times \mathbf{B} = -\mathbf{J}$
imaginary vector:	$\partial_t \mathbf{B} + \boldsymbol{\partial}_r \times \mathbf{E} = 0$

Rearranging the vector equations gives.

$$\boldsymbol{\partial}_r \times \mathbf{B} = \mathbf{J} + \partial_t \mathbf{E}, \quad (332)$$

$$\boldsymbol{\partial}_r \times \mathbf{E} = -\partial_t \mathbf{B}. \quad (333)$$

Wave Operator and d'Alembertian

The wave operator is written out here in full so that its factorization into the d'Alembertian and the component field equations can be read directly. See Eq. (311). The following numbered identities fix the notation and formulas used in this block.

$$\partial = \partial_t + \boldsymbol{\sigma} \cdot \boldsymbol{\partial}_r, \quad (334)$$

$$\partial^* = \partial_t - \boldsymbol{\sigma} \cdot \boldsymbol{\partial}_r. \quad (335)$$

The d'Alembertian is.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \square &:= \partial_t^2 - \boldsymbol{\partial}_r^2, \\ &= \partial \partial^* = \partial^* \partial, \\ &= \langle\langle \partial \rangle\rangle^2. \end{aligned} \quad (336)$$

The final numbered line is the derived relation used in the subsequent discussion.

Note. This expansion uses the sigma-algebra conventions fixed earlier in the manuscript.

$$\partial^2 = \partial_t^2 + \boldsymbol{\partial}_r^2 + 2\partial_t \boldsymbol{\partial}_r \cdot \boldsymbol{\sigma}. \quad (337)$$

This differs from $\square$.

Note. Sign conventions for F vary across the literature; the present manuscript keeps the convention fixed by Eqs. (312) and (315).

Note. Some authors instead use the metric signature $(-, +, +, +)$.

Note. Some authors denote the wave operator with a different box-symbol convention.

The corresponding wave equation is.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}\square F &= \partial^* \partial F, \\ &= \partial^* (\rho - \mathbf{J} \cdot \boldsymbol{\sigma}).\end{aligned}\tag{338}$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned}\square F^* &= \partial(\rho + \mathbf{J} \cdot \boldsymbol{\sigma}), \\ (\square \mathbf{E} + i \square \mathbf{B}) \cdot \boldsymbol{\sigma}, \\ &= (\partial_t \rho + \partial_{\mathbf{r}} \cdot \mathbf{J}) + (-\partial_t \mathbf{J} - \partial_{\mathbf{r}} \rho + i(\partial_{\mathbf{r}} \times \mathbf{J})) \cdot \boldsymbol{\sigma}.\end{aligned}\tag{339}$$

The final numbered line is the derived relation used in the subsequent discussion.

Table 19: Components of $\square F = \partial^* (\rho - \mathbf{J} \cdot \boldsymbol{\sigma})$.

real scalar:	$0 = \partial_t \rho + \partial_{\mathbf{r}} \cdot \mathbf{J}$
imaginary scalar:	$0 = 0$
real vector:	$\square \mathbf{E} = -\partial_t \mathbf{J} - \partial_{\mathbf{r}} \rho$
imaginary vector:	$\square \mathbf{B} = \partial_{\mathbf{r}} \times \mathbf{J}$

The following numbered relation is recorded for later use.

$$\partial_t \rho = -\partial_{\mathbf{r}} \cdot \mathbf{J} \quad \text{conservation of charge.}\tag{340}$$

If $\rho = 0$ and $\mathbf{J} = 0$, then the following relation holds.

$$\square \mathbf{E} = 0 \Rightarrow \partial_t^2 \mathbf{E} = \partial_{\mathbf{r}}^2 \mathbf{E},\tag{341}$$

$$\square \mathbf{B} = 0 \Rightarrow \partial_t^2 \mathbf{B} = \partial_{\mathbf{r}}^2 \mathbf{B}.\tag{342}$$

Potential Fields

The potential-based formulation is expanded here because it is the point at which gauge freedom, source reconstruction, and the emergence of the electric and magnetic fields all become transparent at once. See Eqs. (313) and (314). Here $V, S, \rho, \lambda \in \mathbb{R}$, $\mathbf{A}, \mathbf{E}, \mathbf{B}, \mathbf{J} \in \mathbb{R}^3$, $\Phi \in \mathbb{R} \oplus \mathbb{R}^3$, and $F \in \{0\} \oplus \mathbb{C}^3 \subset \mathbb{C} \oplus \mathbb{C}^3$. The intermediate steps are written out explicitly as follows.

$$\begin{aligned}F^* &= (-\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ \Phi &= V + \mathbf{A} \cdot \boldsymbol{\sigma}, \\ \partial \Phi &= \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}, \\ &\quad + (\partial_t \mathbf{A} + \partial_{\mathbf{r}} V) \cdot \boldsymbol{\sigma}, \\ &\quad + i(\partial_{\mathbf{r}} \times \mathbf{A}) \cdot \boldsymbol{\sigma}.\end{aligned}\tag{343}$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned}S &:= \text{scalar}(\partial \Phi), \\ &= \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}.\end{aligned}\tag{344}$$

The final numbered line is the derived relation used in the subsequent discussion.
The target component identities are.

$$\mathbf{E} = -\partial_t \mathbf{A} - \partial_{\mathbf{r}} V,\tag{345}$$

$$\mathbf{B} = \partial_{\mathbf{r}} \times \mathbf{A}. \quad (346)$$

Consider the following identity.

$$\partial\Phi = S + F^*, \quad (347)$$

$$F^* = \partial\Phi - S, \quad (348)$$

$$\text{scalar}(F^*) = \text{scalar}(\partial\Phi - S) = 0, \quad (349)$$

$$\text{vector}(F^*) = \text{vector}(\partial\Phi). \quad (350)$$

Note. always differentiate first.

$$(\partial\Phi)^{*R} \neq \Phi^{*R}\partial^{*R}. \quad (351)$$

Equivalently, without introducing S , one may write.

$$F^* = (\partial\Phi - (\partial\Phi)^{*R})/2. \quad (352)$$

Table 20: Components of $\partial\Phi = S + F^*$.

real scalar: $\partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A} = S$

imaginary scalar: $0 = 0$

real vector: $\partial_t \mathbf{A} + \partial_{\mathbf{r}} V = -\mathbf{E}$

imaginary vector: $\partial_{\mathbf{r}} \times \mathbf{A} = \mathbf{B}$

Sources Derived from the Potential

The formulas below make sources derived from the potential explicit and collect identities used later in the manuscript. See Eq. (315). Recall the previously established definitions.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} F &= (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ \Phi &= V + \mathbf{A} \cdot \boldsymbol{\sigma}, \\ S &= \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}, \\ F^* &= \partial\Phi - S \Rightarrow F = \partial^* \Phi^* - S, \\ \partial F &= \partial\partial^* \Phi^* - \partial S, \\ &= \square \Phi^* - \partial S, \\ &= (\square V - \partial_t S) + (-\square \mathbf{A} - \partial_{\mathbf{r}} S) \cdot \boldsymbol{\sigma}, \\ &= \rho - \mathbf{J} \cdot \boldsymbol{\sigma}. \end{aligned} \quad (353)$$

The final numbered line is the derived relation used in the subsequent discussion.

The resulting relations are.

$$\rho = \square V - \partial_t S, \quad (354)$$

$$\mathbf{J} = \square \mathbf{A} + \partial_{\mathbf{r}} S. \quad (355)$$

If $S = 0$, then the following relation holds.

Lorenz gauge condition.

$$\partial_t V = -\partial_{\mathbf{r}} \cdot \mathbf{A}, \quad (356)$$

$$\rho = \square V, \quad (357)$$

$$\mathbf{J} = \square \mathbf{A}. \quad (358)$$

Gauge Transform of the Potential

The formulas below make gauge transform of the potential explicit and collect identities used later in the manuscript. See Eq. (320). The following numbered identities fix the notation and formulas used in this block.

$$\text{scalar}(\partial\Phi) = S, \quad (359)$$

$$S = \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}. \quad (360)$$

Let λ denote scalar field.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \Phi' &= \Phi + \partial^* \lambda, \\ \partial\Phi' &= \partial(\Phi + \partial^* \lambda), \\ &= \partial\Phi + \partial(\partial^* \lambda), \\ &= \partial\Phi + \square\lambda. \end{aligned} \quad (361)$$

The final numbered line is the derived relation used in the subsequent discussion.

Note. The quantity $\square\lambda$ is again a scalar field.

$$\text{scalar}(\partial\Phi') = S + \square\lambda, \quad (362)$$

$$\text{vector}(\partial\Phi') = \text{vector}(\partial\Phi), \quad (363)$$

$$F' = F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (364)$$

Proper Force and Lorentz Dynamics

The formulas below make proper force and lorentz dynamics explicit and collect identities used later in the manuscript. See Eq. (321). Here $q, m \in \mathbb{R}$, $F \in \{0\} \oplus \mathbb{C}^3 \subset \mathbb{C} \oplus \mathbb{C}^3$, $U \in \mathbb{R} \oplus \mathbb{R}^3$, and $\mathbf{E}, \mathbf{B}, \mathbf{v}, \mathbf{f} \in \mathbb{R}^3$. The intermediate steps are written out explicitly as follows.

$$\begin{aligned} mU &= mdX/ds, \\ &= m\gamma + m\gamma\mathbf{v} \cdot \boldsymbol{\sigma}. \end{aligned} \quad (365)$$

The final numbered line is the derived relation used in the subsequent discussion.

The following numbered relation is recorded for later use.

$$d/ds = \gamma d/dt. \quad (366)$$

The time component gives the relativistic work law.

$$d/dt(m\gamma) = q\mathbf{v} \cdot \mathbf{E}. \quad (367)$$

Note. Here $\mathbf{E}$ denotes the electric field, not the energy operator E .

Note. The magnetic field contributes no work term in the scalar component.

The spatial component gives the relativistic Lorentz force law.

$$\mathbf{P} := m\gamma\mathbf{v}, \quad (368)$$

$$d\mathbf{P}/dt = q\mathbf{E} + q\mathbf{v} \times \mathbf{B}. \quad (369)$$

These terms combine into the single relation.

$$d/dt(mU) = q\mathbf{v} \cdot \mathbf{E} + (q\mathbf{E} + q\mathbf{v} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (370)$$

Consider the following identity.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
FU/\gamma &= ((\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma})(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), \\
&= \mathbf{E} \cdot \boldsymbol{\sigma} + i\mathbf{B} \cdot \boldsymbol{\sigma} + (\mathbf{E} \cdot \boldsymbol{\sigma})(\mathbf{v} \cdot \boldsymbol{\sigma}), \\
&\quad + i(\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{v} \cdot \boldsymbol{\sigma}), \\
&= \mathbf{E} \cdot \boldsymbol{\sigma} + i\mathbf{B} \cdot \boldsymbol{\sigma} + \mathbf{E} \cdot \mathbf{v}, \\
&\quad + i(\mathbf{E} \times \mathbf{v}) \cdot \boldsymbol{\sigma} + i\mathbf{B} \cdot \mathbf{v} - (\mathbf{B} \times \mathbf{v}) \cdot \boldsymbol{\sigma}.
\end{aligned} \tag{371}$$

The final numbered line is the derived relation used in the subsequent discussion.

Table 21: Components of FU/γ .

real scalar:	$\mathbf{E} \cdot \mathbf{v}$
imaginary scalar:	$\mathbf{B} \cdot \mathbf{v}$
real vector:	$\mathbf{E} + \mathbf{v} \times \mathbf{B}$
imaginary vector:	$\mathbf{B} + \mathbf{E} \times \mathbf{v}$

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
\text{real}(FU) &= \frac{1}{2}(FU + UF^R), \\
&= \gamma(\mathbf{v} \cdot \mathbf{E} + (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \boldsymbol{\sigma}).
\end{aligned} \tag{372}$$

The final numbered line is the derived relation used in the subsequent discussion.

Note. The imaginary part of FU exchanges the electric and magnetic roles and is therefore not the sector used in the physical force law.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
mdU/ds &= q\text{real}(FU), \\
&= \frac{1}{2}q(FU + UF^R).
\end{aligned} \tag{373}$$

The final numbered line is the derived relation used in the subsequent discussion.

The following numbered relation is recorded for later use.

$$mddX/ds^2 = q\text{real}(FdX/ds) \quad \text{4-force.} \tag{374}$$

Variational Derivation of the Lorentz Force

The formulas below make variational derivation of the lorentz force explicit and collect identities used later in the manuscript. See Eq. (322). Here $X, U, P \in \mathbb{R} \oplus \mathbb{R}^3$, $s, t, q, m \in \mathbb{R}$, $\mathbf{r}, \mathbf{v}, \mathbf{A} \in \mathbb{R}^3$, and $V \in \mathbb{R}$. *Recall the previously established definitions.* The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
\Phi &= V + \mathbf{A} \cdot \boldsymbol{\sigma}, \\
U &= dX/ds, \\
\gamma &= dt/ds, \\
\text{action} &= -m \int ds - q \int \text{scalar}(\Phi^* U) ds, \\
&= \int (-m/\gamma - q\text{scalar}(\Phi^* U/\gamma)) dt, \\
&= \int (-m/\gamma, \\
&\quad - q\text{scalar}((V - \mathbf{A} \cdot \boldsymbol{\sigma})(1 + \mathbf{v} \cdot \boldsymbol{\sigma}))) dt, \\
&= \int (-m/\gamma - qV + q\mathbf{v} \cdot \mathbf{A}) dt.
\end{aligned} \tag{375}$$

The final numbered line is the derived relation used in the subsequent discussion.
The following numbered relation is recorded for later use.

$$\mathcal{L} = -m/\gamma - q(V - \mathbf{v} \cdot \mathbf{A}) \quad \text{Lagrangian.} \quad (376)$$

Verification.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \partial_{\mathbf{v}} \mathcal{L} &= m\gamma \mathbf{v} + q\mathbf{A}, \\ &= \mathbf{P} + q\mathbf{A}. \end{aligned} \quad (377)$$

The final numbered line is the derived relation used in the subsequent discussion.
The following numbered identities fix the notation and formulas used in this block.

$$d/dt \partial_{\mathbf{v}} \mathcal{L} = d\mathbf{P}/dt + q\partial_t \mathbf{A} + q(\mathbf{v} \cdot \partial_{\mathbf{r}}) \mathbf{A}, \quad (378)$$

$$\partial_{\mathbf{r}} \mathcal{L} = -q\partial_{\mathbf{r}} V + q\partial_{\mathbf{r}}(\mathbf{v} \cdot \mathbf{A}). \quad (379)$$

Use the following vector-calculus identity.

$$(\mathbf{v}(t) \cdot \partial_{\mathbf{r}}) \mathbf{A}(t, \mathbf{r}) = \partial_{\mathbf{r}}(\mathbf{v} \cdot \mathbf{A}) - \mathbf{v} \times (\partial_{\mathbf{r}} \times \mathbf{A}). \quad (380)$$

Solving $d/dt \partial_{\mathbf{v}} \mathcal{L} = \partial_{\mathbf{r}} \mathcal{L}$ for $d\mathbf{P}/dt$ gives the following relation.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} d\mathbf{P}/dt &= -q\partial_{\mathbf{r}} V - q\partial_t \mathbf{A}, \\ &\quad + q(\partial_{\mathbf{r}}(\mathbf{v} \cdot \mathbf{A}) - (\mathbf{v} \cdot \partial_{\mathbf{r}}) \mathbf{A}), \\ &= q(-\partial_{\mathbf{r}} V - \partial_t \mathbf{A} + \mathbf{v} \times (\partial_{\mathbf{r}} \times \mathbf{A})), \\ &= q(\mathbf{E} + \mathbf{v} \times \mathbf{B}), \\ &= m\text{vector}(dU/ds)/\gamma. \end{aligned} \quad (381)$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \mathcal{H} &= (\partial_{\mathbf{v}} \mathcal{L}) \cdot \mathbf{v} - \mathcal{L}, \\ &= m\gamma \mathbf{v} \cdot \mathbf{v} + q\mathbf{v} \cdot \mathbf{A} + m/\gamma. \end{aligned} \quad (382)$$

The final numbered line is the derived relation used in the subsequent discussion.

Adding the potential term contributes $qV - q\mathbf{v} \cdot \mathbf{A}$.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} &= m\gamma(\mathbf{v}^2 + 1/\gamma^2) + qV, \\ &= m\gamma(\mathbf{v}^2 + 1 - \mathbf{v}^2) + qV, \\ &= m\gamma + qV. \end{aligned} \quad (383)$$

The final numbered line is the derived relation used in the subsequent discussion.

Recall the Hamiltonian property.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} dH/dt &= -\partial_t \mathcal{L}, \\ d\mathcal{H}/dt &= d(m\gamma)/dt + qdV/dt, \\ &= d(m\gamma)/dt + q\partial_t V + q\mathbf{v} \cdot \partial_{\mathbf{r}} V. \end{aligned} \quad (384)$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned}\partial_t H &= q\partial_t V - q\mathbf{v} \cdot \partial_t \mathbf{A}, \\ d(m\gamma)/dt &= q\mathbf{v} \cdot (-\partial_t \mathbf{A} - \partial_t \mathbf{A}), \\ &= q\mathbf{v} \cdot \mathbf{E}.\end{aligned}\tag{385}$$

The final numbered line is the derived relation used in the subsequent discussion.

Electromagnetic Energy-Momentum

The formulas below make electromagnetic energy-momentum explicit and collect identities used later in the manuscript. See Eq. (323). The intermediate steps are written out explicitly as follows.

$$\begin{aligned}F &= (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ F^R &= (\mathbf{E} - i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ T &= \frac{1}{2} F F^R, \\ &= \frac{1}{2} (\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\ &= T_s + T_v \cdot \boldsymbol{\sigma}.\end{aligned}\tag{386}$$

The final numbered line is the derived relation used in the subsequent discussion.
The following numbered relation is recorded for later use.

$$\partial = \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}.\tag{387}$$

This is a frame-dependent energy-density/Poynting paravector, not by itself the full electromagnetic stress-energy tensor.

Table 22: Components of ∂T .

real scalar:	$\partial_t T_s + \partial_{\mathbf{r}} \cdot T_v$
imaginary scalar:	0
real vector:	$\partial_{\mathbf{r}} T_s + \partial_t T_v$
imaginary vector:	$\partial_{\mathbf{r}} \times T_v$

Recall for any paravectors Z and Z' .

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}\partial(ZZ') &= (\partial Z)Z' + Z(\partial Z') + [\boldsymbol{\sigma}, Z] \cdot \partial_{\mathbf{r}} Z', \\ \partial T &= \frac{1}{2} \partial(F F^R), \\ &= \frac{1}{2} (\partial F) F^R + \frac{1}{2} F (\partial F^R) + \frac{1}{2} [\boldsymbol{\sigma}, F] \cdot \partial_{\mathbf{r}} F^R.\end{aligned}\tag{388}$$

The final numbered line is the derived relation used in the subsequent discussion.
The following numbered identities fix the notation and formulas used in this block.

$$D = \rho + \mathbf{J} \cdot \boldsymbol{\sigma} \quad \text{density paravector},\tag{389}$$

$$D^* = \rho - \mathbf{J} \cdot \boldsymbol{\sigma},\tag{390}$$

$$\partial = \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}},\tag{391}$$

$$\partial F = D^*,\tag{392}$$

$$F^2 = (\mathbf{E}^2 - \mathbf{B}^2) + 2i(\mathbf{E} \cdot \mathbf{B}) \quad \text{scalar}.\tag{393}$$

Table 23: Components of ∂F^2 .

real scalar:	$\partial_t (\mathbf{E}^2 - \mathbf{B}^2)$
imaginary scalar:	$\partial_t 2(\mathbf{E} \cdot \mathbf{B})$
real vector:	$\partial_{\mathbf{r}} (\mathbf{E}^2 - \mathbf{B}^2)$
imaginary vector:	$\partial_{\mathbf{r}} 2(\mathbf{E} \cdot \mathbf{B})$

Table 24: Components of FD^* .

real scalar:	$-\mathbf{E} \cdot \mathbf{J}$
imaginary scalar:	$-\mathbf{B} \cdot \mathbf{J}$
real vector:	$\rho\mathbf{E} + \mathbf{B} \times \mathbf{J}$
imaginary vector:	$\rho\mathbf{B} - \mathbf{E} \times \mathbf{J}$

Table 25: Components of D^*F .

real scalar:	$-\mathbf{E} \cdot \mathbf{J}$
imaginary scalar:	$-\mathbf{B} \cdot \mathbf{J}$
real vector:	$\rho\mathbf{E} - \mathbf{B} \times \mathbf{J}$
imaginary vector:	$\rho\mathbf{B} + \mathbf{E} \times \mathbf{J}$

Table 26: Components of $D^*F + FD^*$.

real scalar:	$-2\mathbf{E} \cdot \mathbf{J}$
imaginary scalar:	$-2\mathbf{B} \cdot \mathbf{J}$
real vector:	$2\rho\mathbf{E}$
imaginary vector:	$2\rho\mathbf{B}$

Recall for any paravectors Z and Z' .

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
\partial(ZZ') &= (\partial Z)Z' + Z(\partial Z') + [\boldsymbol{\sigma}, Z] \cdot \partial_{\mathbf{r}}Z', \\
\partial(F^2) &= (\partial F)F + F(\partial F) + [\boldsymbol{\sigma}, F] \cdot \partial_{\mathbf{r}}F, \\
&= D^*F + FD^* + [\boldsymbol{\sigma}, F] \cdot \partial_{\mathbf{r}}F.
\end{aligned} \tag{394}$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
F &= F_{\mathbf{v}} \cdot \boldsymbol{\sigma}, \\
[\boldsymbol{\sigma}, F] &= 2iF_{\mathbf{v}} \times \boldsymbol{\sigma}, \\
[\boldsymbol{\sigma}, F] \cdot \partial_{\mathbf{r}}F &= -2iF_{\mathbf{v}} \cdot (\partial_{\mathbf{r}} \times F_{\mathbf{v}}), \\
&\quad - 2((\partial_{\mathbf{r}} \cdot F_{\mathbf{v}})F_{\mathbf{v}} - \tfrac{1}{2}\partial_{\mathbf{r}}F_{\mathbf{v}}^2) \cdot \boldsymbol{\sigma}.
\end{aligned} \tag{395}$$

The final numbered line is the derived relation used in the subsequent discussion.

The following numbered relation is recorded for later use.

$$\square F^2 = \partial^* \partial(F^2) \quad \text{scalar.} \tag{396}$$

VI Relativistic Wave Equations and Spinor Dynamics

The relativistic wave-equation sector is best read as a second application of the same algebra: first the scalar wave equation, then minimal coupling and factorization, and only then the Dirac block map and Pauli limit.

Klein–Gordon and frequency splitting

For a complex scalar field $\Psi \in \mathbb{C}$,

$$(\partial\partial^* + m^2)\Psi = 0 \tag{397}$$

is the free Klein–Gordon equation [4, 5]. Introduce also

$$\Omega := \langle\langle m + \boldsymbol{\sigma} \cdot \boldsymbol{\nabla} \rangle\rangle = \sqrt{m^2 - \boldsymbol{\nabla}^2}, \quad (\partial_t \pm i\Omega)\Psi(\pm) = 0, \quad \Psi = \Psi(+) + \Psi(-), \tag{398}$$

which is the positive/negative-frequency factorization of (397). This factorization is useful both conceptually and technically. It separates opposite-frequency sectors of the scalar theory, and it also prepares the reader for the later factorizations in which second-order relativistic equations are rewritten as first-order systems.

Minimal coupling and operator factorization

Minimal coupling is introduced here directly at the operator level. Rather than modifying each component equation separately, one defines energy and momentum operators that already incorporate the potential Φ established in Section V.

Here $\Psi(\pm) \in \mathbb{C}$, while Ω is a scalar differential operator built from $m \in \mathbb{R}$ and ∇ . Define also the scalar differential operator E and vector differential operator $\mathbf{P}$ by

$$E := i\partial_t - qV, \quad \mathbf{P} := -i\nabla - q\mathbf{A}. \quad (399)$$

Then

$$(-E^2 + \mathbf{P}^2 + m^2)\Psi = 0 \quad (400)$$

expands to the minimal-coupling Klein–Gordon operator

$$\left(\square + iqS - q^2\langle\langle\Phi\rangle\rangle^2 + m^2 + 2iq(V\partial_t + \mathbf{A} \cdot \nabla)\right)\Psi = 0. \quad (401)$$

Using the commutators above,

$$[E, P_m] = iqE_m, \quad [P_m, P_n] = iq(\partial_m A_n - \partial_n A_m), \quad (402)$$

one gets the useful factorization

$$(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E - \boldsymbol{\sigma} \cdot \mathbf{P}) = E^2 - \mathbf{P}^2 - iqF. \quad (403)$$

Equation (403) is the bridge between the Klein–Gordon and Dirac-type formulas below. The word “bridge” is literal here. The square of a first-order operator built from E and $\mathbf{P}$ produces the second-order relativistic operator together with the field term, which is exactly the structural fact needed before introducing a Dirac-type equation.

Dirac block form, gauge covariance, and Pauli limit

A useful 2×2 block-operator map is

$$W(Z) := \{0, Z; Z^{*\mathbf{R}}, 0\}, \quad W(Z)^2 = \text{diag}(ZZ^{*\mathbf{R}}, Z^{*\mathbf{R}}Z). \quad (404)$$

Here $Z \in \mathbb{C} \oplus \mathbb{C}^3$, and $W(Z)$ lives in the auxiliary block-matrix representation built from paravector entries. The free and coupled Dirac operators act on a two-component block field $\Psi = \{\Psi_1; \Psi_2\}$ in that representation:

$$D_0 := W(i\partial) - m\mathbf{I}, \quad D_0\Psi = 0, \quad (405)$$

$$D := W(i\partial - q\Phi^*) - m\mathbf{I}, \quad D\Psi = 0. \quad (406)$$

Multiplying (406) by $D' := W(i\partial - q\Phi^*) + m\mathbf{I}$ yields

$$D'D\Psi = 0 \Rightarrow \begin{cases} (-E^2 + \mathbf{P}^2 + m^2 - iqF^*)\Psi_1 = 0, \\ (-E^2 + \mathbf{P}^2 + m^2 + iqF)\Psi_2 = 0. \end{cases} \quad (407)$$

The calculation in Eq. (407) explains why the block construction is more than formal packaging. Squaring the first-order system returns Klein–Gordon-type equations with the expected field couplings, so the Dirac form is seen as a controlled factorization of the relativistic wave operator. With the present ordering of $Z = i\partial - q\Phi^* = E - \boldsymbol{\sigma} \cdot \mathbf{P}$ inside $W(Z)$, the upper component carries $-iqF^*$ and the lower component carries $+iqF$. Appendix A11 records a symbolic Python verification of this sign structure.

Gauge covariance combines the electromagnetic gauge law (320) with the block-field phase transformation

$$\Psi = e^{iq\lambda}\Psi', \quad (408)$$

which leaves the form of (406) unchanged.

For the low-energy reduction, using the operators from Eq. (399), one obtains the leading-order Pauli equation for the same block field Ψ ,

$$2mE\Psi = (\mathbf{P}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma})\Psi. \quad (409)$$

Solving Eq. (409) for $i\partial_t\Psi$ gives the more familiar Schrödinger-type form

$$i\partial_t\Psi \approx \left(\frac{(-i\nabla - q\mathbf{A})^2}{2m} + qV\right)\Psi - \frac{q}{2m}(\mathbf{B} \cdot \boldsymbol{\sigma})\Psi. \quad (410)$$

This matches the usual Zeeman term [6, 14], provided one remembers that (410) is a leading-order $1/m$ approximation and that later specialization to uniform static $\mathbf{B}$ is used when decoupling spin-up and spin-down sectors. The nonrelativistic theory therefore appears as an approximation internal to the relativistic one, not as an unrelated starting point. The Pauli equation is the leading

low-energy shadow of the Dirac block system once the rest-energy scale has been separated and higher-order corrections are neglected.

If $\mathbf{B} = \|\mathbf{B}\|\mathbf{u}$ with $\mathbf{B}, \mathbf{u} \in \mathbb{R}^3$, $\|\mathbf{u}\| = 1$, and $\mathbf{B}$ uniform and constant, decompose

$$\Psi = \Psi(+) + \Psi(-), \quad \Psi(\pm) = \Pi(\pm)\Psi, \quad \Pi(\pm) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}). \quad (411)$$

Then Eq. (410) separates into

$$i\partial_t \Psi(\pm) = \left(\frac{\mathbf{P}^2 \mp q\|\mathbf{B}\|}{2m} + qV \right) \Psi(\pm). \quad (412)$$

In this special case the projector language acquires an immediate physical meaning: the field direction selects a preferred axis, and the two projector sectors evolve with opposite Zeeman shifts.

Wave-Operator Expansions and Low-Energy Limits

The following material expands the scalar and spinor wave equations into their intermediate operator identities, explicit component forms, Fourier interpretations, low-energy limits, and spin-sector decompositions. It is placed here so that every use of minimal coupling, factorization, and projector splitting is developed before the manuscript turns to representation theory.

Relativistic Quantum-Mechanical Setting

The formulas below make relativistic quantum-mechanical setting explicit and collect identities used later in the manuscript. The following numbered identities fix the notation and formulas used in this block.

$$\partial = \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}, \quad (413)$$

$$\partial^* = \partial_t - \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}, \quad (414)$$

$$(i\partial)^* = -i\partial^*. \quad (415)$$

Recall the previously established definitions.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \square &:= \partial_t^2 - \partial_{\mathbf{r}}^2, \\ &= \partial\partial^* = \partial^*\partial, \\ &= \langle\langle\partial\rangle\rangle^2. \end{aligned} \quad (416)$$

The final numbered line is the derived relation used in the subsequent discussion.

Note. all differential operators act left to right.

Use the following bra-ket notation.

Assume the state is normalized.

$$\langle\psi|\psi\rangle = \exp(\psi^R\psi) = 1. \quad (417)$$

The expectation value is.

$$\langle\psi|M|\psi\rangle = \exp(\psi^R M\psi). \quad (418)$$

Note. In this block representation, R plays the role of Hermitian conjugation.

Spin-0 Wave Mechanics

The formulas below make spin-0 wave mechanics explicit and collect identities used later in the manuscript. See Eq. (397). Here $\Psi \in \mathbb{C}$ is a complex scalar field, $m \in \mathbb{R}$, and ∂ is the paravector differential operator. The following numbered relation is recorded for later use.

$$\langle\langle mU \rangle\rangle^2 = m^2 \Rightarrow -E^2 + \mathbf{P}^2 + m^2 = 0. \quad (419)$$

For a free particle.

$$E = i\partial_t, \quad (420)$$

$$\mathbf{P} = -i\partial_{\mathbf{r}}. \quad (421)$$

Note. Equivalently, $E + \mathbf{P} \cdot \boldsymbol{\sigma} = i\partial^*$.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} (-E^2 + \mathbf{P}^2 + m^2)\Psi, \\ &= -(i\partial_t)^2 - \partial_{\mathbf{r}}^2 + m^2)\Psi, \\ &= (\partial_t^2 - \partial_{\mathbf{r}}^2 + m^2)\Psi, \\ &= (\partial\partial^* + m^2)\Psi, \\ &= (\square + m^2)\Psi. \end{aligned} \quad (422)$$

The final numbered line is the derived relation used in the subsequent discussion.

The following numbered relation is recorded for later use.

$$(\square + m^2)\Psi = 0 \quad \text{Klein-Gordon equation.} \quad (423)$$

Note. Here Ψ is a complex scalar field.

Positive and Negative Frequency Sectors

The frequency decomposition is one of the central interpretive steps in the scalar wave theory. The following formulas write out the split in a way that makes later Fourier and root-operator remarks easier to follow. See Eq. (398). Here $\Psi, \Psi(\pm), c(\pm) \in \mathbb{C}$, while the Fourier parameters $\omega_n \in \mathbb{R}$ and $\mathbf{k}_n \in \mathbb{R}^3$ appear when the root operator is analyzed mode by mode. The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \Omega &= \sqrt{m^2 - \partial_{\mathbf{r}}^2}, \\ &= \langle\langle m + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}} \rangle\rangle. \end{aligned} \quad (424)$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \square + m^2 &= \partial_t^2 + \Omega^2, \\ &= \partial_t^2 + (m^2 - \partial_{\mathbf{r}}^2), \\ &= (\partial_t - i\Omega)(\partial_t + i\Omega), \\ &= (\partial_t + i\Omega)(\partial_t - i\Omega). \end{aligned} \quad (425)$$

The final numbered line is the derived relation used in the subsequent discussion.

The following numbered identities fix the notation and formulas used in this block.

$$(\partial_t \pm i\Omega)\Psi(\pm) = 0, \quad (426)$$

$$\Psi(\pm) = c(\pm) \exp(\mp i\Omega t), \quad (427)$$

$$\Psi = \Psi(+) + \Psi(-). \quad (428)$$

Root Operators and Fourier Interpretation

Square-root operators can look formal if they are introduced without analytic context. This subsection explains them by switching to Fourier space, where their action reduces to multiplication by an explicit dispersion function. In the Fourier discussion, $\Psi \in \mathbb{C}$, the coefficients $c_n \in \mathbb{C}$, the frequencies $\omega_n \in \mathbb{R}$, and the wave vectors $\mathbf{k}_n \in \mathbb{R}^3$. The following numbered identities fix the notation and formulas used in this block.

$$\Psi = \sum_n c_n f_n, \quad (429)$$

$$f_n = \exp(i(\omega_n t + k_n \cdot \mathbf{r})), \quad (430)$$

$$\square \Psi = \sum_n (k_n^2 - \omega_n^2) c_n f_n, \quad (431)$$

$$\pm \sqrt{\square} \Psi = \sum_n \pm \sqrt{k_n^2 - \omega_n^2} c_n f_n, \quad (432)$$

$$(\square + m^2) \Psi = \sum_n (k_n^2 + m^2 - \omega_n^2) c_n f_n, \quad (433)$$

$$k_n^2 + m^2 = \omega_n^2, \quad (434)$$

$$\square + m^2 = (m \mp i\sqrt{\square})(m \pm i\sqrt{\square}), \quad (435)$$

$$\omega_n = \pm \sqrt{k_n^2 + m^2}, \quad (436)$$

$$\Omega^2 \Psi = \sum_n (m^2 + k_n^2) c_n f_n, \quad (437)$$

$$\Omega \Psi = \sum_n \sqrt{m^2 + k_n^2} c_n f_n. \quad (438)$$

Spin-1/2 Wave Mechanics

The formulas below make spin-1/2 wave mechanics explicit and collect identities used later in the manuscript. See Eqs. (404) and (405). Here $\Psi = \{\Psi_1; \Psi_2\}$ is a two-component block field, $Z \in \mathbb{C} \oplus \mathbb{C}^3$, and $W(Z)$ is a 2×2 block matrix with paravector entries. *Introduce the two-component block field.*

$$\Psi = \{\Psi_1; \Psi_2\}. \quad (439)$$

Define the corresponding 2×2 block operator.

$$W(Z) := \{0, Z; Z^{*R}, 0\} \quad 2 \times 2 \text{ block matrix}, \quad (440)$$

$$W(Z)^2 = \text{diag}(ZZ^{*R}, Z^{*R}Z). \quad (441)$$

If $a, b \in \mathbb{R}$, then

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} W((a + bi)Z) &= (a + bi)W(Z), \\ W(i\partial) &= i0, \partial; \partial^*, 0, \\ &= iW(\partial). \end{aligned} \quad (442)$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} W(i\partial)^2 &= -\text{diag}(\partial\partial^*, \partial^*\partial), \\ &= -\square \mathbf{I}. \end{aligned} \quad (443)$$

The final numbered line is the derived relation used in the subsequent discussion.

Define the Dirac operator matrix.

$$D = W(i\partial) - m\mathbf{I}. \quad (444)$$

Introduce also the mass-conjugate operator.

$$D' = W(i\partial) + m\mathbf{I}. \quad (445)$$

With these definitions fixed, one obtains.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
D'D &= (W + m\mathbf{I})(W - m\mathbf{I}), \\
&= (W^2 + mW - mW - m^2\mathbf{I}), \\
&= -(\square + m^2)\mathbf{I}.
\end{aligned} \tag{446}$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
D\Psi &= 0 \quad \text{Dirac equation,} \\
0 &= \det(D - s\mathbf{I}), \\
&= \det(-(s + m), i\partial; i\partial^*, -(s + m)), \\
&= (s + m)^2 + \square \Rightarrow s = -m \pm i\sqrt{\square}.
\end{aligned} \tag{447}$$

The final numbered line is the derived relation used in the subsequent discussion.

This produces two coupled equations.

$$i\partial\Psi_2 - m\Psi_1 = 0, \tag{448}$$

$$i\partial^*\Psi_1 - m\Psi_2 = 0. \tag{449}$$

Solving the first equation for Ψ_1 gives the following relation.

$$\Psi_1 = i\partial\Psi_2/m. \tag{450}$$

Substituting this into the second equation yields.

$$-\partial^*\partial\Psi_2/m - m\Psi_2 = 0, \tag{451}$$

$$\Rightarrow -(\square + m^2)\Psi_2 = 0. \tag{452}$$

Using a scalar solution Ψ' of the Klein–Gordon equation, one may construct a Dirac solution by
With constant complex scalars a and b .

$$\Psi_1 = (ma + ib\partial)\Psi'/m, \tag{453}$$

$$\Psi_2 = (ia\partial^* + mb)\Psi'/m. \tag{454}$$

The resulting relations are.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
i\partial\Psi_2 - m\Psi_1, \\
&= (-\partial a\partial^* + im\partial b - m^2a - imb\partial)\Psi'/m, \\
&= (-a/m)(\square + m^2)\Psi' = 0.
\end{aligned} \tag{455}$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
i\partial^*\Psi_1 - m\Psi_2, \\
&= (mi\partial^*a - \partial^*b\partial - mia\partial^* - m^2b)\Psi'/m, \\
&= (-b/m)(\square + m^2)\Psi' = 0.
\end{aligned} \tag{456}$$

The final numbered line is the derived relation used in the subsequent discussion.

With $a = 1$ and $b = 0$, the formulas reduce to

$$\Psi_1 = \Psi' \quad \text{scalar}, \quad (457)$$

$$\Psi_2 = (i/m)\partial^* \Psi'. \quad (458)$$

Klein-Gordon Equation with Potentials

This subsection expands the minimally coupled Klein–Gordon equation in the presence of electromagnetic potentials. The point is to show explicitly how the compact operator form reproduces the familiar scalar equation with gauge-coupled derivatives and field-dependent commutators. See Eqs. (400) and (401). Here $\Psi \in \mathbb{C}$, $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3$, $F \in \{0\} \oplus \mathbb{C}^3 \subset \mathbb{C} \oplus \mathbb{C}^3$, $V, S, q, m \in \mathbb{R}$, and $\mathbf{A}, \mathbf{E}, \mathbf{B} \in \mathbb{R}^3$. The following numbered relation is recorded for later use.

$$\mathcal{L} = -m/\gamma - q(V - \mathbf{v} \cdot \mathbf{A}). \quad (459)$$

Note. E = energy (not electric field).

The Hamiltonian is.

$$i\partial_t = \mathcal{H} = E + qV. \quad (460)$$

The canonical momentum is.

$$-i\partial_{\mathbf{r}} = \partial_{\mathbf{v}} \mathcal{L} = \mathbf{P} + q\mathbf{A}, \quad (461)$$

$$E = i\partial_t - qV \quad \text{scalar operator}, \quad (462)$$

$$\mathbf{P} = -i\partial_{\mathbf{r}} - q\mathbf{A} \quad \text{vector operator}. \quad (463)$$

Note. The following supporting identities are used in the surrounding derivation.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \partial_t V \Psi &= (\partial_t V) \Psi + V \partial_t \Psi, \\ \partial_m A_n \Psi &= (\partial_m A_n) \Psi + A_n \partial_m \Psi, \\ E^2 \Psi &= E(i\partial_t \Psi - qV \Psi), \\ &= -\partial_t^2 \Psi - iq\partial_t(V \Psi) - iqV \partial_t \Psi + q^2 V^2 \Psi, \\ &= (-\partial_t^2 - iq(\partial_t V) - 2iqV \partial_t + q^2 V^2) \Psi. \end{aligned} \quad (464)$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \mathbf{P}^2 \Psi &= \mathbf{P} \cdot (-i\partial_{\mathbf{r}} \Psi - q\mathbf{A} \Psi), \\ &= -\partial_{\mathbf{r}}^2 \Psi + iq\partial_{\mathbf{r}} \cdot (\mathbf{A} \Psi) + iq\mathbf{A} \cdot \partial_{\mathbf{r}} \Psi + q^2 \mathbf{A}^2 \Psi, \\ &= (-\partial_{\mathbf{r}}^2 + iq(\partial_{\mathbf{r}} \cdot \mathbf{A}) + 2iq\mathbf{A} \cdot \partial_{\mathbf{r}} + q^2 \mathbf{A}^2) \Psi. \end{aligned} \quad (465)$$

The final numbered line is the derived relation used in the subsequent discussion.

Recall the previously established definitions.

$$\square = \partial_t^2 - \partial_{\mathbf{r}}^2, \quad (466)$$

$$\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}, \quad (467)$$

$$S = \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}, \quad (468)$$

$$\langle\langle \partial \rangle\rangle^2 = \square \quad \text{scalar operator}, \quad (469)$$

$$\langle\langle \Phi \rangle\rangle^2 = V^2 - \mathbf{A}^2 \quad \text{scalar}. \quad (470)$$

Klein-Gordon equation.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
0 &= (-E^2 + \mathbf{P}^2 + m^2)\Psi, \\
&= (\partial_t^2 - \boldsymbol{\partial}_r^2, \\
&\quad + iq(\partial_t V + \boldsymbol{\partial}_r \cdot \mathbf{A}), \\
&\quad + 2iq(V\partial_t + \mathbf{A} \cdot \boldsymbol{\partial}_r), \\
&\quad - q^2(V^2 - \mathbf{A}^2) + m^2)\Psi, \\
&= (\square + iqS - q^2\langle\langle\Phi\rangle\rangle^2 + m^2, \\
&\quad + 2iq(V\partial_t + \mathbf{A} \cdot \boldsymbol{\partial}_r))\Psi.
\end{aligned} \tag{471}$$

The final numbered line is the derived relation used in the subsequent discussion.

Recall the previously established definitions.

$$\mathbf{E} = -\partial_t \mathbf{A} - \boldsymbol{\partial}_r V, \tag{472}$$

$$\mathbf{B} = \boldsymbol{\partial}_r \times \mathbf{A}, \tag{473}$$

$$F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}. \tag{474}$$

Commutators.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
&[E, P_m]\Psi, \\
&= (i\partial_t - qV)(-i\partial_m \Psi - qA_m \Psi), \\
&\quad - (-i\partial_m - qA_m)(i\partial_t \Psi - qV \Psi), \\
&= -iq((\partial_t A_m)\Psi + A_m \partial_t \Psi), \\
&\quad - iq((\partial_m V)\Psi + V \partial_m \Psi), \\
&\quad + iqV \partial_m \Psi + iqA_m \partial_t \Psi, \\
&= -iq(\partial_t A_m + \partial_m V)\Psi, \\
&= iq\mathbf{E}_m \Rightarrow [E, \mathbf{P}] = iq\mathbf{E}.
\end{aligned} \tag{475}$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
&[P_m, P_n]\Psi, \\
&= (-i\partial_m - qA_m)(-i\partial_n \Psi - qA_n \Psi), \\
&\quad - (-i\partial_n - qA_n)(-i\partial_m \Psi - qA_m \Psi), \\
&= iq(\partial_m A_n - \partial_n A_m)\Psi.
\end{aligned} \tag{476}$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
\mathbf{P} \times \mathbf{P} &= ([P_2, P_3], -[P_1, P_3], [P_1, P_2]), \\
&= iq(\boldsymbol{\partial}_r \times \mathbf{A}), \\
&= iq\mathbf{B}.
\end{aligned} \tag{477}$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E - \boldsymbol{\sigma} \cdot \mathbf{P}),$$

$$\begin{aligned}
&= E^2 - \mathbf{P}^2 + \boldsymbol{\sigma} \cdot (-[E, \mathbf{P}] - i\mathbf{P} \times \mathbf{P}), \\
&= E^2 - \mathbf{P}^2 - iqF.
\end{aligned} \tag{478}$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
&(E - \boldsymbol{\sigma} \cdot \mathbf{P})(E + \boldsymbol{\sigma} \cdot \mathbf{P}), \\
&= E^2 - \mathbf{P}^2 + \boldsymbol{\sigma} \cdot ([E, \mathbf{P}] - i\mathbf{P} \times \mathbf{P}), \\
&= E^2 - \mathbf{P}^2 - iqF^*.
\end{aligned} \tag{479}$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
&E + \boldsymbol{\sigma} \cdot \mathbf{P} = i\partial^* - q\Phi, \\
&E - \boldsymbol{\sigma} \cdot \mathbf{P} = i\partial - q\Phi^* \quad (E + \boldsymbol{\sigma} \cdot \mathbf{P})^{*\mathbf{R}}, \\
&\langle\langle E + \boldsymbol{\sigma} \cdot \mathbf{P} \rangle\rangle^2, \\
&= E^2 - \mathbf{P}^2 + \boldsymbol{\sigma} \cdot (\pm[E, \mathbf{P}] - i\mathbf{P} \times \mathbf{P}).
\end{aligned} \tag{480}$$

The final numbered line is the derived relation used in the subsequent discussion.
Note. operators are not unique scalars.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
&-(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E + \boldsymbol{\sigma} \cdot \mathbf{P})^{*\mathbf{R}}, \\
&= (\partial^* + iq\Phi)(\partial + iq\Phi^*), \\
&-(E + \boldsymbol{\sigma} \cdot \mathbf{P})^{*\mathbf{R}}(E + \boldsymbol{\sigma} \cdot \mathbf{P}), \\
&= (\partial + iq\Phi^*)(\partial^* + iq\Phi).
\end{aligned} \tag{481}$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
&(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E - \boldsymbol{\sigma} \cdot \mathbf{P})\Psi, \\
&= -(\partial^* + iq\Phi)(\partial + iq\Phi^*)\Psi, \\
&= -\square\Psi - iq\partial^*\Phi^*\Psi - iq\Phi\partial\Psi + q^2\langle\langle\Phi\rangle\rangle^2\Psi, \\
&= (E^2 - \mathbf{P}^2 - iqF)\Psi, \\
&= -\square\Psi - iqS\Psi + q^2\langle\langle\Phi\rangle\rangle^2\Psi, \\
&\quad - 2iq(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}})\Psi - iqF\Psi.
\end{aligned} \tag{482}$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
&\partial^*\Phi^*\Psi + \Phi\partial\Psi, \\
&= S\Psi + 2(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}})\Psi + F\Psi.
\end{aligned} \tag{483}$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
&(E - \boldsymbol{\sigma} \cdot \mathbf{P})(E + \boldsymbol{\sigma} \cdot \mathbf{P})\Psi, \\
&= -(\partial + iq\Phi^*)(\partial^* + iq\Phi)\Psi,
\end{aligned}$$

$$\begin{aligned}
&= -\square\Psi - iq\partial\Phi\Psi - iq\Phi^*\partial^*\Psi + q^2\langle\langle\Phi\rangle\rangle^2\Psi, \\
&= (E^2 - \mathbf{P}^2 - iqF^*)\Psi, \\
&= -\square\Psi - iqS\Psi + q^2\langle\langle\Phi\rangle\rangle^2\Psi, \\
&\quad - 2iq(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}})\Psi - iqF^*\Psi.
\end{aligned} \tag{484}$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
&\partial\Phi\Psi + \Phi^*\partial^*\Psi, \\
&= S\Psi + 2(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}})\Psi + F^*\Psi.
\end{aligned} \tag{485}$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
&(\partial^*\Phi^*\Psi + \Phi\partial\Psi) - (\partial\Phi\Psi + \Phi^*\partial^*\Psi), \\
&= [\Phi, \partial]\Psi + [\partial, \Phi]^*\Psi, \\
&= (F - F^*)\Psi, \\
&= 2(\mathbf{E} \cdot \boldsymbol{\sigma})\Psi.
\end{aligned} \tag{486}$$

The final numbered line is the derived relation used in the subsequent discussion.
Collecting the definitions just obtained, one has.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
E &:= i\partial_t - qV && \text{scalar operator,} \\
\mathbf{P} &:= -i\partial_{\mathbf{r}} - q\mathbf{A} && \text{vector operator,} \\
\partial &:= \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}, \\
\partial^* &:= \partial_t - \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}, \\
\Phi &:= V + \mathbf{A} \cdot \boldsymbol{\sigma}, \\
F &:= (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\
S &:= \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}, \\
\mathbf{E} &:= -\partial_t \mathbf{A} - \partial_{\mathbf{r}} V, \\
\mathbf{B} &:= \partial_{\mathbf{r}} \times \mathbf{A}, \\
& (E \pm \boldsymbol{\sigma} \cdot \mathbf{P})(E \mp \boldsymbol{\sigma} \cdot \mathbf{P}), \\
&= E^2 - (\boldsymbol{\sigma} \cdot \mathbf{P})^2 \mp [E, \boldsymbol{\sigma} \cdot \mathbf{P}], \\
&= E^2 - \mathbf{P}^2 - iq(\pm\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}.
\end{aligned} \tag{487}$$

The final numbered line is the derived relation used in the subsequent discussion.
Klein-Gordon equation.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
0 &= (-E^2 + \mathbf{P}^2 + m^2)\Psi, \\
&= (\square + iqS - q^2\langle\langle\Phi\rangle\rangle^2 + m^2, \\
&\quad + 2iq(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}}))\Psi, \\
&= (\partial^* + iq\Phi)(\partial + iq\Phi^*)\Psi - iqF\Psi + m^2\Psi, \\
&= (\partial + iq\Phi^*)(\partial^* + iq\Phi)\Psi - iqF^*\Psi + m^2\Psi, \\
&= \text{scalar}((\partial + iq\Phi^*)(\partial^* + iq\Phi))\Psi + m^2\Psi.
\end{aligned} \tag{488}$$

The final numbered line is the derived relation used in the subsequent discussion.

Dirac Equation with Potentials

The formulas below make dirac equation with potentials explicit and collect identities used later in the manuscript. See Eq. (406). Here $\Psi = \{\Psi_1; \Psi_2\}$ is a two-component block field, $\Phi \in \mathbb{R} \oplus \mathbb{R}^3$, and D is a block operator built from paravector entries. The following numbered relation is recorded for later use.

$$\Psi = \{\Psi_1; \Psi_2\}. \quad (489)$$

Define the following object.

$$W(Z) := \{0, Z; Z^{*\mathbf{R}}, 0\}, \quad (490)$$

$$W(Z)^2 = \{ZZ^{*\mathbf{R}}, 0; 0, Z^{*\mathbf{R}}Z\}, \quad (491)$$

$$W(Z)^* = W(Z^*), \quad (492)$$

$$W(Z)^{\mathbf{R}} = W(Z^{\mathbf{R}}). \quad (493)$$

Introduce the Dirac operator in the presence of a potential field.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} D &= W(i\partial - q\Phi^*) - m\mathbf{I}, \\ &= iW(\partial) - qW(\Phi^*) - m\mathbf{I}. \end{aligned} \quad (494)$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} W(\partial)^2 &= \square\mathbf{I}, \\ W(\partial)W(\Phi^*), &= \{0, \partial; \partial^*, 0\}0, \Phi^*; \Phi, 0, \\ &= \text{diag}(\partial\Phi, \partial^*\Phi^*). \end{aligned} \quad (495)$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} W(\Phi^*)W(\partial), &= \{0, \Phi^*; \Phi, 0\}0, \partial; \partial^*, 0, \\ &= \text{diag}(\Phi^*\partial^*, \Phi\partial). \end{aligned} \quad (496)$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} W(\partial)W(\Phi^*) + W(\Phi^*)W(\partial), &= \text{diag}(\partial\Phi + \Phi^*\partial^*, \partial^*\Phi^* + \Phi\partial). \end{aligned} \quad (497)$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} W(\Phi^*)^2, &= \{0, \Phi^*; \Phi, 0\}0, \Phi^*; \Phi, 0, \\ &= \text{diag}(\Phi^*\Phi, \Phi\Phi^*), \\ &= \langle\langle\Phi\rangle\rangle^2\mathbf{I}. \end{aligned} \quad (498)$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
W(i\partial - q\Phi^*)^2, \\
&= -W(\partial)^2, \\
&\quad -iq(W(\partial)W(\Phi^*) + W(\Phi^*)W(\partial)), \\
&\quad +q^2W(\Phi^*)^2, \\
&= -\square\mathbf{I}, \\
&\quad -iq\text{diag}(\partial\Phi + \Phi^*\partial^*, \partial^*\Phi^* + \Phi\partial), \\
&\quad +q^2\langle\langle\Phi\rangle\rangle^2\mathbf{I}, \\
&= -\text{diag}(.
\end{aligned} \tag{499}$$

The final numbered line is the derived relation used in the subsequent discussion.

The following numbered identities fix the notation and formulas used in this block.

$$(\partial + iq\Phi^*)(\partial^* + iq\Phi), \tag{500}$$

$$(\partial^* + iq\Phi)(\partial + iq\Phi^*), \tag{501}$$

$$D\Psi = 0 \quad \text{Dirac equation.} \tag{502}$$

This produces two coupled equations.

$$(-m)\Psi_1 + (i\partial - q\Phi^*)\Psi_2 = 0, \tag{503}$$

$$(i\partial^* - q\Phi)\Psi_1 + (-m)\Psi_2 = 0, \tag{504}$$

$$\text{solve } \Psi_1 \text{ from first equation,} \tag{505}$$

$$\Psi_1 = (i\partial - q\Phi^*)\Psi_2/m. \tag{506}$$

Substituting this into the second equation yields.

$$-((\partial^* + iq\Phi)(\partial + iq\Phi^*) + m^2)\Psi_2 = 0. \tag{507}$$

Introduce also the mass-conjugate operator.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
D' &= W(i\partial - q\Phi^*) + m\mathbf{I}, \\
0 &= D'D\Psi, \\
&= (W(i\partial - q\Phi^*)^2 - m^2\mathbf{I})\Psi, \\
&= -\text{diag}(.
\end{aligned} \tag{508}$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
&(\partial + iq\Phi^*)(\partial^* + iq\Phi) + m^2, \\
&(\partial^* + iq\Phi)(\partial + iq\Phi^*) + m^2, \\
&)\Psi, \\
0 &= ((\partial + iq\Phi^*)(\partial^* + iq\Phi) + m^2)\Psi_1, \\
&= (-E^2 + \mathbf{P}^2 + m^2 + iqF^*)\Psi_1.
\end{aligned} \tag{509}$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
0 &= ((\partial^* + iq\Phi)(\partial + iq\Phi^*) + m^2)\Psi_2, \\
&= (-E^2 + \mathbf{P}^2 + m^2 + iqF)\Psi_2.
\end{aligned} \tag{510}$$

The final numbered line is the derived relation used in the subsequent discussion.

Gauge Covariance in the Dirac System

The formulas below make gauge covariance in the dirac system explicit and collect identities used later in the manuscript. See Eq. (408). Let λ denote real scalar.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
\Phi' &= \Phi + \partial^* \lambda, \\
\partial \lambda &= (\Phi' - \Phi)^*, \\
\Psi &= \exp(iq\lambda) \Psi', \\
\partial \exp(iq\lambda) \Psi' &= \exp(iq\lambda) (iq(\partial \lambda) + \partial) \Psi'.
\end{aligned} \tag{511}$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
0 &= (W(i\partial - q\Phi^*) - mI) \Psi, \\
&= (W(i\partial - q\Phi^*) - mI) \exp(iq\lambda) \Psi', \\
&= \exp(iq\lambda) (W(i\partial - q(\partial \lambda) - q\Phi^*) - mI) \Psi', \\
&= \exp(iq\lambda) (W(i\partial - q(\Phi' - \Phi)^* - q\Phi^*) - mI) \Psi', \\
&= \exp(iq\lambda) (W(i\partial - q\Phi'^*) - mI) \Psi'.
\end{aligned} \tag{512}$$

The final numbered line is the derived relation used in the subsequent discussion.

The following numbered relation is recorded for later use.

$$(W(i\partial - q\Phi'^*) - mI) \Psi' = 0. \tag{513}$$

Dirac Equation in Components

The block Dirac equation becomes most transparent when its component equations are written out explicitly. Doing so makes it easier to compare the STA formulation with the familiar coupled two-component form. See Eq. (407). Here $\psi = \psi_s + \psi_v \cdot \sigma \in \mathbb{C} \oplus \mathbb{C}^3$, $\Phi \in \mathbb{R} \oplus \mathbb{R}^3$, and $F = F_v \cdot \sigma \in \{0\} \oplus \mathbb{C}^3$ with $F_v \in \mathbb{C}^3$. The following numbered identities fix the notation and formulas used in this block.

$$\partial = \partial_t + \sigma \cdot \partial_r, \tag{514}$$

$$\Phi = V + \mathbf{A} \cdot \sigma \in \mathbb{R} \oplus \mathbb{R}^3, \tag{515}$$

$$\psi = \psi_s + \psi_v \cdot \sigma \in \mathbb{C} \oplus \mathbb{C}^3, \tag{516}$$

$$F = F_v \cdot \sigma, \tag{517}$$

$$F_s = 0, \tag{518}$$

$$F_v = \mathbf{E} + i\mathbf{B} \in \mathbb{C}^3, \tag{519}$$

$$\partial^* = \partial_t - \sigma \cdot \partial_r, \tag{520}$$

$$\Phi^* = V - \mathbf{A} \cdot \sigma, \tag{521}$$

$$F^* = -F_v^* \cdot \sigma. \tag{522}$$

Table 27: Components of $\partial\psi$.

scalar: $\partial_t \psi_s + \partial_r \cdot \psi_v$

vector: $\partial_t \psi_v + \partial_r \psi_s + i(\partial_r \times \psi_v)$

Table 28: Components of $\partial^* \psi$.

scalar: $\partial_t \psi_s - \partial_r \cdot \psi_v$

vector: $\partial_t \psi_v - \partial_r \psi_s - i(\partial_r \times \psi_v)$

Table 29: Components of $\Phi\psi$.

$$\begin{aligned} \text{scalar: } & V\psi_s + \mathbf{A} \cdot \psi_v \\ \text{vector: } & V\psi_v + \psi_s \mathbf{A} + i(\mathbf{A} \times \psi_v) \end{aligned}$$

Table 30: Components of $\Phi^*\psi$.

$$\begin{aligned} \text{scalar: } & V\psi_s - \mathbf{A} \cdot \psi_v \\ \text{vector: } & V\psi_v - \psi_s \mathbf{A} - i(\mathbf{A} \times \psi_v) \end{aligned}$$

Table 31: Components of $iqF\psi$.

$$\begin{aligned} \text{scalar: } & iq(F_v \cdot \psi_v) \\ \text{vector: } & iq\psi_s F_v - q(F_v \times \psi_v) \end{aligned}$$

Table 32: Components of $iqF^*\psi$.

$$\begin{aligned} \text{scalar: } & -iq(F_v^* \cdot \psi_v) \\ \text{vector: } & -iq\psi_s F_v^* + q(F_v^* \times \psi_v) \end{aligned}$$

Low-Energy Spin-0 Limit

The formulas below make low-energy spin-0 limit explicit and collect identities used later in the manuscript. See Eq. (400). Here $\Psi \in \mathbb{C}$, while E and $\mathbf{P}$ remain the scalar and vector differential operators defined in the main text. The following numbered identities fix the notation and formulas used in this block.

$$E = i\partial_t - qV, \quad (523)$$

$$\mathbf{P} = -i\partial_{\mathbf{r}} - q\mathbf{A}. \quad (524)$$

Factor out the rest-mass oscillation.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \Psi' &= \exp(-imt)\Psi, \\ E^2\Psi' &= \exp(-imt)(E + m)^2\Psi, \\ \mathbf{P}^2\Psi' &= \exp(-imt)\mathbf{P}^2\Psi, \\ 0 &= (-E^2 + \mathbf{P}^2 + m^2)\Psi', \\ &= \exp(-imt)(-(E + m)^2 + \mathbf{P}^2 + m^2)\Psi. \end{aligned} \quad (525)$$

The final numbered line is the derived relation used in the subsequent discussion.

The following numbered relation is recorded for later use.

$$0 = (-E^2 - 2mE + \mathbf{P}^2)\Psi. \quad (526)$$

Recursively iterate E .

$$E = \frac{1}{2}(\mathbf{P}^2 - E^2)/m, \quad (527)$$

$$E = \frac{1}{2}\mathbf{P}^2/m - \frac{1}{8}\mathbf{P}^4/m^3 + \dots. \quad (528)$$

At leading order in the low-energy expansion, this reduces to.

$$E\Psi \approx (\frac{1}{2}\mathbf{P}^2/m)\Psi. \quad (529)$$

The resulting Schrödinger equation is.

$$i\partial_t\Psi = (\frac{1}{2}(i\partial_{\mathbf{r}} + q\mathbf{A})^2/m + qV)\Psi. \quad (530)$$

If $\mathbf{A} = 0$, then the following relation holds.

$$i\partial_t\Psi = (-\frac{1}{2}\partial_{\mathbf{r}}^2/m + qV)\Psi. \quad (531)$$

Low-Energy Spin-1/2 Limit

The formulas below make low-energy spin-1/2 limit explicit and collect identities used later in the manuscript. See Eqs. (409) and (410). Here Ψ_1 and Ψ_2 denote the block components of the Dirac field, and the low-energy reduction keeps the same real fields $V \in \mathbb{R}$ and $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$. The following numbered relation is recorded for later use.

$$\Psi' = \{\Psi'_1; \Psi'_2\}. \quad (532)$$

Begin with the Dirac equation in the presence of a potential field.

$$(W(i\partial - q\Phi^*) - m\mathbf{I})\Psi' = 0. \quad (533)$$

This produces two coupled equations.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} (i\partial - q\Phi^*)\Psi'_2 - m\Psi'_1 &= 0, \\ (i\partial^* - q\Phi)\Psi'_1 - m\Psi'_2 &= 0, \\ \partial &= \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}, \\ \Phi &= V + \mathbf{A} \cdot \boldsymbol{\sigma}, \\ (i\partial_t - qV)\Psi'_2 + ((i\partial_{\mathbf{r}} + q\mathbf{A}) \cdot \boldsymbol{\sigma})\Psi'_2, \\ &- m\Psi'_1 = 0. \end{aligned} \quad (534)$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} (i\partial_t - qV)\Psi'_1 + (-(i\partial_{\mathbf{r}} + q\mathbf{A}) \cdot \boldsymbol{\sigma})\Psi'_1, \\ - m\Psi'_2 = 0. \end{aligned} \quad (535)$$

The final numbered line is the derived relation used in the subsequent discussion.

The following numbered identities fix the notation and formulas used in this block.

$$E = i\partial_t - qV, \quad (536)$$

$$\mathbf{P} = -i\partial_{\mathbf{r}} - q\mathbf{A}. \quad (537)$$

This produces two coupled equations.

$$(E - \mathbf{P} \cdot \boldsymbol{\sigma})\Psi'_2 - m\Psi'_1 = 0, \quad (538)$$

$$(E + \mathbf{P} \cdot \boldsymbol{\sigma})\Psi'_1 - m\Psi'_2 = 0. \quad (539)$$

Next factor out the rest-mass oscillation.

$$\Psi'_n = \exp(-imt)\Psi_n, \quad (540)$$

$$E\Psi'_n = \exp(-imt)(E + m)\Psi_n. \quad (541)$$

After removing the rest phase, the system becomes.

$$(E + m - \mathbf{P} \cdot \boldsymbol{\sigma})\Psi_2 - m\Psi_1 = 0, \quad (542)$$

$$(E + m + \mathbf{P} \cdot \boldsymbol{\sigma})\Psi_1 - m\Psi_2 = 0, \quad (543)$$

$$\text{solve } \Psi_1 \text{ from first equation,} \quad (544)$$

$$\Psi_1 = (E + m - \mathbf{P} \cdot \boldsymbol{\sigma})\Psi_2/m. \quad (545)$$

Substituting this into the second equation yields.

After multiplying through by m , one obtains.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
0 &= ((E + m + \mathbf{P} \cdot \boldsymbol{\sigma})(E + m - \mathbf{P} \cdot \boldsymbol{\sigma}) - m^2)\Psi_2, \\
(E + m + \mathbf{P} \cdot \boldsymbol{\sigma})(E + m - \mathbf{P} \cdot \boldsymbol{\sigma}) - m^2, \\
&= (E + m)^2 - m^2 - (\mathbf{P} \cdot \boldsymbol{\sigma})^2 + [\mathbf{P} \cdot \boldsymbol{\sigma}, E], \\
&\quad - (\mathbf{P} \cdot \boldsymbol{\sigma})^2 = -\mathbf{P}^2 - i(\mathbf{P} \times \mathbf{P}) \cdot \boldsymbol{\sigma}.
\end{aligned} \tag{546}$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
[P_m, P_n] &= [-i\partial_m - qA_m, -i\partial_n - qA_n], \\
&= iq(\partial_m A_n - \partial_n A_m).
\end{aligned} \tag{547}$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
\mathbf{P} \times \mathbf{P} &= \{[P_2, P_3], -[P_1, P_3], [P_1, P_2]\}, \\
&= iq\boldsymbol{\partial}_r \times \mathbf{A}, \\
&= iq\mathbf{B}, \\
&\quad -i(\mathbf{P} \times \mathbf{P}) \cdot \boldsymbol{\sigma} = q\mathbf{B} \cdot \boldsymbol{\sigma}.
\end{aligned} \tag{548}$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
[\mathbf{P} \cdot \boldsymbol{\sigma}, E + m] &= [\mathbf{P}, E] \cdot \boldsymbol{\sigma}, \\
&= iq(\boldsymbol{\partial}_r V + \partial_t \mathbf{A}) \cdot \boldsymbol{\sigma}, \\
&= -iq\mathbf{E} \cdot \boldsymbol{\sigma}, \\
&\quad -iqF = q\mathbf{B} \cdot \boldsymbol{\sigma} - iq\mathbf{E} \cdot \boldsymbol{\sigma}.
\end{aligned} \tag{549}$$

The final numbered line is the derived relation used in the subsequent discussion.

Keeping the full operator structure gives the exact equation.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
0 &= ((E + m)^2 - m^2 - \mathbf{P}^2 + q\mathbf{B} \cdot \boldsymbol{\sigma}, \\
&\quad + [\mathbf{P} \cdot \boldsymbol{\sigma}, E])\Psi_2, \\
&= (E^2 + 2mE - \mathbf{P}^2 + q\mathbf{B} \cdot \boldsymbol{\sigma} - iq\mathbf{E} \cdot \boldsymbol{\sigma})\Psi_2, \\
&= (E^2 + 2mE - \mathbf{P}^2 - iqF)\Psi_2.
\end{aligned} \tag{550}$$

The final numbered line is the derived relation used in the subsequent discussion.

Recursively iterate E .

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
E &= (\mathbf{P}^2 - E^2 - q\mathbf{B} \cdot \boldsymbol{\sigma} - [\mathbf{P} \cdot \boldsymbol{\sigma}, E])/(2m), \\
&= \frac{1}{2}(\mathbf{P}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma})/m + O(1/m^2).
\end{aligned} \tag{551}$$

The final numbered line is the derived relation used in the subsequent discussion.

At leading order in the low-energy expansion, this reduces to.

$$0 \approx (2mE - \mathbf{P}^2 + q\mathbf{B} \cdot \boldsymbol{\sigma})\Psi_2. \tag{552}$$

The resulting Pauli equation is.

The intermediate steps are written out explicitly as follows.

$$i\partial_t \Psi_2 = ((-i\partial_{\mathbf{r}} - q\mathbf{A})^2/(2m) + qV)\Psi_2, \\ - q(\mathbf{B} \cdot \boldsymbol{\sigma})/(2m)\Psi_2. \quad (553)$$

The final numbered line is the derived relation used in the subsequent discussion.

Spin via Rotors

The formulas below make spin via rotors explicit and collect identities used later in the manuscript. See Eq. (234). Here $\mathbf{u}, \mathbf{n} \in \mathbb{R}^3$ are unit directions, and R is a rotor algebra element. The following numbered identities fix the notation and formulas used in this block.

$$i = \sigma_1 \sigma_2 \sigma_3, \quad (554)$$

$$i\sigma_k = \sigma_n \sigma_m n \neq m \neq k. \quad (555)$$

Choose unit vectors as follows.

The intermediate steps are written out explicitly as follows.

$$\mathbf{n} = (n_1, n_2, n_3), \mathbf{n}^2 = 1, \\ \mathbf{u} = (u_1, u_2, u_3), \mathbf{u}^2 = 1, \\ R = \exp(\tfrac{1}{2}\varphi(\mathbf{n} \cdot i\boldsymbol{\sigma})) \quad \text{rotor}, \\ = \cos(\tfrac{1}{2}\varphi) + \sin(\tfrac{1}{2}\varphi)(\mathbf{n} \cdot i\boldsymbol{\sigma}). \quad (556)$$

The final numbered line is the derived relation used in the subsequent discussion.

Note. The rotor changes sign under a full 2π rotation: $R(\varphi + 2\pi) = -R(\varphi)$.

$$R^R = \cos(\tfrac{1}{2}\varphi) - \sin(\tfrac{1}{2}\varphi)(\mathbf{n} \cdot i\boldsymbol{\sigma}), \quad (557)$$

$$R^R R = \cos(\tfrac{1}{2}\varphi)^2 + \sin(\tfrac{1}{2}\varphi)^2 = 1, \quad (558)$$

$$\Rightarrow R^{-1} = R^R = R^{*R}, \quad (559)$$

$$R^R(\mathbf{u} \cdot \boldsymbol{\sigma})R = \mathbf{u}' \cdot \boldsymbol{\sigma}. \quad (560)$$

Applying the rotor action gives Rodrigues' formula,.

The intermediate steps are written out explicitly as follows.

$$\mathbf{u}' = \mathbf{u} \cos(\varphi) + (\mathbf{n} \times \mathbf{u}) \sin(\varphi), \\ + (\mathbf{n} \cdot \mathbf{u})(1 - \cos(\varphi))\mathbf{n}. \quad (561)$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\psi = \psi_s + \psi_{\mathbf{v}} \cdot \boldsymbol{\sigma}, \\ \psi_s = \text{scalar}(\psi) \quad \text{complex}, \\ \psi_{\mathbf{v}} = \text{vector}(\psi) \quad \text{complex}, \\ \psi^R = \psi_s^R - \psi_{\mathbf{v}}^R \cdot \boldsymbol{\sigma}, \\ = \psi_s^* - \psi_{\mathbf{v}}^* \cdot \boldsymbol{\sigma}. \quad (562)$$

The final numbered line is the derived relation used in the subsequent discussion.

The following numbered relation is recorded for later use.

$$\psi' = R\psi \quad \text{spinor.} \quad (563)$$

Note. In this block representation, R plays the role of Hermitian conjugation. The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \langle \psi' | \mathbf{u} \cdot \boldsymbol{\sigma} | \psi' \rangle &= \langle R\psi | \mathbf{u} \cdot \boldsymbol{\sigma} | R\psi \rangle, \\ &= \psi^R (R^R (\mathbf{u} \cdot \boldsymbol{\sigma}) R) \psi, \\ &= \psi^R (\mathbf{u}' \cdot \boldsymbol{\sigma}) \psi, \\ &= \langle \psi | \mathbf{u}' \cdot \boldsymbol{\sigma} | \psi \rangle. \end{aligned} \quad (564)$$

The final numbered line is the derived relation used in the subsequent discussion. The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \psi \psi^R &= \text{scalar}(\psi \psi^R) + \text{vector}(\psi \psi^R) \cdot \boldsymbol{\sigma}, \\ \text{scalar}(\psi^R (\mathbf{u}' \cdot \boldsymbol{\sigma}) \psi) &= \text{scalar}((\mathbf{u}' \cdot \boldsymbol{\sigma}) \psi \psi^R) \quad \text{cyclic property,} \\ &= \mathbf{u}' \cdot \text{vector}(\psi \psi^R), \\ &= \mathbf{u} \cdot \text{vector}(\psi' \psi'^R). \end{aligned} \quad (565)$$

The final numbered line is the derived relation used in the subsequent discussion.

Spin Projectors

The formulas below make spin projectors explicit and collect identities used later in the manuscript. See Eq. (15). Here $\mathbf{u}, \mathbf{u}' \in \mathbb{R}^3$ are unit directions and $\Pi(\pm)$ are the corresponding algebraic projectors. *Define the projector by.* The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \Pi(\pm) &= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \\ \Pi(\pm)^2 &= \Pi(\pm), \\ \Pi(\pm)\Pi(\mp) &= 0, \\ \Pi(\pm) + \Pi(\mp) &= 1, \\ (\mathbf{u} \cdot \boldsymbol{\sigma})\Pi(\pm) &= \pm \Pi(\pm) \quad \pm 1 \text{ eigenvalue,} \\ \Pi(\pm)^R &= \Pi(\pm), \\ \Pi(\pm)^* &= \Pi(\mp), \\ \mathbf{u} \cdot \boldsymbol{\sigma} &= (+1)\Pi(+1) + (-1)\Pi(-1), \\ \Pi(\pm)' &= \frac{1}{2}(1 \pm \mathbf{u}' \cdot \boldsymbol{\sigma}), \\ \mathbf{u}' \cdot \boldsymbol{\sigma} &= R^R (\mathbf{u} \cdot \boldsymbol{\sigma}) R, \\ &= (+1)\Pi(+1)' + (-1)\Pi(-1)'. \end{aligned} \quad (566)$$

The final numbered line is the derived relation used in the subsequent discussion. The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \Psi(\pm) &= \Pi(\pm)\Psi, \\ (\mathbf{u} \cdot \boldsymbol{\sigma})\Psi(\pm) &= \pm \Psi(\pm), \\ \Psi &= \Psi(+) + \Psi(-) \quad \text{spin decomposition,} \\ \Psi(\pm)^R &= \Psi^R \Pi(\pm)^R, \\ &= \Psi^R \Pi(\pm). \end{aligned} \quad (567)$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
\Psi(\pm)^R \Psi(\pm) &= \Psi^{*R} \Pi(\pm) \Pi(\pm) \Psi = \Psi^{*R} \Pi(\pm) \Psi, \\
\Psi(\pm)^R \Psi(\mp) &= \Psi^{*R} \Pi(\pm) \Pi(\mp) \Psi = 0, \\
\Psi(\mp)^R \Psi(\pm) &= \Psi^{*R} \Pi(\mp) \Pi(\pm) \Psi = 0, \\
\langle \psi | \mathbf{u}' \cdot \boldsymbol{\sigma} | \psi \rangle &= \langle \Psi_+ + \Psi_- | \mathbf{u}' \cdot \boldsymbol{\sigma} | \Psi_+ + \Psi_- \rangle, \\
&= \Psi(+)^R (\mathbf{u}' \cdot \boldsymbol{\sigma}) \Psi(+) + \Psi(+)^R (\mathbf{u}' \cdot \boldsymbol{\sigma}) \Psi(-), \\
&\quad + \Psi(-)^R (\mathbf{u}' \cdot \boldsymbol{\sigma}) \Psi(+) + \Psi(-)^R (\mathbf{u}' \cdot \boldsymbol{\sigma}) \Psi(-).
\end{aligned} \tag{568}$$

The final numbered line is the derived relation used in the subsequent discussion.
The following numbered identities fix the notation and formulas used in this block.

$$(\mathbf{u}' \cdot \boldsymbol{\sigma})(\mathbf{u} \cdot \boldsymbol{\sigma}) = \mathbf{u}' \cdot \mathbf{u} + i(\mathbf{u}' \times \mathbf{u}) \cdot \boldsymbol{\sigma}, \tag{569}$$

$$(\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma}) = \mathbf{u}' \cdot \mathbf{u} - i(\mathbf{u}' \times \mathbf{u}) \cdot \boldsymbol{\sigma}, \tag{570}$$

$$(\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma})(\mathbf{u} \cdot \boldsymbol{\sigma}) = (2(\mathbf{u}' \cdot \mathbf{u})\mathbf{u} - \mathbf{u}') \cdot \boldsymbol{\sigma}, \tag{571}$$

$$\text{par}(\mathbf{u}') := (\mathbf{u} \cdot \mathbf{u}')\mathbf{u}, \tag{572}$$

$$\text{prp}(\mathbf{u}') := \mathbf{u}' - \text{par}(\mathbf{u}'). \tag{573}$$

Same-side projector.

$$\Pi(\pm)(\mathbf{u}' \cdot \boldsymbol{\sigma})\Pi(\pm) = \pm(\mathbf{u} \cdot \mathbf{u}')\Pi(\pm). \tag{574}$$

Cross-side projector.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
&\Pi(\pm)(\mathbf{u}' \cdot \boldsymbol{\sigma})\Pi(\mp), \\
&= \frac{1}{2}([\text{prp}(\mathbf{u}') \pm i(\mathbf{u} \times \mathbf{u}')] \cdot \boldsymbol{\sigma})\Pi(\mp).
\end{aligned} \tag{575}$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
\mathbf{u}' \cdot \mathbf{u} &=: \cos(\theta), \\
&= (\mathbf{u} \cdot \mathbf{u}) \cos(\varphi) + (\mathbf{u} \cdot (\mathbf{n} \times \mathbf{u})) \sin(\varphi), \\
&\quad + (\mathbf{n} \cdot \mathbf{u})(1 - \cos(\varphi))(\mathbf{n} \cdot \mathbf{u}), \\
&= (\mathbf{n} \cdot \mathbf{u})^2 + (1 - (\mathbf{n} \cdot \mathbf{u})^2) \cos(\varphi), \\
&= \cos(\varphi) \text{ if } \mathbf{n} \cdot \mathbf{u} = 0.
\end{aligned} \tag{576}$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
\frac{1}{2}(1 + \mathbf{u}' \cdot \mathbf{u}) &= \frac{1}{2}(1 + \cos(\theta)), \\
&= \cos^2(\theta/2), \\
&= \cos^2(\varphi/2) \text{ if } \mathbf{n} \cdot \mathbf{u} = 0.
\end{aligned} \tag{577}$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
\frac{1}{2}(1 - \mathbf{u}' \cdot \mathbf{u}) &= \frac{1}{2}(1 - \cos(\theta)), \\
&= \sin^2(\theta/2), \\
&= \sin^2(\varphi/2) \text{ if } \mathbf{n} \cdot \mathbf{u} = 0.
\end{aligned} \tag{578}$$

The final numbered line is the derived relation used in the subsequent discussion.

Spin Uncertainty and Commutator Bounds

The commutator identities relevant to the uncertainty discussion are collected here in the same notation used elsewhere in the manuscript. The displayed inequality itself must still be read together with the red caution note that follows. For $\mathbf{u}, \mathbf{u}' \in \mathbb{R}^3$, the following formulas apply.

$$\mathbf{u} \cdot \mathbf{u} = \mathbf{u}' \cdot \mathbf{u}' = 1, \quad (579)$$

$$(\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma}) = \mathbf{u} \cdot \mathbf{u}' + i(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \quad (580)$$

$$[\mathbf{u} \cdot \boldsymbol{\sigma}, \mathbf{u}' \cdot \boldsymbol{\sigma}] = 2i(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \quad (581)$$

$$(\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma}) = \mathbf{u} \cdot \mathbf{u}' + \frac{1}{2}[\mathbf{u} \cdot \boldsymbol{\sigma}, \mathbf{u}' \cdot \boldsymbol{\sigma}]. \quad (582)$$

Using the paravector metric $\langle\langle Z \rangle\rangle^2 = ZZ^{*\mathbf{R}}$, one finds

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \langle\langle (\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma}) \rangle\rangle^2 &= (\mathbf{u} \cdot \mathbf{u}')^2 - (i(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma})^2, \\ &= (\mathbf{u} \cdot \mathbf{u}')^2 + (\mathbf{u} \times \mathbf{u}')^2. \end{aligned} \quad (583)$$

The final numbered line is the derived relation used in the subsequent discussion.

Hence the lower bound is governed by $(\mathbf{u} \times \mathbf{u}')^2$.

$$= \|\frac{1}{2}[\mathbf{u} \cdot \boldsymbol{\sigma}, \mathbf{u}' \cdot \boldsymbol{\sigma}]\|^2. \quad (584)$$

Robertson inequality.

This is not the standard Robertson or Robertson–Schrödinger uncertainty relation; the standard result is a state-variance inequality rather than the displayed paravector comparison.

$$\langle\langle (d\mathbf{u} \cdot \boldsymbol{\sigma})(d\mathbf{u}' \cdot \boldsymbol{\sigma}) \rangle\rangle \geq \frac{1}{2}\| [d\mathbf{u} \cdot \boldsymbol{\sigma}, d\mathbf{u}' \cdot \boldsymbol{\sigma}] \|. \quad (585)$$

This displayed inequality should not be presented as the standard Robertson relation.

Raising and Lowering About the z Axis

The formulas below make raising and lowering about the z axis explicit and collect identities used later in the manuscript. Here $\mathbf{u} = e_3 \in \mathbb{R}^3$, and $a(\pm)$ are algebra elements built from σ_1 and σ_2 . (*ladder operators*).

$$\mathbf{u} = e_3 = \{0, 0, 1\}, \quad (586)$$

$$\Pi(\pm) = \frac{1}{2}(1 \pm \sigma_3), \quad (587)$$

$$a(\pm) := \frac{1}{2}(\sigma_1 \pm i\sigma_2) \quad \text{ladder operator.} \quad (588)$$

Commutator.

$$[\sigma_3, a(\pm)] = \pm 2a(\pm). \quad (589)$$

Projector products.

$$a(\pm)a(\mp) = \Pi(\pm). \quad (590)$$

Nilpotence.

$$a(\pm)^2 = 0. \quad (591)$$

Action with projectors.

$$a(+)\Pi(+) = 0, \quad (592)$$

$$a(+)\Pi(-) = a(+), \quad (593)$$

$$\Pi(+)a(+) = a(+), \quad (594)$$

$$\Pi(-)a(+) = 0, \quad (595)$$

$$a(-)\Pi(-) = 0, \quad (596)$$

$$a(-)\Pi(+) = a(-), \quad (597)$$

$$\Pi(-)a(-) = a(-), \quad (598)$$

$$\Pi(+)a(-) = 0. \quad (599)$$

Reconstruction.

$$\sigma_1 = a(+) + a(-), \quad (600)$$

$$\sigma_2 = (a(+) - a(-))/i, \quad (601)$$

$$\sigma_3 = \Pi(+) - \Pi(-). \quad (602)$$

Spin-Up and Spin-Down Decomposition

The formulas below make spin-up and spin-down decomposition explicit and collect identities used later in the manuscript. See Eqs. (411) and (412). Here $\mathbf{B}, \mathbf{u} \in \mathbb{R}^3$, $\|\mathbf{u}\| = 1$, and $\Psi(\pm)$ are the projector-selected spin sectors of Ψ . *Let the following notation be fixed.*

$$\mathbf{u} = \mathbf{B}/\|\mathbf{B}\|. \quad (603)$$

Define the projection operators by.

$$\Pi(\pm) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \quad (604)$$

$$\Pi(\pm)^2 = \Pi(\pm), \quad (605)$$

$$\Pi(\pm)\Pi(\mp) = 0, \quad (606)$$

$$\Pi(\pm) + \Pi(\mp) = 1, \quad (607)$$

$$(\mathbf{u} \cdot \boldsymbol{\sigma})\Pi(\pm) = \pm\Pi(\pm) \quad \pm 1 \text{ eigenvalue}, \quad (608)$$

$$\Pi(\pm)^R = \Pi(\pm), \quad (609)$$

$$\Pi(\pm)^* = \Pi(\mp), \quad (610)$$

$$\mathbf{u} \cdot \boldsymbol{\sigma} = (+1)\Pi(+) + (-1)\Pi(-), \quad (611)$$

$$\Psi(\pm) = \Pi(\pm)\Psi, \quad (612)$$

$$(\mathbf{u} \cdot \boldsymbol{\sigma})\Psi(\pm) = \pm\Psi(\pm), \quad (613)$$

$$\Psi = \Psi(+) + \Psi(-) \quad \text{spin decomposition.} \quad (614)$$

The resulting Pauli equation is.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \mathbf{P} &= -i\partial_{\mathbf{r}} - q\mathbf{A}, \\ i\partial_t\Psi &= (\mathbf{P}^2/(2m) + qV)\Psi, \\ &\quad - q\|\mathbf{B}\|(\mathbf{u} \cdot \boldsymbol{\sigma})/(2m)\Psi. \end{aligned} \quad (615)$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} i\partial_t\Psi &= H\Psi, \\ i\partial_t\Psi(\pm) &= i(\partial_t\Pi(\pm))\Psi + \Pi(\pm)(i\partial_t\Psi), \\ &= i(\partial_t\Pi(\pm))\Psi + \Pi(\pm)(H\Psi). \end{aligned} \quad (616)$$

The final numbered line is the derived relation used in the subsequent discussion.

Add a vanishing term to expose the desired decomposition.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}\Pi(\pm)(H\Psi) &= \Pi(\pm)(H\Psi) + (H\Pi(\pm)\Psi - H\Pi(\pm)\Psi), \\ &= H\Psi(\pm) + [\Pi(\pm), H]\Psi.\end{aligned}\tag{617}$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned}i\partial_t\Psi(\pm) &= H\Psi(\pm) + (i(\partial_t\Pi(\pm)) + [\Pi(\pm), H])\Psi, \\ &=: H\Psi(\pm) + S(\pm) \quad \text{coupled.}\end{aligned}\tag{618}$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned}i\partial_t\Psi(\pm) &= (\tfrac{1}{2}(1/m)\mathbf{P}^2 + qV)\Psi(\pm), \\ &\quad - \tfrac{1}{2}(q/m)\|\mathbf{B}\|(\mathbf{u} \cdot \boldsymbol{\sigma})\Psi(\pm) + S(\pm).\end{aligned}\tag{619}$$

The final numbered line is the derived relation used in the subsequent discussion.
The following numbered relation is recorded for later use.

$$S(\pm) = ((i\partial_t\Pi(\pm)) + [\mathbf{P}^2, \Pi(\pm)]/(2m))\Psi.\tag{620}$$

If *uniform static* $\mathbf{B}$, then the following relation holds.

$$S(+) = S(-) = 0 \quad \text{decoupled,}\tag{621}$$

$$(-\mathbf{B} \cdot \boldsymbol{\sigma})\Psi(\pm) = \mp \|\mathbf{B}\|\Psi(\pm).\tag{622}$$

For a uniform static magnetic field, the Zeeman splitting becomes.

$$i\partial_t\Psi(\pm) = (\tfrac{1}{2}(\mathbf{P}^2 \mp q\|\mathbf{B}\|)/m + qV)\Psi(\pm).\tag{623}$$

Four-Sector Decomposition of the Dirac System

The formulas below make four-sector decomposition of the dirac system explicit and collect identities used later in the manuscript. See Eqs. (641) and (642). Here $\epsilon, s \in \{+, -\}$, $\Lambda(\epsilon)$ and $\Pi(s)$ are the energy and spin projectors, and $\Psi(\epsilon, s)$ is the corresponding block-field sector. *Introduce the unit momentum paravector.*

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}U &= dX/ds, \\ &= (E + \mathbf{P} \cdot \boldsymbol{\sigma})/m.\end{aligned}\tag{624}$$

The final numbered line is the derived relation used in the subsequent discussion.
On the mass shell one has.

$$\langle\langle U \rangle\rangle^2 = (E^2 - \mathbf{P}^2)/m^2 = 1,\tag{625}$$

$$W(U)^2 = \text{diag}(\langle\langle U \rangle\rangle^2, \langle\langle U \rangle\rangle^2) = \mathbf{I}.\tag{626}$$

Define the energy projectors by.

$$\text{positiveenergy} \quad \text{matter},\tag{627}$$

$$\text{negativeenergy} \quad \text{antimatter},\tag{628}$$

$$\Lambda(\pm) := \frac{1}{2}(\mathbf{I} \pm W(U)). \quad (629)$$

These operators satisfy the following properties.

$$\Lambda(\pm)^2 = \Lambda(\pm), \quad (630)$$

$$\Lambda(+)\Lambda(-) = 0, \quad (631)$$

$$\Lambda(+) + \Lambda(-) = \mathbf{I}. \quad (632)$$

Define the spin projectors by.

$$\Pi(\pm) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}). \quad (633)$$

Introduce the normalized momentum operator.

$$U \Rightarrow i\partial/m, \quad (634)$$

$$(W(U) - \mathbf{I})\Psi = 0 \quad \text{normalized Dirac.} \quad (635)$$

This is an on-shell projector condition after normalization by mass, not the general Dirac equation.

The resulting solution condition is.

$$\Psi = \Psi(+, +) + \Psi(+, -) + \Psi(-, +) + \Psi(-, -). \quad (636)$$

From these identities, the resulting parameter formulas are.

$$\Psi(\epsilon, s) := \Lambda(\epsilon)\Pi(s)\Psi, \quad (637)$$

$$\epsilon \in +, - \quad \text{energy sector,} \quad (638)$$

$$s \in +, - \quad \text{spin along } \mathbf{u}. \quad (639)$$

VII Spin Structure and Representation Links

After the wave-equation sector, the remaining material branches into spin splitting, matrix realization, and quaternion correspondences. These are best read as representation tools and structural side-results rather than prerequisites for the earlier sections.

Spin projectors, ladder operators, and four sectors

The spin projectors along a unit direction $\mathbf{u} \in \mathbb{R}^3$ are again (15). About the z axis, the ladder operators are algebra elements

$$a(\pm) := \frac{1}{2}(\sigma_1 \pm i\sigma_2), \quad [\sigma_3, a(\pm)] = \pm 2a(\pm), \quad a(\pm)^2 = 0. \quad (640)$$

Using the projector family (15), the proper-velocity paravector $U \in \mathbb{R} \oplus \mathbb{R}^3$ from Eq. (176), and the block map (404), define for $\epsilon, s \in \{+, -\}$

$$\Lambda(\pm) := \frac{1}{2}(\mathbf{I} \pm W(U)), \quad \Psi(\epsilon, s) := \Lambda(\epsilon)\Pi(s)\Psi. \quad (641)$$

This packages positive/negative energy and spin up/down into one algebraic splitting. Writing the four-sector decomposition explicitly helps organize the later interpretation of states. Once energy sign and spin projection are both represented by projectors, the algebra can distinguish the sectors cleanly without introducing a separate layer of notation for each one.

With the same U and W , one also has

$$(W(U) - \mathbf{I})\Psi = 0. \quad (642)$$

Editorial note: Eq. (642) is not the general Dirac equation. It is better read as an on-shell projector relation for a free state satisfying $W(U)^2 = \mathbf{I}$, after momentum has already been normalized by m .

Pauli matrices and matrix realization

Consistent with Eq. (1), the matrix realization is the usual Pauli basis,

$$\sigma_1 = \{0, 1; 1, 0\}, \quad \sigma_2 = \{0, -i; i, 0\}, \quad \sigma_3 = \{1, 0; 0, -1\}, \quad (643)$$

Specializing the real paravector $X = a + \mathbf{A} \cdot \boldsymbol{\sigma}$ from Eq. (9) to $a = t$ and $\mathbf{A} = \mathbf{r}$ gives

$$X = t + \mathbf{r} \cdot \boldsymbol{\sigma} = \{t + r_3, r_1 - ir_2; r_1 + ir_2, t - r_3\}, \quad \det X = t^2 - \mathbf{r}^2. \quad (644)$$

This matrix realization is included to anchor the paravector formalism to a familiar representation. The determinant reproduces the same Minkowski quadratic form already seen in Eq. (175), so the matrix picture confirms rather than replaces the intrinsic STA interpretation.

Quaternion correspondence

The quaternion identification is

$$q_1 = -\sigma_2\sigma_3, \quad q_2 = -\sigma_3\sigma_1, \quad q_3 = -\sigma_1\sigma_2, \quad \mathbf{q} = -i\boldsymbol{\sigma}. \quad (645)$$

For $a \in \mathbb{R}$ and $\mathbf{b} \in \mathbb{R}^3$, let $Q = a + \mathbf{b} \cdot \mathbf{q} = a - i\mathbf{b} \cdot \boldsymbol{\sigma}$. Then

$$QQ^R = a^2 + \mathbf{b}^2, \quad Q^2 = a^2 - \mathbf{b}^2 - 2ia\mathbf{b} \cdot \boldsymbol{\sigma}. \quad (646)$$

The purpose of this correspondence is structural. Quaternion algebra appears here as a restricted sector of the same STA language after multiplication by $-i$, which makes it easier to compare quaternion rotations with the rotor formalism introduced earlier. **A typographical error sometimes appears in the intermediate expansion $Q^2 = a^2 - 2i(\mathbf{b} \cdot \boldsymbol{\sigma}) + (-i)^2(\mathbf{b} \cdot \boldsymbol{\sigma})^2$. The linear term must carry the factor a .**

Representation-Theoretic Complements

The final local development collects the quaternion, generalized-metric, and matrix-realization material that extends the main STA narrative into adjacent representational viewpoints. It remains at the end because these formulas clarify and compare the earlier constructions rather than serving as prerequisites for them.

Quaternion Correspondence

The quaternion correspondence is written out explicitly here to show how it sits inside the same multiplication table already developed for the sigma algebra. See Eqs. (645) and (646). Here $a \in \mathbb{R}$, $\mathbf{b} \in \mathbb{R}^3$, and $Q = a + \mathbf{b} \cdot \mathbf{q} = a - i\mathbf{b} \cdot \boldsymbol{\sigma} \in \mathbb{C} \oplus \mathbb{C}^3$ is the quaternion-paravector correspondence used in this section. The intermediate steps are written out explicitly as follows.

$$\begin{aligned} q_1 &= -\sigma_2\sigma_3, \\ q_2 &= -\sigma_3\sigma_1, \\ q_3 &= -\sigma_1\sigma_2, \\ q_1q_2q_3 &= -1, \\ &= q_1^2, \\ &= q_2^2, \\ &= q_3^2. \end{aligned} \quad (647)$$

The final numbered line is the derived relation used in the subsequent discussion.

Introduce the following vectors.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \mathbf{q} &= (q_1, q_2, q_3), \\ \mathbf{b} &= (b_1, b_2, b_3), \\ \mathbf{q} &= -i\boldsymbol{\sigma}, \\ q^* &= q, \\ q^R &= -q, \\ Q &= a + \mathbf{b} \cdot \mathbf{q}, \\ &= a - i\mathbf{b} \cdot \boldsymbol{\sigma}. \end{aligned} \quad (648)$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned} Q^* &= Q, \\ QQ^* &= Q^2, \\ &= a^2 - 2i(\mathbf{b} \cdot \boldsymbol{\sigma}) + (-i)^2(\mathbf{b} \cdot \boldsymbol{\sigma})^2, \\ &= a^2 - \mathbf{b}^2 - 2ia\mathbf{b} \cdot \boldsymbol{\sigma}. \end{aligned} \quad (649)$$

The final numbered line is the derived relation used in the subsequent discussion.
The following numbered identities fix the notation and formulas used in this block.

$$Q^R = a + i\mathbf{b} \cdot \boldsymbol{\sigma}, \quad (650)$$

$$QQ^R = a^2 + \mathbf{b}^2. \quad (651)$$

The linear term is missing a factor of a ; it should be $-2ia(\mathbf{b} \cdot \boldsymbol{\sigma})$.

The corresponding rotation statement is.

$$\mathbf{r}' \cdot \mathbf{q} = Q(\mathbf{r} \cdot \mathbf{q})Q^R, \quad \text{cf. Eq. (234).} \quad (652)$$

Generalized Spacetime Forms

The formulas below make generalized spacetime forms explicit and collect identities used later in the manuscript. The intermediate steps are written out explicitly as follows.

$$\begin{aligned} X &= f(t, \mathbf{r}) + \mathbf{g}(t, \mathbf{r}) \cdot \boldsymbol{\sigma}, \\ dX &= df + d\mathbf{g} \cdot \boldsymbol{\sigma}, \\ &= (\partial_t f + \partial_t \mathbf{g} \cdot \boldsymbol{\sigma})dt + (\partial_{\mathbf{r}} f + \partial_{\mathbf{r}}(\mathbf{g} \cdot \boldsymbol{\sigma})) \cdot d\mathbf{r}, \\ &=: \mathbf{A}dt + \mathbf{B} \cdot d\mathbf{r}. \end{aligned} \quad (653)$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned} dX/dt &= (\partial_t f + \partial_t \mathbf{g} \cdot \boldsymbol{\sigma}) + (\partial_{\mathbf{r}} f + \partial_{\mathbf{r}}(\mathbf{g} \cdot \boldsymbol{\sigma})) \cdot d\mathbf{r}/dt, \\ &= \mathbf{A} + \mathbf{B} \cdot d\mathbf{r}/dt. \end{aligned} \quad (654)$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned} dX^* &= df - d\mathbf{g} \cdot \boldsymbol{\sigma}, \\ &= (\partial_t f - \partial_t \mathbf{g} \cdot \boldsymbol{\sigma})dt + (\partial_{\mathbf{r}} f - \partial_{\mathbf{r}}(\mathbf{g} \cdot \boldsymbol{\sigma})) \cdot d\mathbf{r}, \\ &=: \mathbf{A}^*dt + \mathbf{B}^* \cdot d\mathbf{r}. \end{aligned} \quad (655)$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned} dX^*/dt &= (\partial_t f - \partial_t \mathbf{g} \cdot \boldsymbol{\sigma}) + (\partial_{\mathbf{r}} f - \partial_{\mathbf{r}}(\mathbf{g} \cdot \boldsymbol{\sigma})) \cdot d\mathbf{r}/dt, \\ &= \mathbf{A}^* + \mathbf{B}^* \cdot d\mathbf{r}/dt. \end{aligned} \quad (656)$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned} \langle\langle dX \rangle\rangle^2 &= (dX/dt)(dX^*/dt)dt^2, \\ &= (\mathbf{A} + \mathbf{B} \cdot d\mathbf{r}/dt)(\mathbf{A}^* + \mathbf{B}^* \cdot d\mathbf{r}/dt)dt^2, \\ &= AA^*dt^2 + (AB^* + BA^*) \cdot d\mathbf{r}dt. \end{aligned} \quad (657)$$

The final numbered line is the derived relation used in the subsequent discussion.
and the remaining quadratic contribution is $(\mathbf{B} \cdot d\mathbf{r})(\mathbf{B}^* \cdot d\mathbf{r})$.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
&= df^2 - dg^2, \\
ds &= \sqrt{df^2 - dg^2}, \\
&= \sqrt{1 - (dg/df)^2} df.
\end{aligned} \tag{658}$$

The final numbered line is the derived relation used in the subsequent discussion.

The relativistic velocity is.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
U &= dX/ds, \\
&= (df + d\mathbf{g} \cdot \boldsymbol{\sigma}) / \sqrt{df^2 - d\mathbf{g}^2}, \\
&= (1 + dg/d\mathbf{f} \cdot \boldsymbol{\sigma}) / \sqrt{1 - (dg/d\mathbf{f})^2}, \\
&= (1 + \mathbf{v} \cdot \boldsymbol{\sigma}) / \sqrt{1 - \mathbf{v}^2}, \\
&= \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}).
\end{aligned} \tag{659}$$

The final numbered line is the derived relation used in the subsequent discussion.

The geometric velocity is.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
\mathbf{v} &= dg/df, \\
&= (dg/dt)/(df/dt), \\
&= (dg/ds)/(df/ds).
\end{aligned} \tag{660}$$

The final numbered line is the derived relation used in the subsequent discussion.

The geometric Lorentz factor is.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
\gamma &= df/ds, \\
&= df / \sqrt{df^2 - dg^2}, \\
&= (df/dt)/(ds/dt), \\
&= 1/\sqrt{1 - \mathbf{v}^2}.
\end{aligned} \tag{661}$$

The final numbered line is the derived relation used in the subsequent discussion.

Note. The timelike condition requires $\mathbf{v}^2 < 1$.

Note. 4 momentum equation is unchanged.

The geometric acceleration is.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
a &= d/ds \mathbf{v}, \\
&= d/ds (dg/df), \\
&= d/df (df/ds) (dg/df), \\
&= \gamma d/df (dg/df), \\
&= \gamma d^2 g / df^2.
\end{aligned} \tag{662}$$

The final numbered line is the derived relation used in the subsequent discussion.

The relativistic acceleration is.

$$dU/ds = \gamma^4 (\mathbf{v} \cdot \mathbf{a}) + (\gamma^4 (\mathbf{v} \cdot \mathbf{a}) \mathbf{v} + \gamma^2 \mathbf{a}) \cdot \boldsymbol{\sigma}. \tag{663}$$

Let the following notation be fixed.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
f' &= df/dt, g' = dg/dt, \\
f'' &= df'/dt, g'' = dg'/dt, \\
d\mathbf{v}/dt &= d/dt(g'/f'), \\
&= (g''f' - g'f'')/(f')^2.
\end{aligned} \tag{664}$$

The final numbered line is the derived relation used in the subsequent discussion.

Consider a plane curve $(x(t), y(t))$.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
k &= |x'y'' - y'x''|/\sqrt{x'^2 + y'^2}^3, \\
[U, dU/ds] &= [U, dU/dt](dt/ds), \\
&= \gamma[U, dU/dt], \\
&= 2\gamma^3(\mathbf{v} \times \mathbf{a}) \cdot (i\boldsymbol{\sigma}).
\end{aligned} \tag{665}$$

The final numbered line is the derived relation used in the subsequent discussion.

For inertial motion, one has

No change in energy or momentum.

$$mdU/ds = 0 \quad \text{geodesic equation,} \tag{666}$$

$$i.e., ddX/ds^2 = 0, \tag{667}$$

$$scalar\ component : \gamma^3(\mathbf{v} \cdot \mathbf{a}) = 0, \tag{668}$$

$$\Rightarrow \mathbf{v} \text{ is perpendicular to } \mathbf{a}, \text{ or } \mathbf{a} = 0, \tag{669}$$

$$vector\ component : \gamma^3(\mathbf{v} \cdot \mathbf{a})\mathbf{v} + \gamma\mathbf{a} = 0. \tag{670}$$

In matrix form.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
M &= \mathbf{I} + \gamma^2 \mathbf{v} \mathbf{v}^T, \\
&= \mathbf{I} + \mathbf{v} \mathbf{v}^T / (1 - \mathbf{v}^2), \mathbf{v}^2 < 1.
\end{aligned} \tag{671}$$

The final numbered line is the derived relation used in the subsequent discussion.

The following numbered identities fix the notation and formulas used in this block.

$$\Rightarrow \text{positive definite,} \tag{672}$$

$$\Rightarrow \gamma \mathbf{M} \cdot \mathbf{a} = 0 \text{ if and only if } \mathbf{a} = 0, \tag{673}$$

$$\Rightarrow d^2g/df^2 = 0 \Rightarrow g(t, \mathbf{r}) = c_1 f(t, \mathbf{r}) + c_2, \tag{674}$$

$$\Rightarrow \text{linear in } (f, g) \text{ coordinates.} \tag{675}$$

Let $\Rightarrow \mathbf{v} = dg/df \Rightarrow c_1 = \mathbf{v}$ denote a constant.

Metric and Diagonalization Notes

For comparison with standard linear-algebra notation, the paravector quadratic form is also written below in matrix language. The final diagonalization claim must still be read together with the red qualification that follows. Here $X \in \mathbb{R} \oplus \mathbb{R}^3$, $dx \in \mathbb{R} \oplus \mathbb{R}^3$, and $Q(x)$ is the matrix-valued bilinear form used for comparison with the intrinsic paravector metric. The intermediate steps are written out explicitly as follows.

$$x = \{t, r_1, r_2, r_3\},$$

$$\begin{aligned}
ds^2 &= \langle\langle dX \rangle\rangle^2, \\
&= dx^T \cdot Q(x) \cdot dx \quad \text{matrix product.}
\end{aligned} \tag{676}$$

The final numbered line is the derived relation used in the subsequent discussion.

The eigenvalue decomposition is.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
Q &= Q^T, \\
\det(Q - \lambda_n \mathbf{I}) &= 0, \\
(Q - \lambda_n \mathbf{I})u_n &= 0, \\
D &= \text{diag}(\lambda_0, \lambda_1, \lambda_2, \lambda_3), \\
&\text{four real eigenvalues,} \\
N &= \{u_0, u_1, u_2, u_3\}, \\
&\text{orthonormal matrix,} \\
N^T N &= N N^T = \mathbf{I} \Rightarrow N^{-1} = N^T, \\
Q^{-1} &= N D^{-1} N^T, \\
Q &= N D N^T, \\
&= \sum_n \lambda_n u_n u_n^T.
\end{aligned} \tag{677}$$

The final numbered line is the derived relation used in the subsequent discussion.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
ds^2 &= d\mathbf{x}^T \cdot \mathbf{Q} \cdot d\mathbf{x}, \\
&= \sum_n \lambda_n (u_n \cdot dx)^2.
\end{aligned} \tag{678}$$

The final numbered line is the derived relation used in the subsequent discussion.

The following numbered identities fix the notation and formulas used in this block.

$$dx'_n = \sqrt{\lambda_n} (u_n \cdot dx), \tag{679}$$

$$ds^2 = d\mathbf{x}' \cdot d\mathbf{x}'. \tag{680}$$

This diagonalized form suppresses the Lorentzian signature; one must keep the indefinite sign structure explicit.

Matrix Realization

The formulas below make matrix realization explicit and collect identities used later in the manuscript. See Eqs. (643) and (644).

The following numbered identities fix the notation and formulas used in this block.

$$\sigma_1 = \{0, 1; 1, 0\}, \tag{681}$$

$$\sigma_2 = \{0, -i; i, 0\}, \tag{682}$$

$$\sigma_3 = \{1, 0; 0, -1\}, \tag{683}$$

$$\sigma_n^2 = \mathbf{I}, \tag{684}$$

$$\sigma_m \sigma_n = -\sigma_n \sigma_m \text{ if } n \neq m, \tag{685}$$

$$\sigma_1 \sigma_2 \sigma_3 = i\mathbf{I}. \tag{686}$$

The spacetime interpretation is.

The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
X &= t + r_1\sigma_1 + r_2\sigma_2 + r_3\sigma_3, \\
&= \{t + r_3, r_1 - ir_2; r_1 + ir_2, t - r_3\}.
\end{aligned} \tag{687}$$

The final numbered line is the derived relation used in the subsequent discussion.
The following numbered identities fix the notation and formulas used in this block.

$$\text{tr}(X) = 2t, \tag{688}$$

$$\det(X) = t^2 - \mathbf{r}^2, \tag{689}$$

$$\lambda^2 - \text{tr}(X)\lambda + \det(X) = 0 \text{ if } \lambda = t \pm \|\mathbf{r}\|. \tag{690}$$

The complex-conjugation rules are.

$$(i)^c = -i, \tag{691}$$

$$\sigma_1^c = +\sigma_1, \tag{692}$$

$$\sigma_2^c = -\sigma_2, \tag{693}$$

$$\sigma_3^c = +\sigma_3. \tag{694}$$

The conjugate-transpose rules are.

$$\sigma_n^{cT} = \sigma_n. \tag{695}$$

For $n \neq m$, the mixed matrix products satisfy
The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
(\sigma_n\sigma_m)^{cT} &= \sigma_m^{cT}\sigma_n^{cT}, \\
&= \sigma_m\sigma_n, \\
&= -\sigma_n\sigma_m.
\end{aligned} \tag{696}$$

The final numbered line is the derived relation used in the subsequent discussion.
The intermediate steps are written out explicitly as follows.

$$\begin{aligned}
(i\mathbf{I})^{cT} &= -i\mathbf{I}, \\
((\sigma_1\sigma_2\sigma_3)\sigma_n)^{cT} &= \sigma_n^{cT}(\sigma_1\sigma_2\sigma_3)^{cT}, \\
&= \sigma_n(i\mathbf{I})^{cT} = -i\sigma_n.
\end{aligned} \tag{697}$$

The final numbered line is the derived relation used in the subsequent discussion.
Reverse operation.

$$\mathbf{I}^R = \mathbf{I}, \tag{698}$$

$$\sigma_n^R = \sigma_n, \tag{699}$$

$$(i\mathbf{I})^R = -i\mathbf{I}, \tag{700}$$

$$(i\sigma_n)^R = -i\sigma_n, \tag{701}$$

$$()^R = ()^{cT} \quad \text{Hermitian adjoint.} \tag{702}$$

Sigma conjugation.

$$\mathbf{I}^* = \mathbf{I}, \tag{703}$$

$$\sigma_n^* = -\sigma_n, \tag{704}$$

$$(i\mathbf{I})^* = -i\mathbf{I}, \quad (705)$$

$$(i\sigma_n)^* = i\sigma_n, \quad (706)$$

$$M = \{0, 1; -1, 0\}, \quad (707)$$

$$M^{-1} = -M, \quad (708)$$

$$()^* = M()^c M^{-1}. \quad (709)$$

Reverse sigma conjugation.

$$()^{*R} = ()^{R*} = M()^T M^{-1}. \quad (710)$$

VIII Literature Verification and Formula Status

The following table records how the principal statements of the manuscript compare with standard literature and with the symbolic checks collected in Appendix A11. Its purpose is to distinguish established reformulations from formulas that require qualification, stronger hypotheses, or explicit correction.

STA statement	Literature correspondence	Status	Notes
Eq. (7)	Pauli/GA product split [9, 11]	Verified	Standard scalar+bivector decomposition.
Eq. (175)	Minkowski interval in paravector form [9, 10]	Verified	Equivalent to $dt^2 - d\mathbf{r}^2$.
Eq. (315)	Compact Maxwell equation in STA [9, 10, 12]	Verified	Sign depends on the chosen F and metric conventions.
Eq. (321)	Lorentz force law in STA packaging [9, 12]	Verified	Combines power and force components into one paravector equation.
Eq. (397)	Klein–Gordon equation [4, 5]	Verified	Standard free relativistic scalar wave equation.
Eq. (400)	Minimal-coupling KG equation [12, 13]	Verified	Operator-domain assumptions should be stated when commuting derivatives and fields.
Eq. (406)	Dirac equation in a 2×2 STA block representation [7, 9, 11]	Verified	Representation-dependent, but equivalent in content once the map to spinor components is fixed.
Eq. (407)	Squared Dirac system after STA factorization [7, 9, 11]	Verified	The upper component carries $-iqF^*$ and the lower component carries $+iqF$ with the operator ordering used here.
Eq. (410)	Leading-order Pauli reduction [6, 14]	Verified with caveat	The displayed form is low-energy and omits higher-order electric-field terms.
Eq. (323)	Electromagnetic energy-momentum density	Needs qualification	Useful frame-dependent object, but not the full stress-energy tensor alone.
Robertson inequality variant	Robertson–Schrödinger uncertainty relation [8]	Incorrect	The displayed expression is not the standard state-variance inequality.
Metric diagonalization line $ds^2 = d\mathbf{x}' \cdot d\mathbf{x}'$	Diagonalization of indefinite quadratic form	Incorrect	Diagonalization does not erase the signature signs; one should keep $\pm$ factors explicit.
Eq. (642)	Free on-shell projector condition	Overstated	It should not be labeled as the general Dirac equation.
Quaternion intermediate line	Direct algebra check	Typo	Missing factor of a in the linear term.

IX Historical Commentary

The physical theories brought together here are historically layered rather than contemporaneous, so the following dates matter for interpretation. The chronology is not decorative: it clarifies that STA is a later unifying language applied to results that

originally emerged in different physical settings and historical moments.

1. 1904: Lorentz published moving-frame transformation formulas for electromagnetism [1].
2. 1905: Einstein recast kinematics in terms of the relativity principle and invariant light speed [2].
3. 1908: Minkowski delivered the “Raum und Zeit” lecture; the standard publication appeared in 1909 [3].
4. 1926: Klein and Gordon independently established the relativistic scalar wave equation [4, 5].
5. 1927: Pauli introduced the nonrelativistic spin-magnetic coupling equation [6].
6. 1928: Dirac produced the relativistic spin- $\frac{1}{2}$ equation, leading to the standard matrix formalism later reexpressed in STA language [7].
7. 1966 and after: Hestenes, Baylis, and Doran–Lasenby gave modern geometric-algebra presentations closely aligned with the spirit of this manuscript [9–11].

X Cautions and Corrections

This section gathers the principal formulas that require correction, reinterpretation, or stronger hypotheses than the surrounding exposition might otherwise suggest. It is intended as part of the mathematical record of the paper rather than as commentary external to it.

1. **The displayed “Robertson inequality” is not the standard Robertson or Robertson–Schrödinger uncertainty relation.**
2. **The metric diagonalization line $ds^2 = dx' \cdot dx'$ drops the indefinite signature and should not be presented as a Lorentzian metric identity.**
3. **The quaternion square expansion has a missing factor of a in the linear term.**
4. **The line $T = \frac{1}{2} F F^R$ should be read as a frame-conditioned energy-density/Poynting paravector, not the full electromagnetic stress-energy tensor.**
5. **The label “normalized Dirac” for $(W(U) - I)\Psi = 0$ is too strong; it is an on-shell projector statement, not the general Dirac equation.**
6. **An earlier sign variant of Eq. (407) with $+iqF^*$ in the upper component is inconsistent with the operator ordering in $W(i\partial - q\Phi^*)$; the corrected sign is $-iqF^*$.**
7. **Several operator equalities stated in differential form implicitly assume smooth test functions and benign boundary behavior; those hypotheses should be stated whenever commutators of differential operators are used rigorously.**

A Derivation Notes Referencing Main Equations

The appendix records derivations that support the main development without interrupting its pace. Each item points back to a labeled equation already introduced in context, so these notes should be read as expansions of the exposition rather than as a second independent presentation.

A1. Product identity for Eq. (7)

Expand

$$(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) = \sum_{n,m} A_n B_m \sigma_n \sigma_m$$

and substitute Eq. (2). Diagonal terms produce $\mathbf{A} \cdot \mathbf{B}$, while antisymmetric off-diagonal terms combine into $i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}$.

A2. Quadratic forms and interval for Eqs. (12) and (175)

For a real paravector $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$,

$$XX^* = (t + \mathbf{r} \cdot \boldsymbol{\sigma})(t - \mathbf{r} \cdot \boldsymbol{\sigma}) = t^2 - \mathbf{r}^2.$$

The same algebra applied to $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ gives Eq. (12).

A3. Exponentials and projectors for Eqs. (13)–(16)

If $\mathbf{u}^2 = 1$, then $(\mathbf{u} \cdot \boldsymbol{\sigma})^2 = 1$ and $(-i\mathbf{u} \cdot \boldsymbol{\sigma})^2 = -1$. Splitting the exponential series into even and odd powers gives the hyperbolic and trigonometric forms. The projector formula follows from the minimal polynomial $(x - 1)(x + 1) = 0$.

A4. *Maxwell split for Eq. (315)*

With $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$,

$$\begin{aligned}\partial F &= (\partial_t + \boldsymbol{\sigma} \cdot \boldsymbol{\nabla})(\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma} \\ &= (\boldsymbol{\nabla} \cdot \mathbf{E}) + i(\boldsymbol{\nabla} \cdot \mathbf{B}) \\ &\quad + (\partial_t \mathbf{E} - \boldsymbol{\nabla} \times \mathbf{B} + i(\partial_t \mathbf{B} + \boldsymbol{\nabla} \times \mathbf{E})) \cdot \boldsymbol{\sigma}.\end{aligned}$$

Matching scalar and vector parts with $\rho - \mathbf{J} \cdot \boldsymbol{\sigma}$ yields Eqs. (316)–(317).

A5. *Lorentz force for Eq. (321)*

Take $U = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma})$ and multiply by $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$. Using Eq. (7),

$$\frac{FU}{\gamma} = \mathbf{E} \cdot \mathbf{v} + (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \boldsymbol{\sigma} + i(\mathbf{B} \cdot \mathbf{v} + (\mathbf{B} + \mathbf{E} \times \mathbf{v}) \cdot \boldsymbol{\sigma}).$$

Keeping only the non- i part produces Eq. (321).

A6. *Minimal coupling and factorization for Eqs. (401) and (403)*

Applying $E = i\partial_t - qV$ and $\mathbf{P} = -i\boldsymbol{\nabla} - q\mathbf{A}$ to a test function gives

$$\begin{aligned}E^2\Psi &= (-\partial_t^2 - iq\partial_t V - 2iqV\partial_t + q^2V^2)\Psi, \\ \mathbf{P}^2\Psi &= (-\boldsymbol{\nabla}^2 + iq\boldsymbol{\nabla} \cdot \mathbf{A} + 2iq\mathbf{A} \cdot \boldsymbol{\nabla} + q^2\mathbf{A}^2)\Psi,\end{aligned}$$

which is Eq. (401). The commutators in Eq. (402) then yield Eq. (403).

A7. *Dirac square and gauge covariance for Eqs. (407) and (408)*

From Eq. (404),

$$W(i\partial - q\Phi^*)^2 = -\text{diag}\left((\partial + iq\Phi^*)(\partial^* + iq\Phi), (\partial^* + iq\Phi)(\partial + iq\Phi^*)\right).$$

Writing $Z = i\partial - q\Phi^* = E - \boldsymbol{\sigma} \cdot \mathbf{P}$ and $Z^{*R} = i\partial^* - q\Phi = E + \boldsymbol{\sigma} \cdot \mathbf{P}$, one has

$$ZZ^{*R} = E^2 - \mathbf{P}^2 + iqF^*, \quad Z^{*R}Z = E^2 - \mathbf{P}^2 - iqF.$$

Since $Z\Psi_2 = m\Psi_1$ and $Z^{*R}\Psi_1 = m\Psi_2$, elimination gives

$$(-E^2 + \mathbf{P}^2 + m^2 - iqF^*)\Psi_1 = 0, \quad (-E^2 + \mathbf{P}^2 + m^2 + iqF)\Psi_2 = 0.$$

This is Eq. (407). For gauge covariance, differentiate $e^{iq\lambda}\Psi'$ and use $\Phi' = \Phi + \partial^*\lambda$ to recover the same operator form.

A8. *Pauli limit for Eq. (410)*

After factoring out the rest oscillation e^{-imt} and eliminating the small component of the Dirac pair, one obtains

$$0 \approx (2mE - \mathbf{P}^2 + q\mathbf{B} \cdot \boldsymbol{\sigma})\Psi.$$

Solving for $i\partial_t\Psi$ gives Eq. (410). Terms omitted at this order are $O(1/m^2)$ and include electric-field corrections.

A9. *Rotors, boosts, and field transforms for Eqs. (234)–(236)*

Equation (234) follows from the bivector exponential (14); Eq. (235) follows from the vector exponential (13). Conjugating F by the boost then reproduces the standard parallel/perpendicular transformation of $\mathbf{E}$ and $\mathbf{B}$.

A10. *Quaternion square for Eq. (646)*

Using $\mathbf{q} = -i\boldsymbol{\sigma}$,

$$Q^2 = (a - i\mathbf{b} \cdot \boldsymbol{\sigma})^2 = a^2 - 2ia\mathbf{b} \cdot \boldsymbol{\sigma} - (\mathbf{b} \cdot \boldsymbol{\sigma})^2 = a^2 - \mathbf{b}^2 - 2ia\mathbf{b} \cdot \boldsymbol{\sigma}.$$

This exposes the missing factor of a in the intermediate line.

A11. Python Verification of Main Algebraic and Operator Identities

The following checks were carried out symbolically in Python using SymPy and the standard Pauli-matrix realization of the sigma basis. Algebraic identities were reduced directly in matrix form, while differential identities were checked by acting on generic test functions $\psi(t, x, y, z)$ with symbolic potentials and field components. The verification script is `tools/verify_sta_results.py`.

Check	Status	Detail
Eqs. (1)–(2)	Verified	All nine basis products satisfy $\sigma_n \sigma_m = \delta_{nm} + \epsilon_{nmk} i \sigma_k$ in the Pauli realization.
Eq. (7)	Verified	The product $(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma})$ reduces to $\mathbf{A} \cdot \mathbf{B} + i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}$ for symbolic components A_n, B_n .
Eqs. (10)–(12)	Verified	The complex paravector product, the Minkowski-type form $ZZ^{*\mathbf{R}} = S^2 - \mathbf{V}^2$, and the Euclidean scalar part $\text{scalar}(ZZ^{\mathbf{R}}) = S ^2 + \ \mathbf{V}\ ^2$ all reduce correctly.
Eqs. (15), (176), (177)	Verified	The idempotents $\Pi(\pm)$ satisfy the projector identities, $UU^{*\mathbf{R}} = 1$, and $E^2 - \mathbf{P}^2 = m^2$ follows from $mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}$.
Eqs. (313), (314), (315)	Verified	The potential identity gives $S = \partial_t V + \nabla \cdot \mathbf{A}$, $\mathbf{E} = -\partial_t \mathbf{A} - \nabla V$, and $\mathbf{B} = \nabla \times \mathbf{A}$, while ∂F splits into Gauss, Faraday, and Ampère–Maxwell component equations with the manuscript’s signs.
Eqs. (320) and (321)	Verified	The gauge law $\Phi' = \Phi + \partial^* \lambda$ gives $V' = V + \partial_t \lambda$ and $\mathbf{A}' = \mathbf{A} - \nabla \lambda$, and the product FU splits into the expected power term $\gamma \mathbf{v} \cdot \mathbf{E}$ and force term $\gamma(\mathbf{E} + \mathbf{v} \times \mathbf{B})$.
Eqs. (401), (402), and (403)	Verified	Acting on a generic test function reproduces the expanded Klein–Gordon operator, the commutators $[E, P_m] = iqE_m$ and $[P_m, P_n] = iq(\partial_m A_n - \partial_n A_m)$, and the factorization $(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E - \boldsymbol{\sigma} \cdot \mathbf{P}) = E^2 - \mathbf{P}^2 - iqF$.
Eqs. (407), (409), and (412)	Verified	Eliminating one Dirac component yields the two squared equations; with the manuscript’s operator ordering, the upper component carries $-iqF^*$ after everything is moved to the left-hand side, while the lower component carries $+iqF$. The exact Pauli identity $(\boldsymbol{\sigma} \cdot \mathbf{P})^2 = \mathbf{P}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma}$ holds on a test function; and for uniform $\mathbf{B} = B_0 \mathbf{u}$ the projectors satisfy $(\mathbf{B} \cdot \boldsymbol{\sigma})\Pi(\pm) = \pm B_0 \Pi(\pm)$.
Eq. (646)	Verified	The quaternion correspondence satisfies $QQ^{\mathbf{R}} = a^2 + \mathbf{b}^2$ and $Q^2 = a^2 - \mathbf{b}^2 - 2ia \mathbf{b} \cdot \boldsymbol{\sigma}$; the omitted factor of a in the older intermediate line is indeed an error.

These Python checks do not replace the literature comparison in Section VIII, but they do separate purely algebraic and operator-theoretic identities from the physically interpretive claims that still require convention-aware reading.

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